Biochemistry



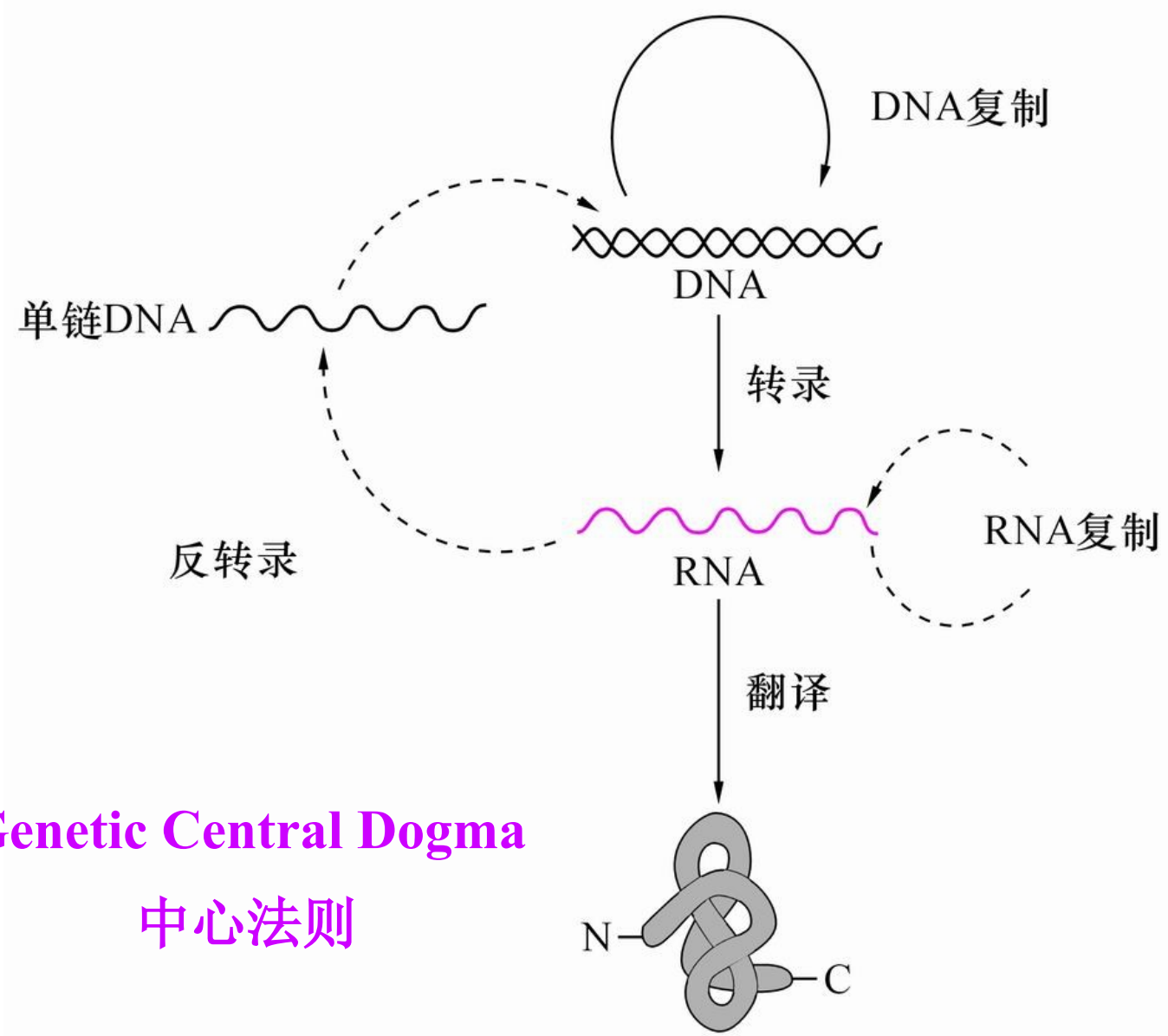
Section F
DNA structure and replication

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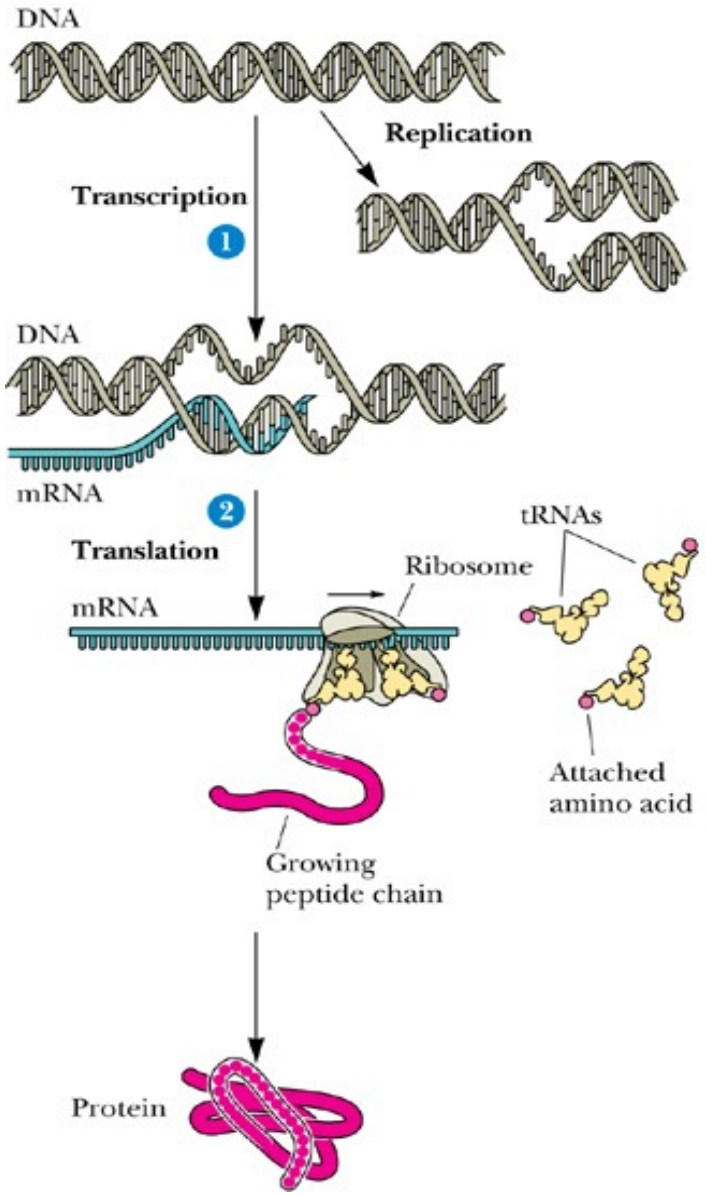
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13854808762



Genetic Central Dogma

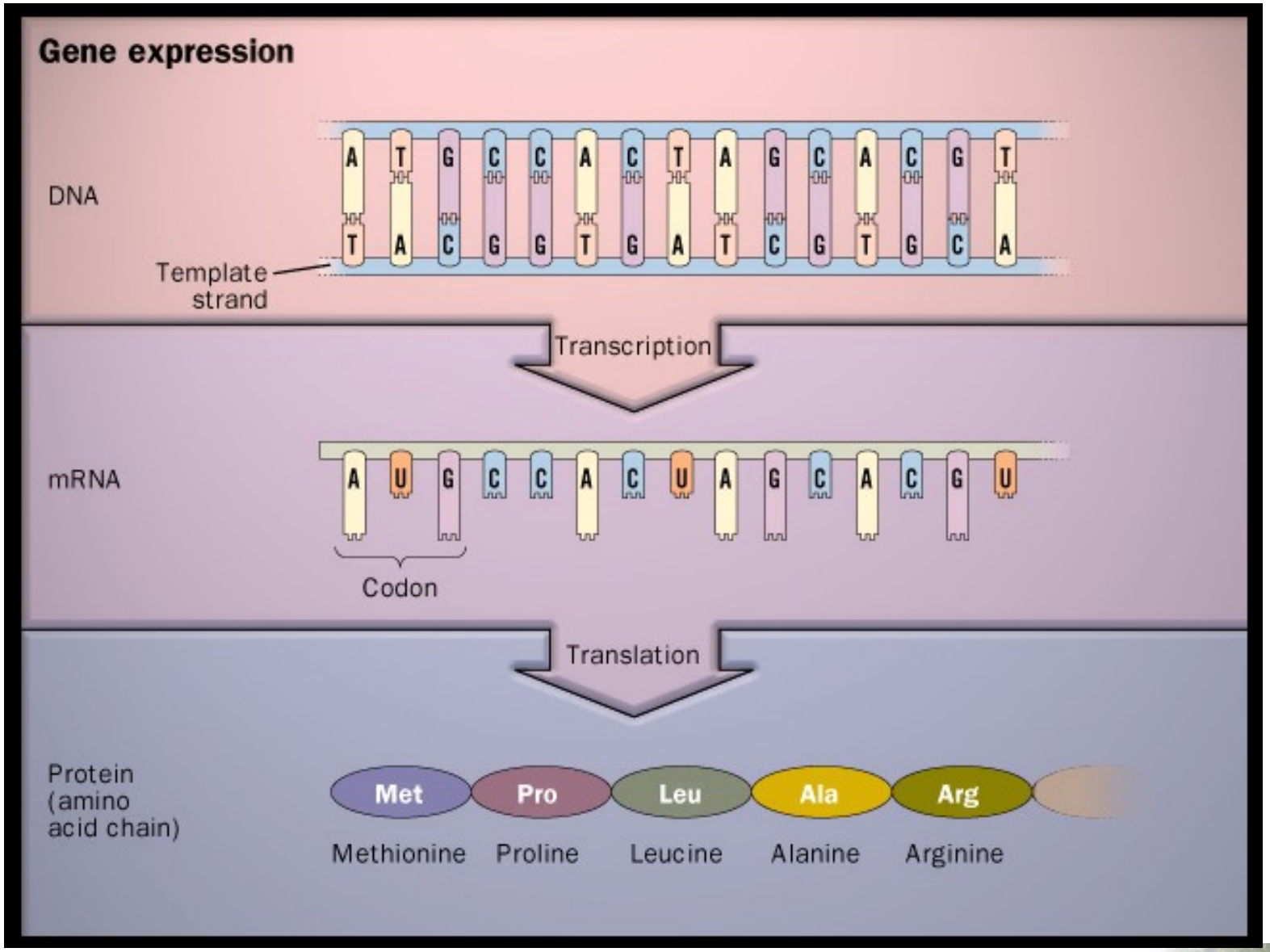
中心法则



Replication
DNA replication yields two DNA molecules identical to the original one, ensuring transmission of genetic information to daughter cells with exceptional fidelity.

Transcription
The sequence of bases in DNA is recorded as a sequence of complementary bases in a single-stranded mRNA molecule.

Translation
Three-base codons on the mRNA corresponding to specific amino acids direct the sequence of building a protein. These codons are recognized by tRNAs (transfer RNAs) carrying the appropriate amino acids. Ribosomes are the "machinery" for protein synthesis.


Gene expression
DNA
Template strand
A T G C C A C T A G C A C G T
T A C G G T G A T C G T G C A
Transcription
mRNA
A U G C C A C U A G C A C G U
Codon
Translation
Protein (amino acid chain)
Met
Pro
Leu
Ala
Arg
Methionine
Proline
Leucine
Alanine
Arginine

Section F DNA structure and replication

F1 DNA (RNA) structure
F2 Chromosomes
F3 DNA replication in bacteria
F4 DNA replication in eukaryote



Section F1 and G1
DNA (RNA) structure

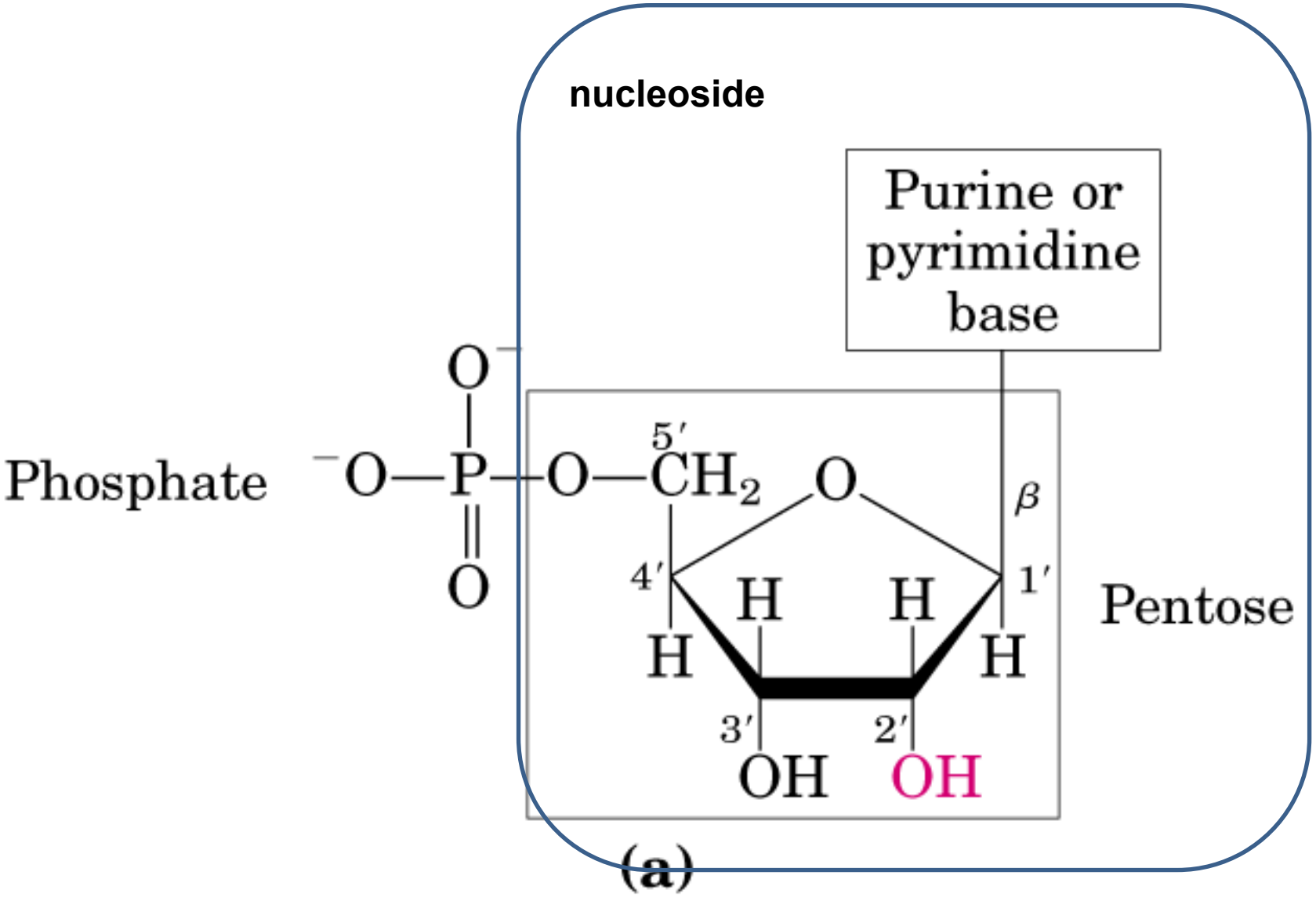


1. The nucleic acids, deoxyribonucleic acid (DNA) and ribonucleic acid (RNA), are polymers of nucleotide units.

1.1 DNA consists of four kinds of deoxyribonucleotide units linked together through covalent bonds

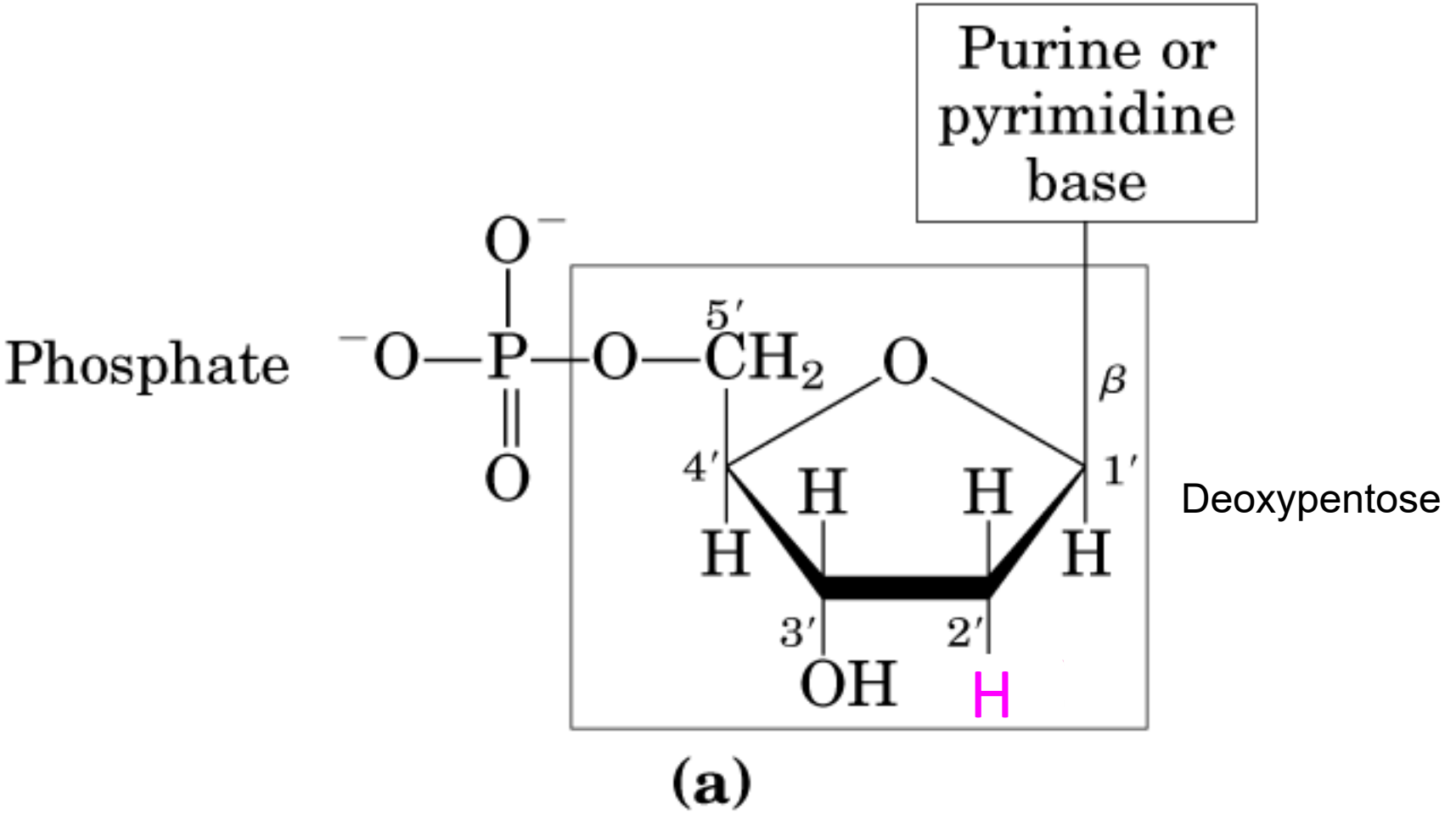
1.1.1 Each nucleotide unit is made of a nitrogenous base (the various part in the four different deoxyribonucleotides), a pentose sugar, and a phosphate group.





ribonucleotide

nucleic acids

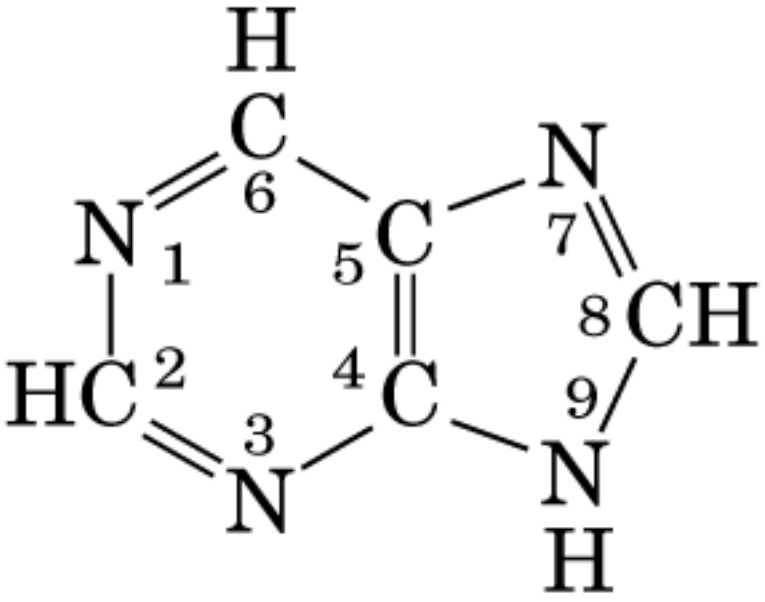


deoxyribonucleotide

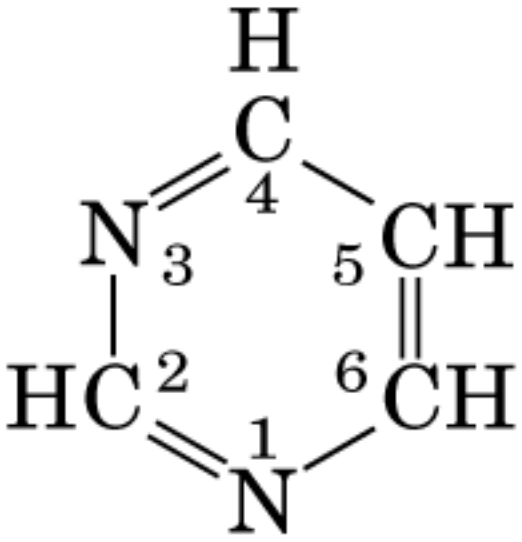
1.1.2 The nitrogenous base can be adenine (A), guanine (G), cytosine (C), or thymine (T) (uracil (U) in RNA).

1.1.3 The nitrogenous bases are derivatives of two parent compounds, pyrimidine （嘧啶） and purine （嘌呤） .

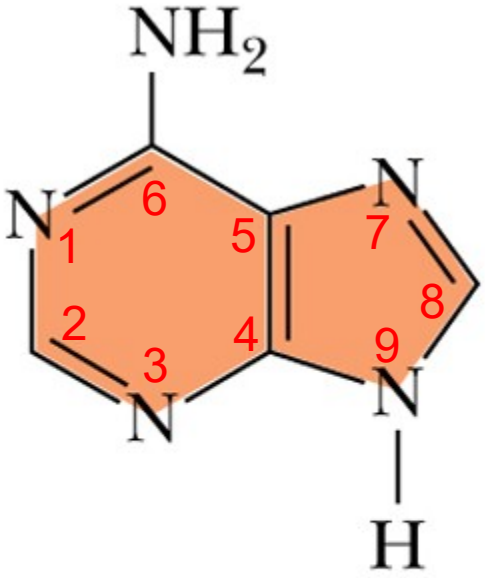
1.1.4 The carbon and nitrogen atoms in the purine and pyrimidine rings are numbered.



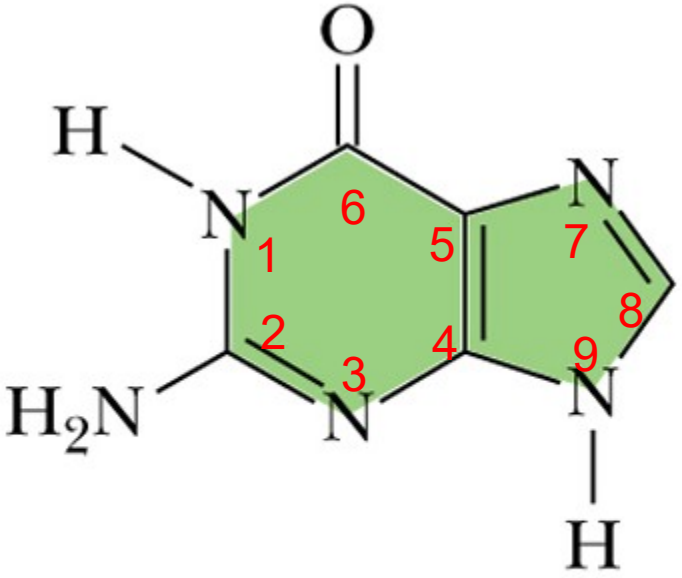
Purine



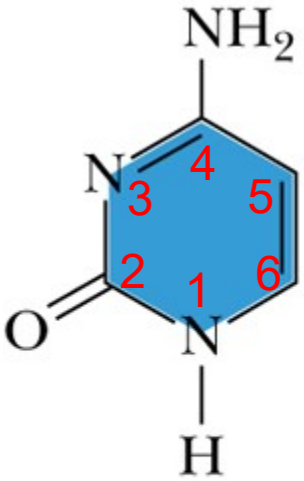
Pyrimidine



Adenine
(6-amino purine)

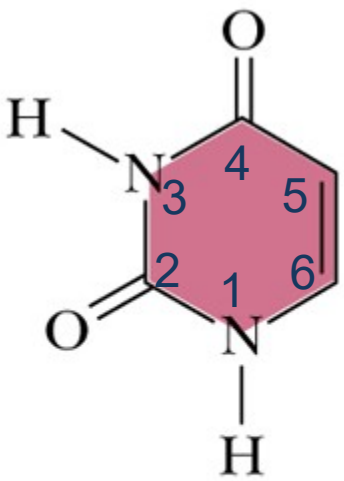


Guanine
(2-amino-6-oxy purine)



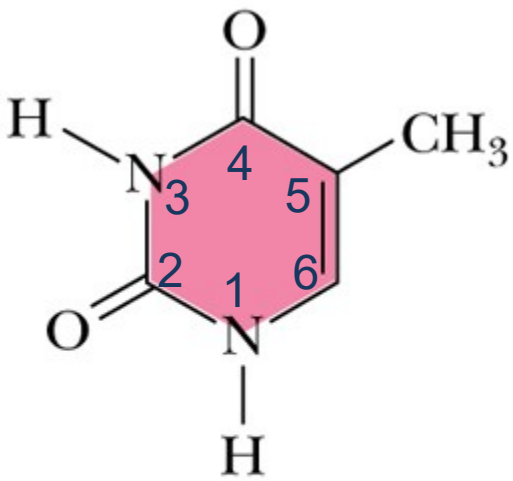
Cytosine (2-oxy-4-amino pyrimidine)

胞嘧啶



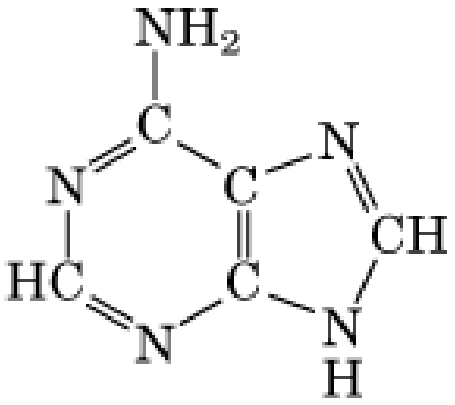
Uracil (2-oxy-4-oxy pyrimidine)

脲嘧啶

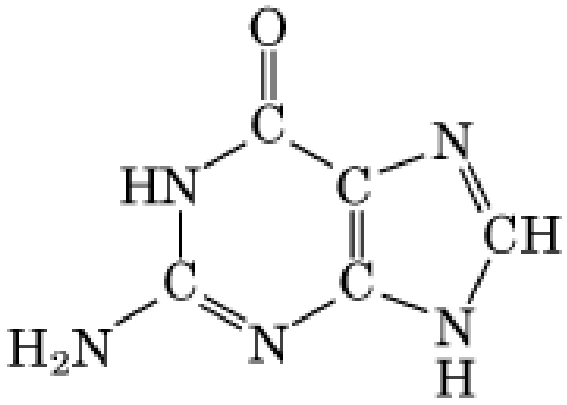


Thymine (2-oxy-4-oxy 5-methyl pyrimidine)

胸腺嘧啶

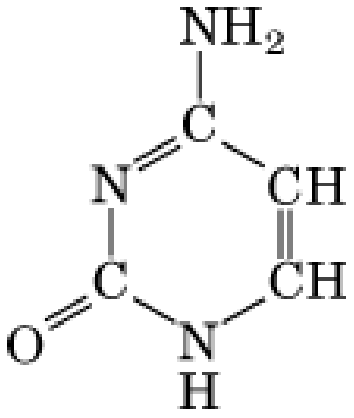


Adenine

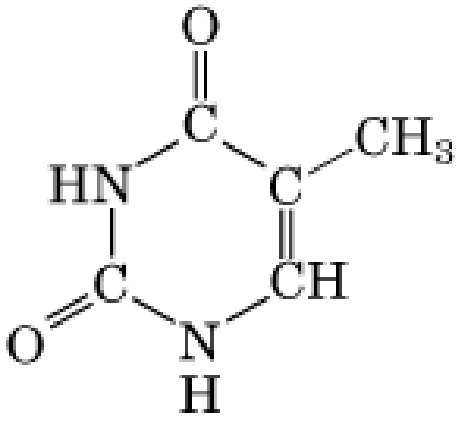


Guanine

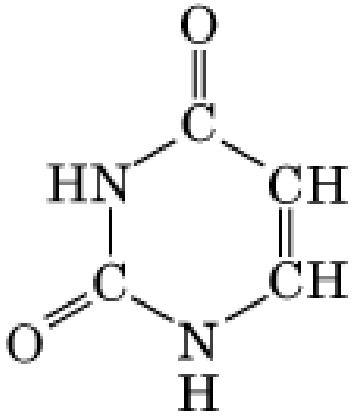
Purines



Cytosine



Thymine
(DNA)



Uracil
(RNA)

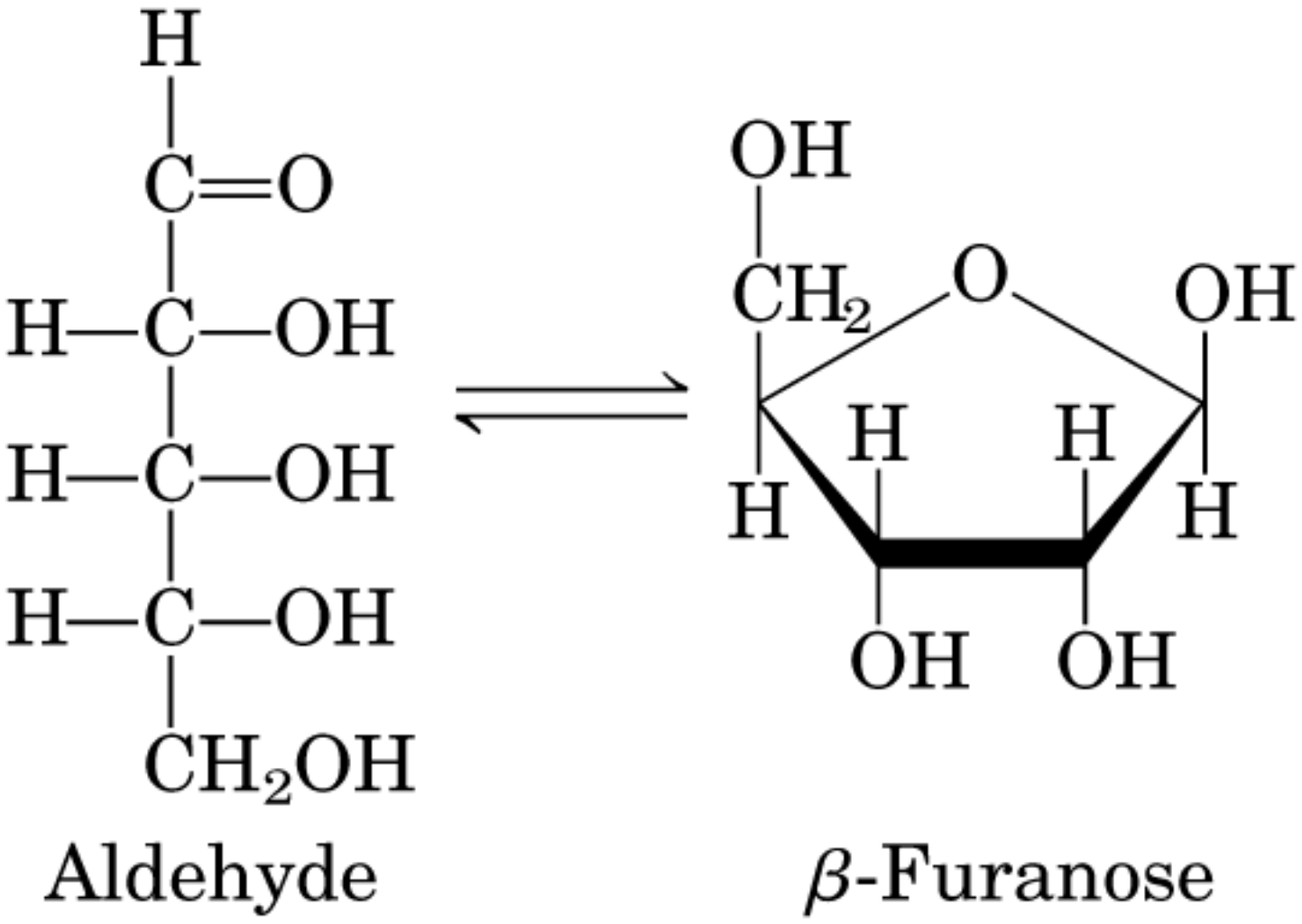
Pyrimidines

1.1.5 The pentose in a deoxyribonucleotide is a deoxyribose, which lacks an oxygen atom at the 2'-position that is present in ribose, the parent compound.

1.1.6 The deoxyribose is in its β-furanose form (a closed five-member ring).

1.1.7 Only D-deoxyribose is found in DNA.

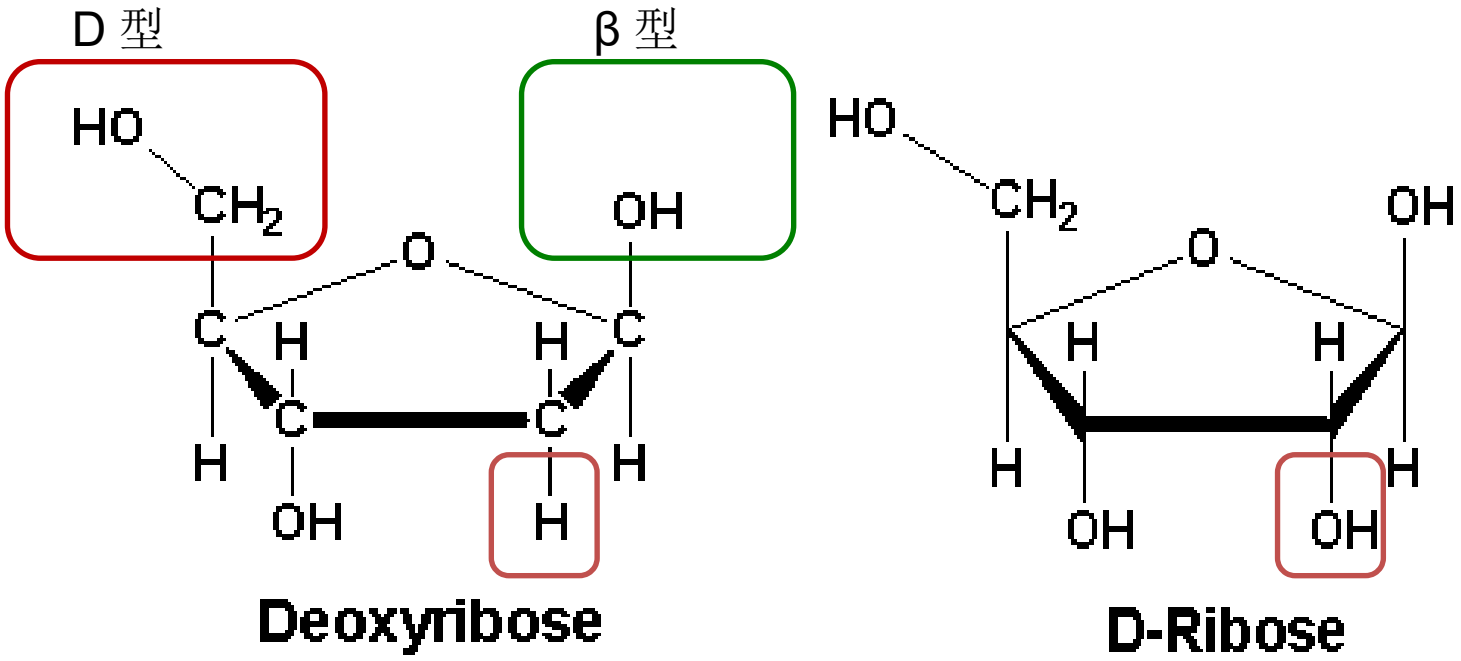



H
C=O
H—C—OH
H—C—OH
H—C—OH
CH2OH
Aldehyde
OH
CH2
O
OH
H
H
H
H
OH
OH
β-Furanose

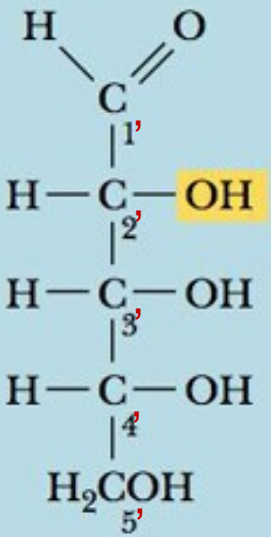
(a)

呋喃糖

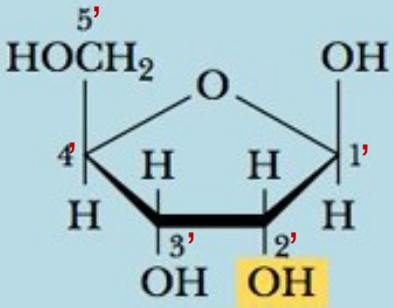
Deoxyribose and Ribose



The atoms in the 5-carbon sugar are also numbered.

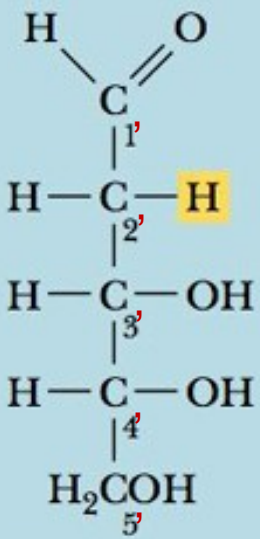


D-Ribose

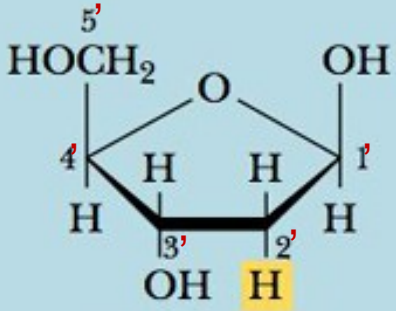


Furanose form of D-Ribose

β-D-Ribofuranose



D-2-Deoxyribose

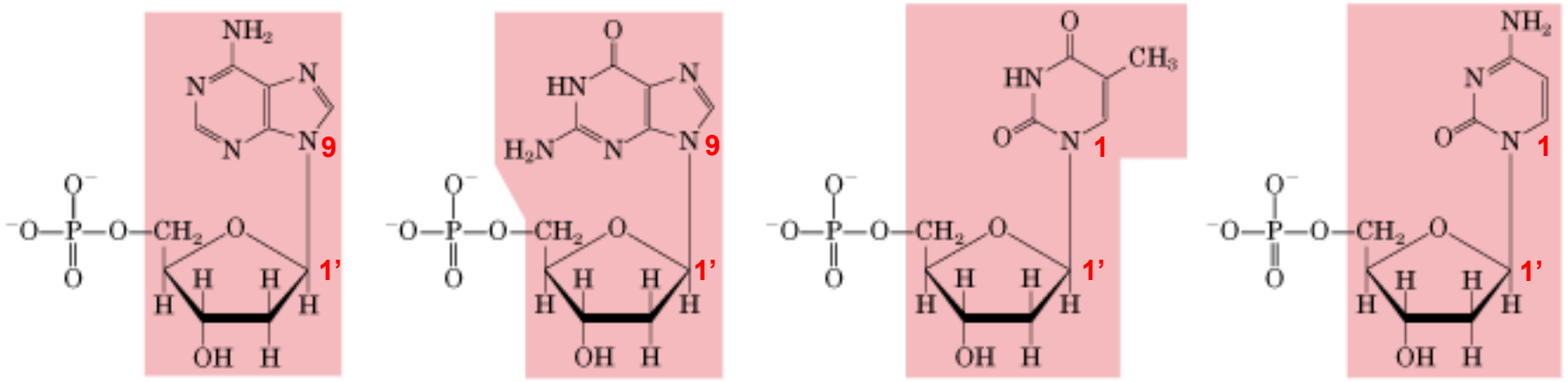


Furanose form of D-2-Deoxyribose

β-D-2-Deoxyribofuranose

1.1.8 Each pyrimidine is covalently linked (through a N-glycosidic bond) to the 1’ carbon of the deoxyribose at N-1 of the pyrimidine, and each purine is covalently linked to the 1’ carbon of the deoxyribose at N-9 of the purine.





nucleoside（核苷）

1.1.9 The phosphate group is esterified to the -OH group on the 5’ carbon of the deoxyribose ring.

1.1.10 A nucleotide lacking the phosphate part is called a **nucleoside**（核苷）.

1.1.11 The four nucleoside units in DNA are called **deoxyadenosine, deoxyguanosine, deoxythymidine, and deoxycytidine**.

1.1.12 The nitrogenous base can be **adenine (A), guanine (G), cytosine (C), or thymine (T) (uracil (U) in RNA).**



1.1.13 The four nucleotide units in DNA are called deoxyadenosine 5'-monophosphate (dAMP, or deoxyadenylate), deoxyguanosine 5'-monophosphate (dGMP, or deoxyguanylate), deoxythymidine 5'-monophosphate (dTMP, or deoxythymidylate), and deoxycytidine 5'-monophosphate (dCMP, or deoxycytidylate).

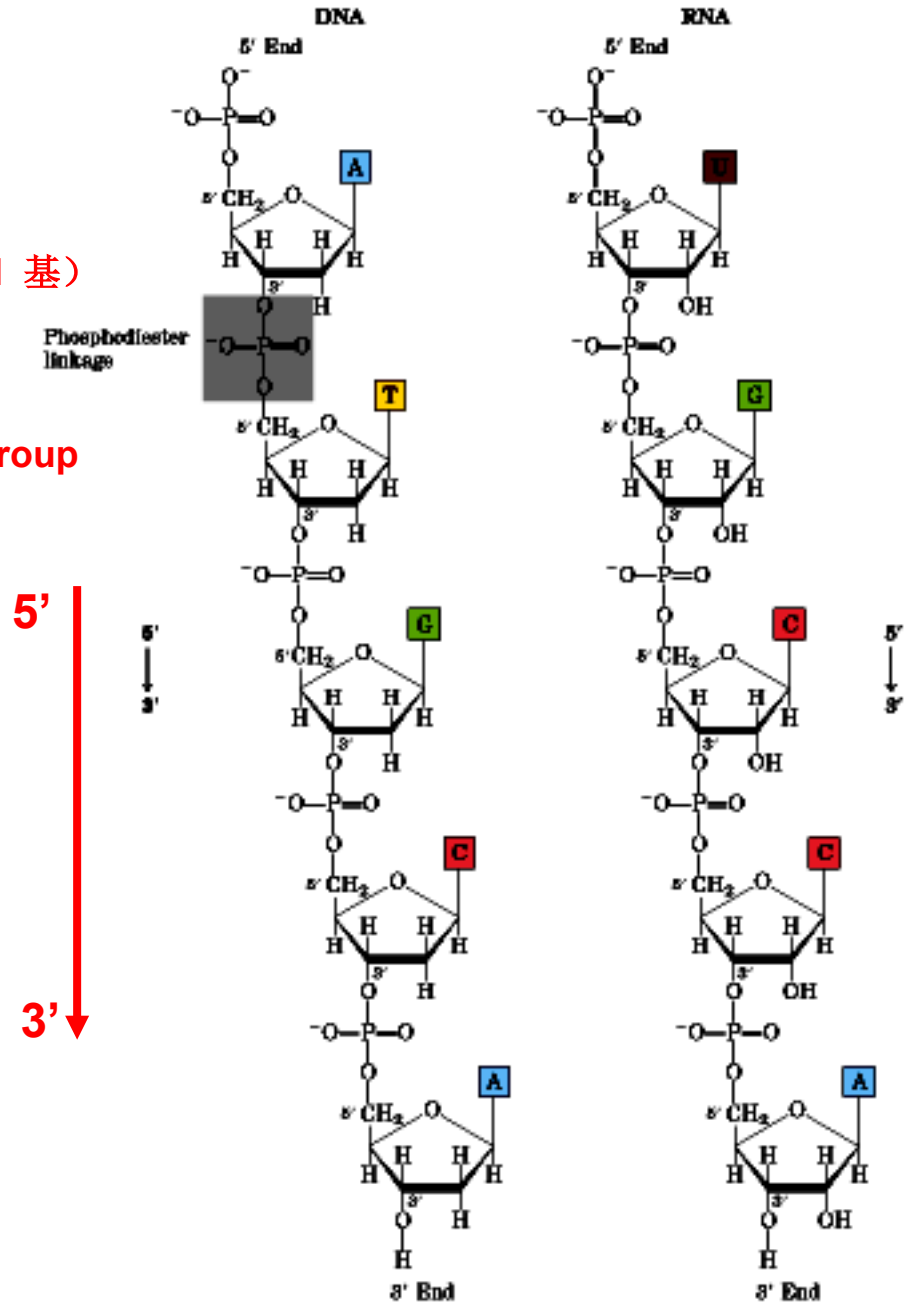
1.1.14 In DNA the nucleotides are covalently joined together by 3',5'-phosphodiester bonds to form a repetive sugar-phoshate chain which is the backbone to which the bases are attached.



3’ hydroxyl group (3’-OH 基）

phosphodiester

5’phosphate group



Biochemistry

1.2 RNA also consists of four different kinds of ribonucleotides.

1.2.1 Each ribonucleotide unit is also made of three parts: a nitrogenous base, a pentose, and a phosphate group.

1.2.2 The base part is adenine, guanine, cytosine or uracil.

1.2.3 Uracil exists only in RNA, and thymine only in DNA.

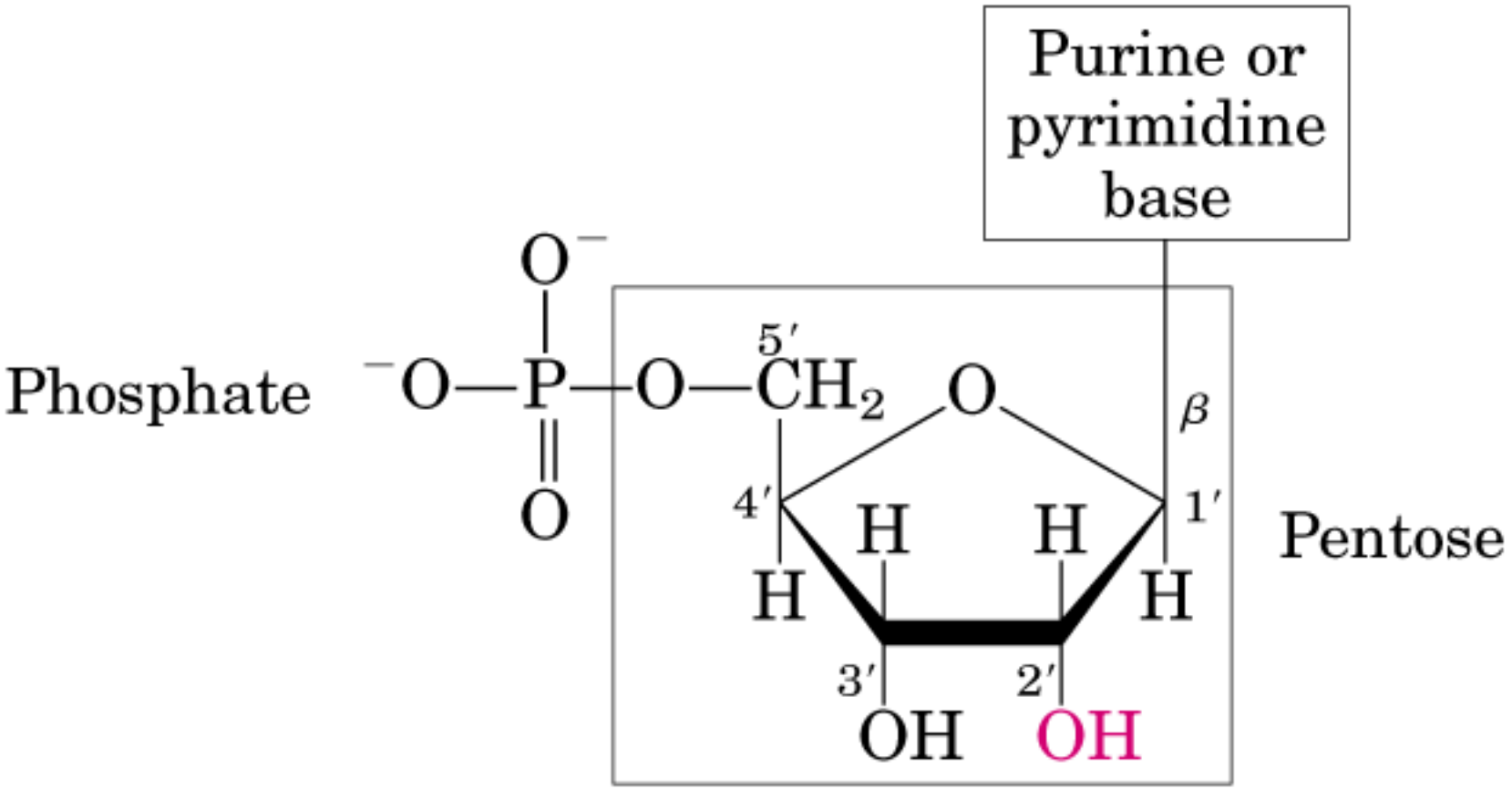


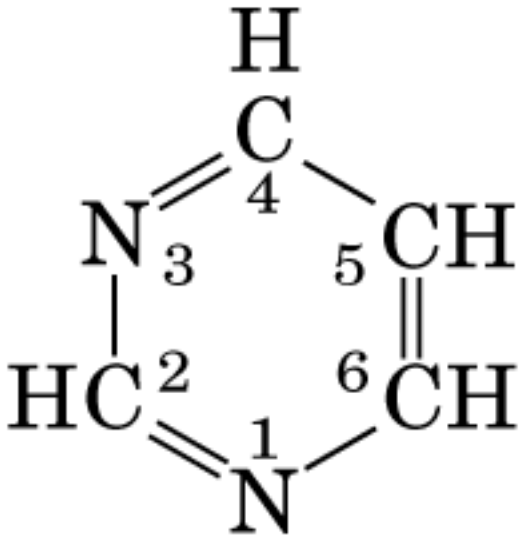
1.2.4 The pentose part is a ribose (without being deoxygenated at the 2' position) in its β-furanose form (as deoxyribose in deoxyribonucleotides).

1.2.5 The bases and the phosphate group are covalently linked to the ribose ring in the same ways as in deoxyribonucleotides.

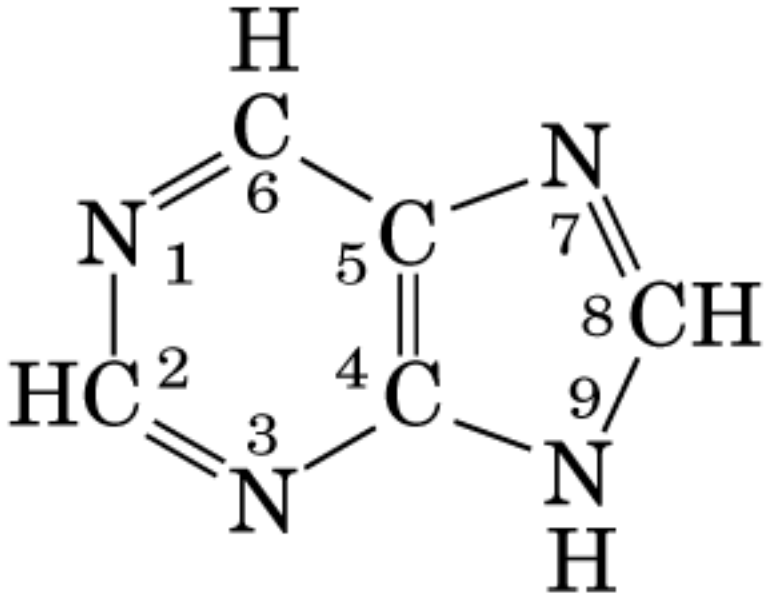
1.2.6 The four nucleoside units in RNA are called adenosine, guanosine, cytidine, and uridine (without deoxy- suffix); and the nucleotide units are AMP, GMP, CMP, and UMP.




Purine or pyrimidine base
Phosphate
Pentose
β
(a)

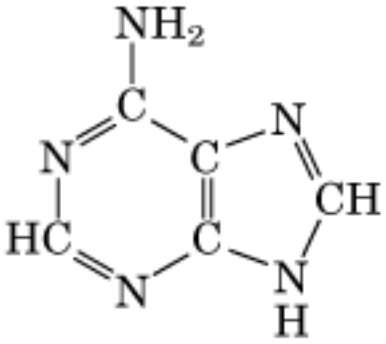


Pyrimidine

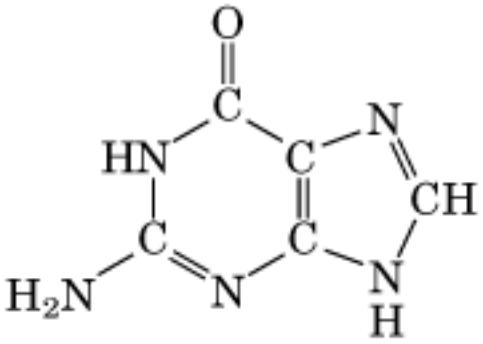


Purine

(b)

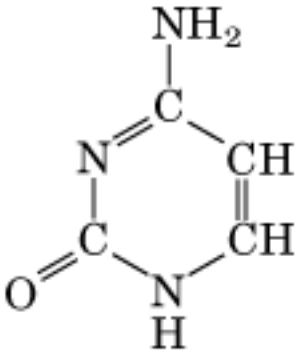


Adenine

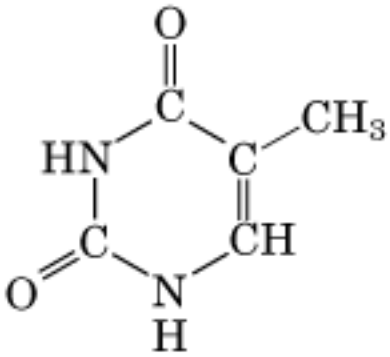


Guanine

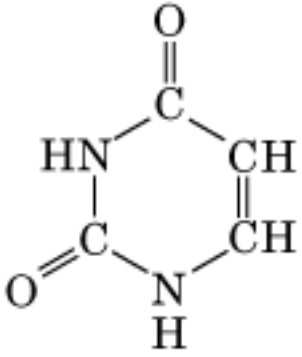
Purines



Cytosine



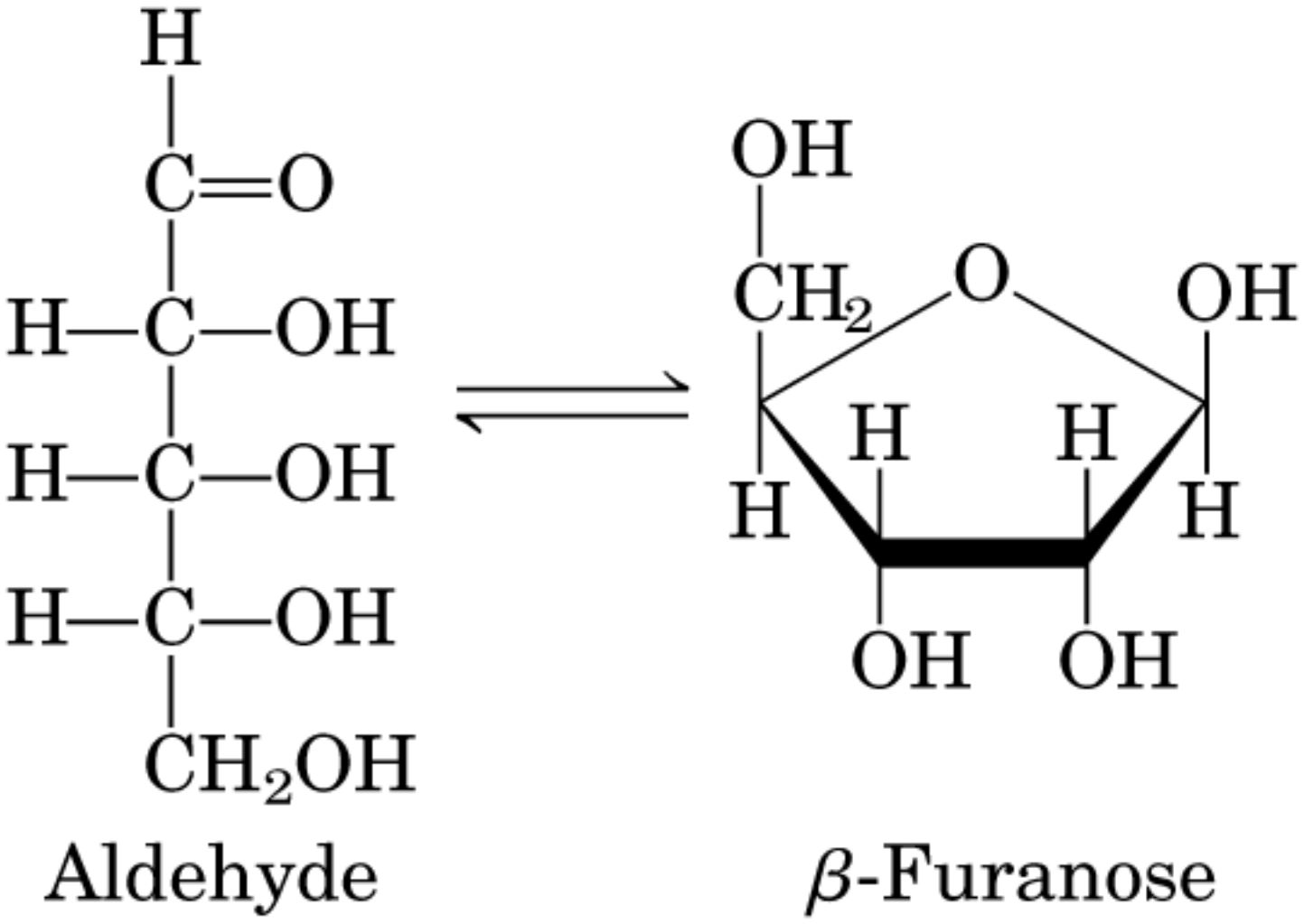
Thymine
(DNA)



Uracil
(RNA)

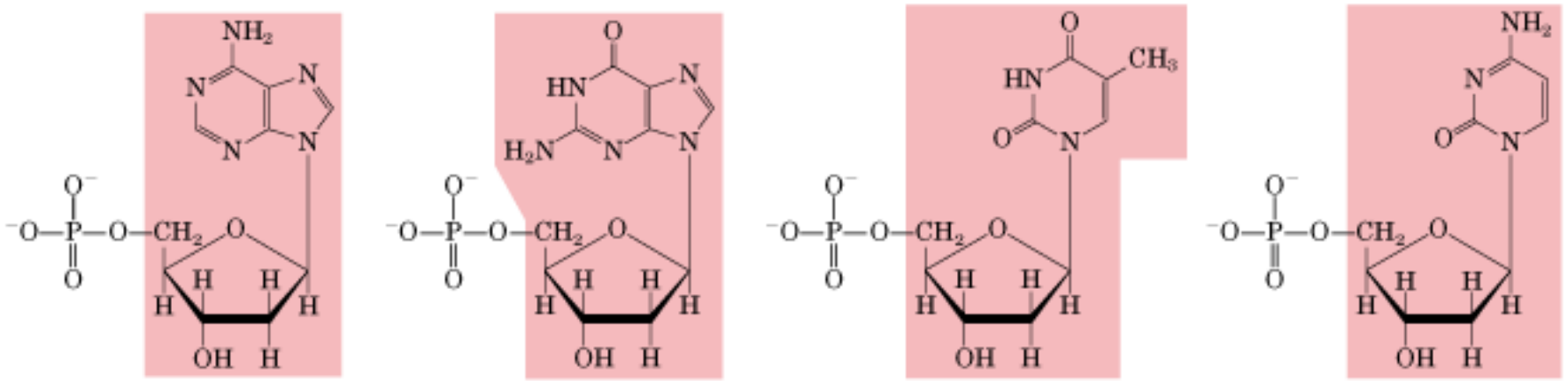
Pyrimidines





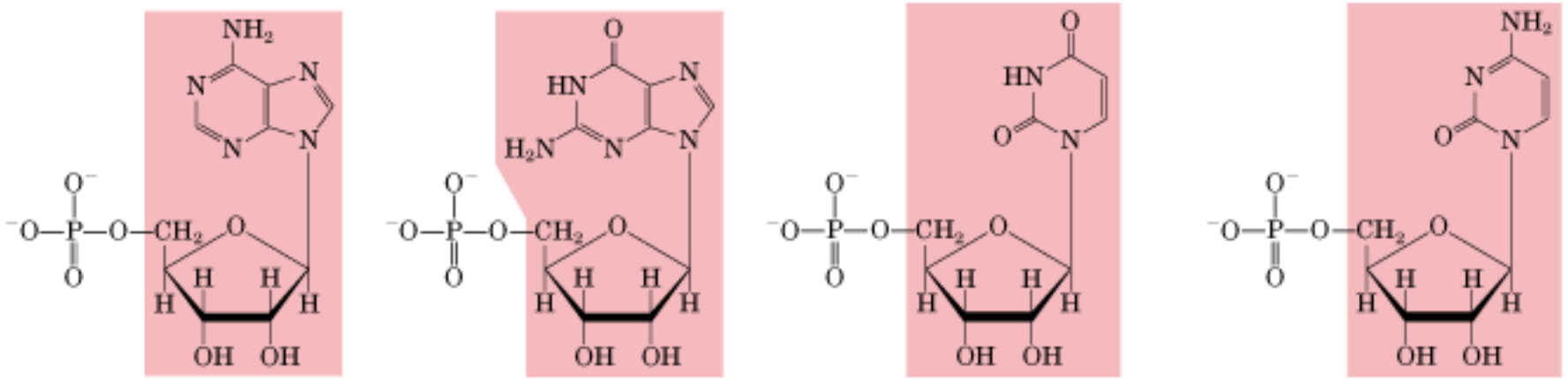
Aldehyde $\rightleftharpoons$ β-Furanose

(a)

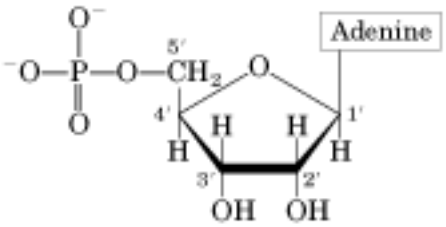


	Deoxyadenylate (deoxyadenosine 5′-monophosphate)	Deoxyguanylate (deoxyguanosine 5′-monophosphate)	Deoxythymidylate (deoxythymidine 5′-monophosphate)	Deoxycytidylate (deoxycytidine 5′-monophosphate)
Nucleotide:				
Symbols:	A, dA, dAMP	G, dG, dGMP	T, dT, dTMP	C, dC, dCMP
Nucleoside:	Deoxyadenosine	Deoxyguanosine	Deoxythymidine	Deoxycytidine

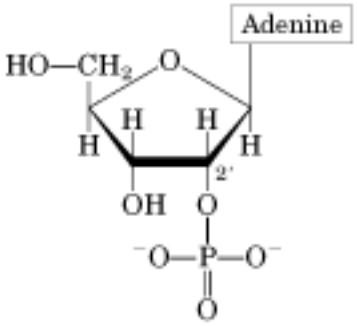
(a) Deoxyribonucleotides



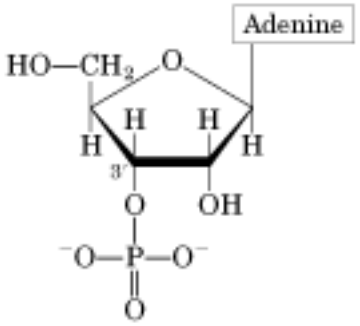
(b) Ribonucleotides



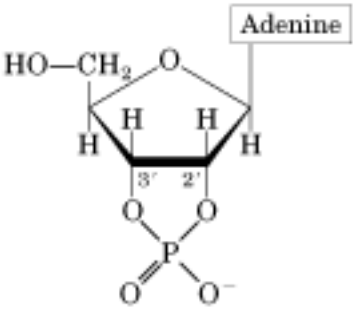
Adenosine 5′-monophosphate



Adenosine 2′-monophosphate

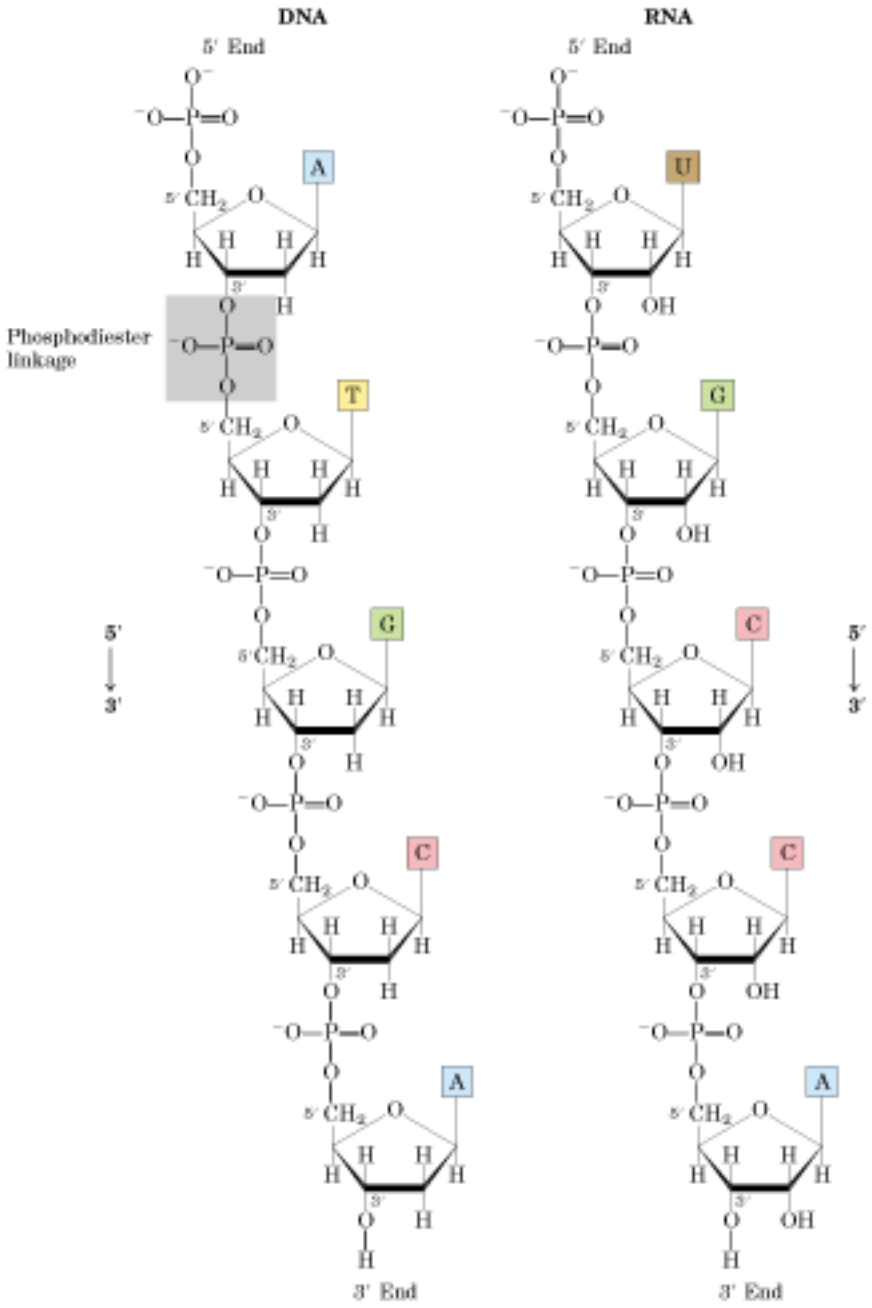


Adenosine 3′-monophosphate



Adenosine 2′,3′-cyclic monophosphate

phosphodiester



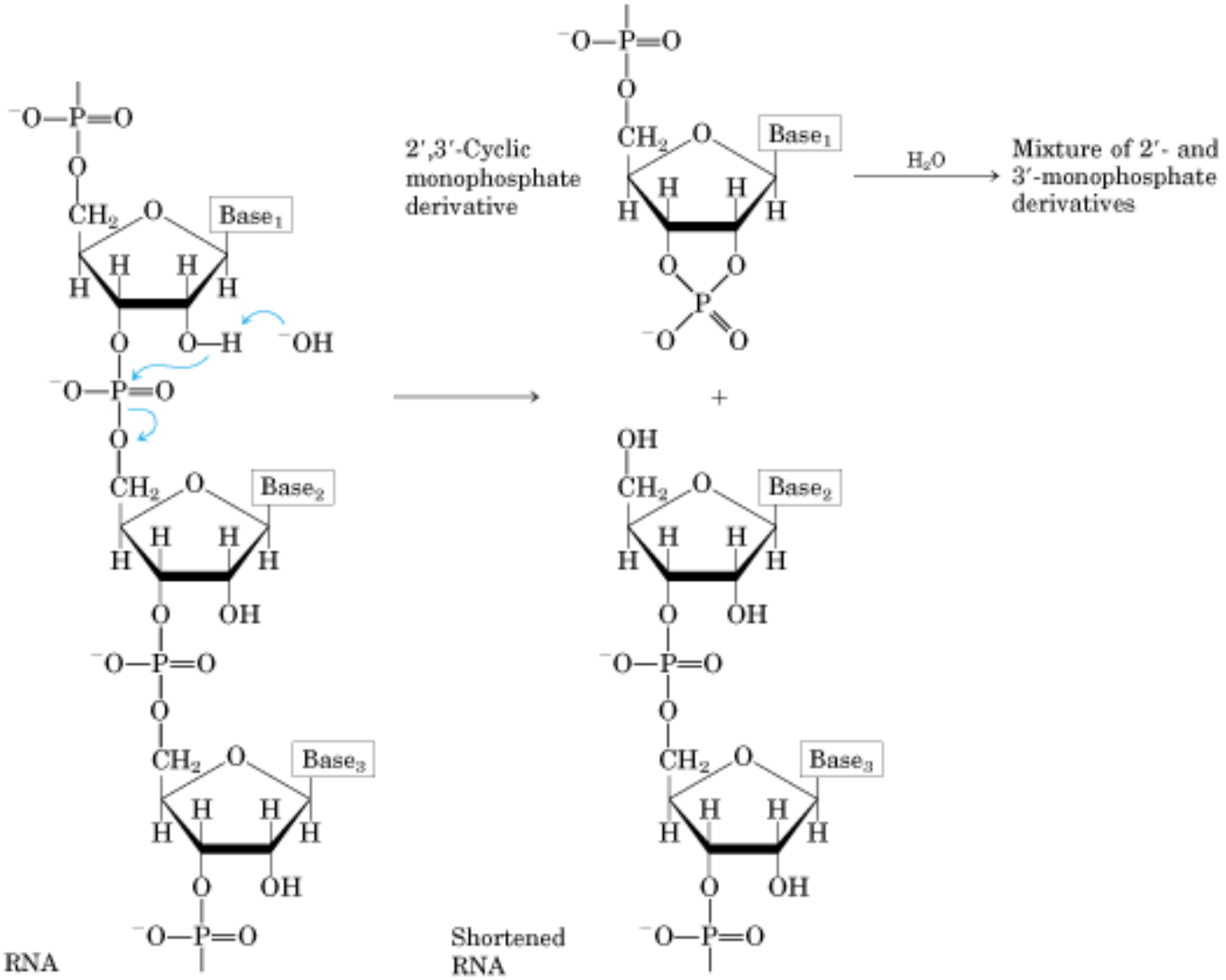

2′,3′-Cyclic monophosphate derivative
H_2O
Mixture of 2′- and 3′-monophosphate derivatives
RNA
Shortened RNA
Base1
Base2
Base3
OH

table 10–1

Nucleotide and Nucleic Acid Nomenclature

Base	Nucleoside*	Nucleotide*	Nucleic acid
Purines			
Adenine	Adenosine	Adenylate	RNA
	Deoxyadenosine	Deoxyadenylate	DNA
Guanine	Guanosine	Guanylate	RNA
	Deoxyguanosine	Deoxyguanylate	DNA
Pyrimidines			
Cytosine	Cytidine	Cytidylate	RNA
	Deoxycytidine	Deoxycytidylate	DNA
Thymine	Thymidine or deoxythymidine	Thymidylate or deoxythymidylate	DNA
Uracil	Uridine	Uridylate	RNA



nucleic acids
deoxyribonucleic acid (DNA)
ribonucleic acid (RNA)

nucleotide , ribonucleotide, deoxyribonucleotide

nitrogenous base, ribose, phosphate

ribose: pentose sugar/ribose/deoxyribose; β-D-ribose; β-D-furanose, β-D-ribofuranose
β-D-2-deoxyribofuranose
nitrogenous base: adenine (A), guanine (G), cytosine (C), or thymine (T), uracil (U) .
pyrimidine, purine

N-glycosidic bond:
1' carbon of the deoxyribose is covalently linked N-1 of the pyrimidine,
1' carbon of the deoxyribose is covalently linked N-9 of the purine

Nucleoside:
deoxyadenosine, deoxyguanosine, deoxythymidine, and deoxycytidine
adenosine, guanosine, uridine, and cytidine

3',5'-phosphodiester
Nucleotide:
deoxyadenylate deoxyguanylate deoxycytidylate deoxythymidylate
Adenylate guanylate cytidylate uridylate

1.3 DNA stores genetic information.

1.3.1 The amino acid sequence of every protein and the nucleotide sequence of every RNA molecule in a cell are all specified by the nucleotide sequence of that cell's DNA molecule.



1.3.2 A segment of DNA that contains the information required for the synthesis of a functional protein or RNA is referred as a gene.

1.3.3 DNA is large biomacromolecule. In bacteria, all the genetic information is stored in a single DNA molecule; in a eukaryotic cell each chromosome contains one single DNA molecule.



1.4 RNA can be divided into several classes of different functions.

1.4.1 Ribosomal RNAs (rRNA) are structural components of ribosomes (the protein synthesis machine in cells).

1.4.2 Messenger RNA (mRNA) are copies of DNA (synthesized by DNA transcription), that carry the information of one or a few genes to the ribosomes, where the corresponding protein(s) is(are) synthesized.



1.4.3 Transfer RNA (tRNA) are adapter molecules that faithfully translate the information in a mRNA molecule into the specific amino acid sequences in a polypeptide chain.

1.4.4 Some RNA molecules, named as Ribozymes, have catalytic activities functioning in the processing (cleavage) of precursor RNA molecules (Thomas Cech and Sidney Altman won the Nobel Prize in Chemistry in 1989 for discovering ribozymes).



1.5 Nucleotides have roles other than being monomeric units of nucleic acids.

1.5.1 Nucleoside triphosphates are used as source of chemical energy to drive a wide variety of biochemical reactions.

1.5.2 ATP is the "energy currency" in cells (UTP, GTP, and CTP are also used in specific reactions as energy sources)

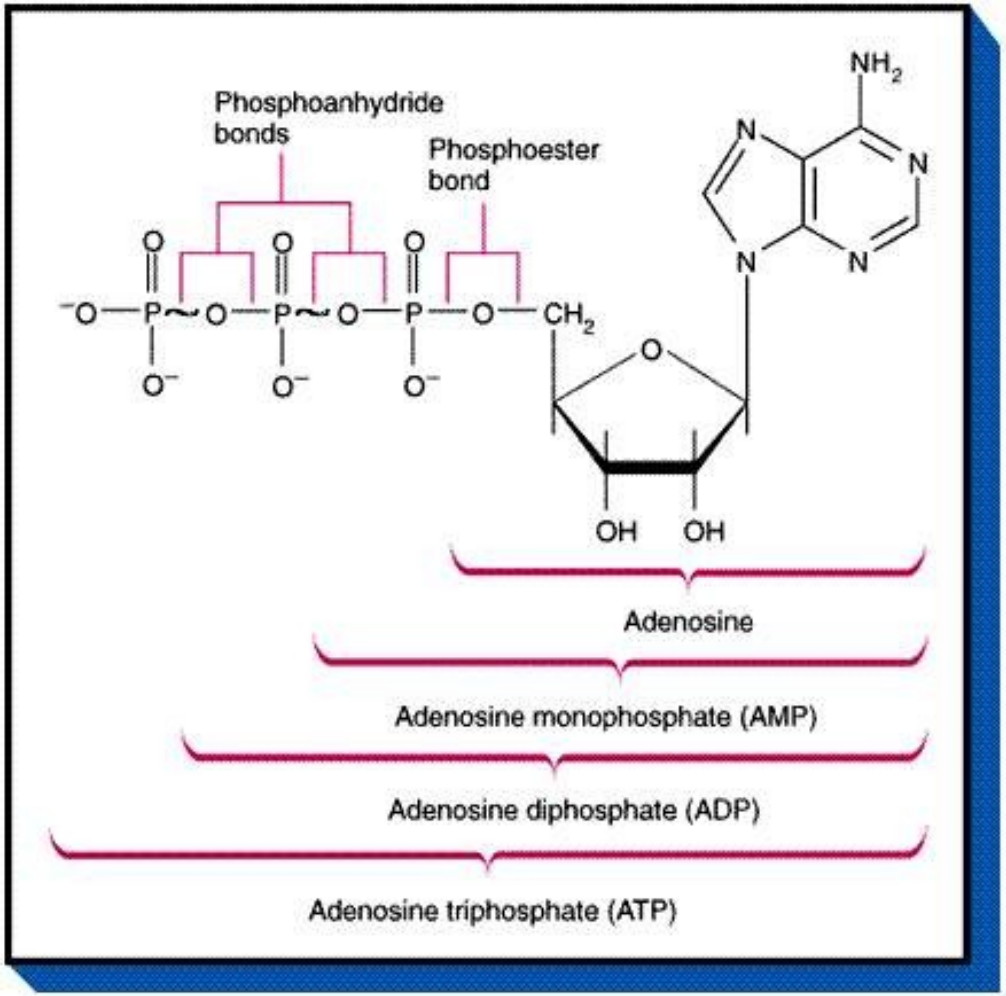


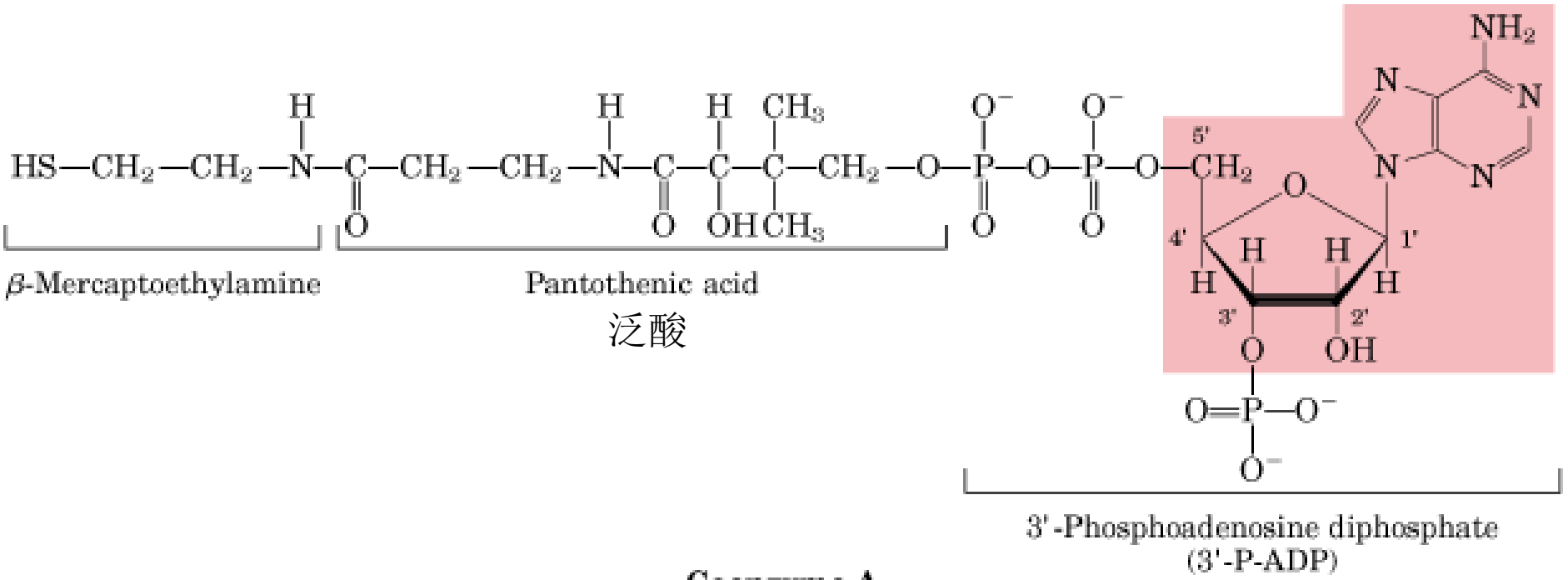
1.5.3 Adenosine diphosphate (ADP) is part of many coenzymes, e.g., coenzyme A, nicotinamide adenine dinucleotide (NAD^+), flavin adenine dinucleotide (FAD).



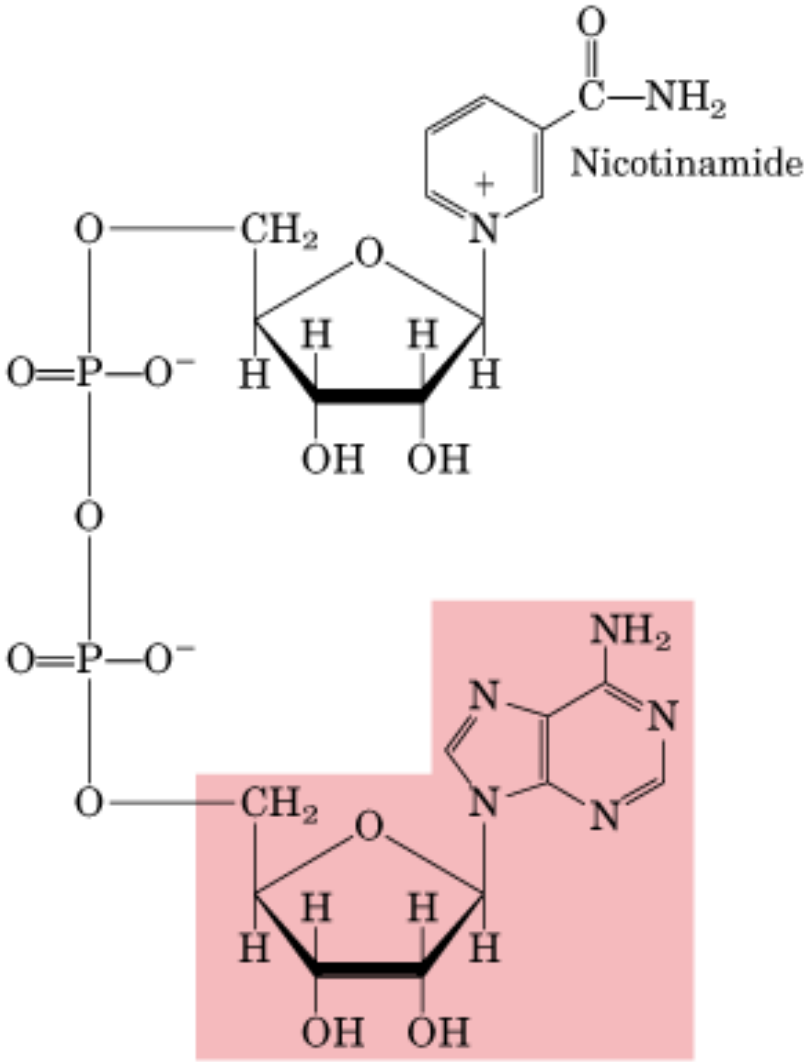
Biochemistry by McKee/McKee, ©1996 McGraw-Hill Higher Education

Structure of ATP.
M Figure 4.10

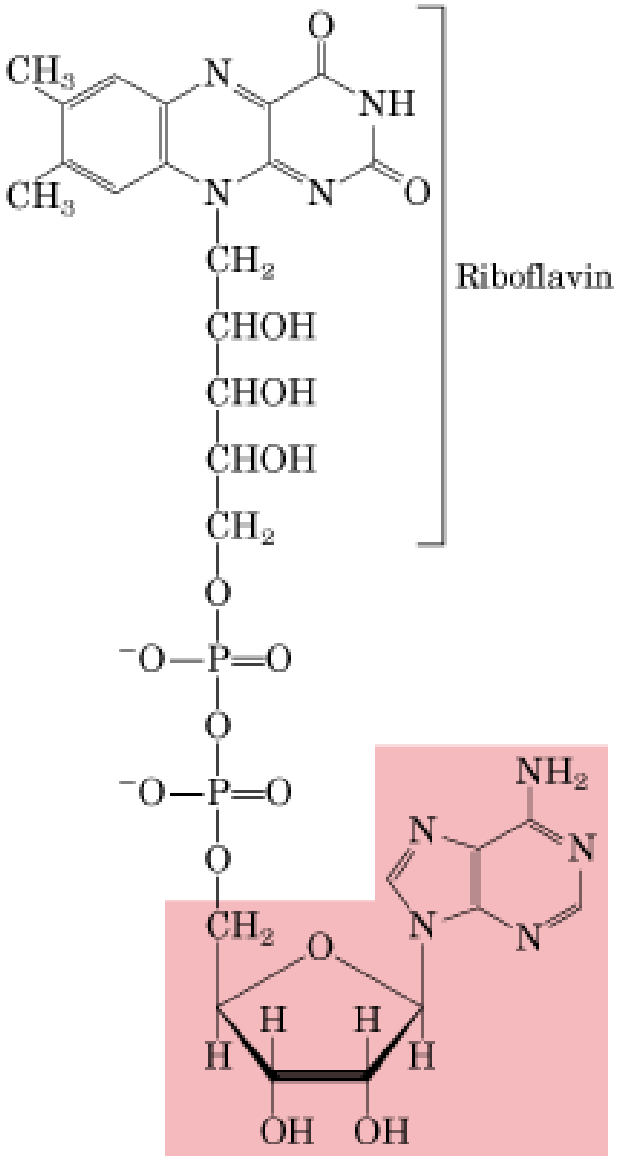



β-Mercaptoethylamine
Pantothenic acid
泛酸
3'-Phosphoadenosine diphosphate
(3'-P-ADP)

Coenzyme A



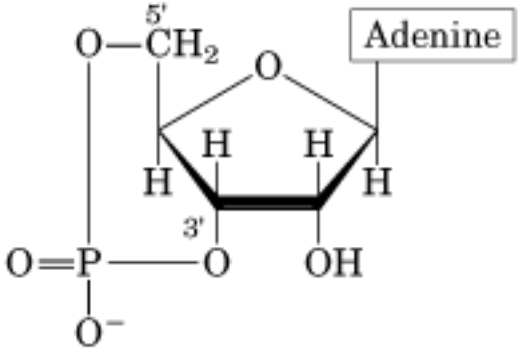
Nicotinamide adenine dinucleotide (NAD^+)



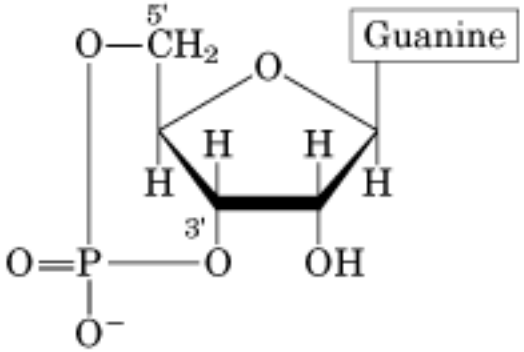
Flavin adenine dinucleotide (FAD)

1.5.4 Adenosine 3',5'-cyclic monophosphate (cAMP), guanosine 3',5'-cyclic monophosphate (cGMP) function as secondary messengers in cell signal transductions.

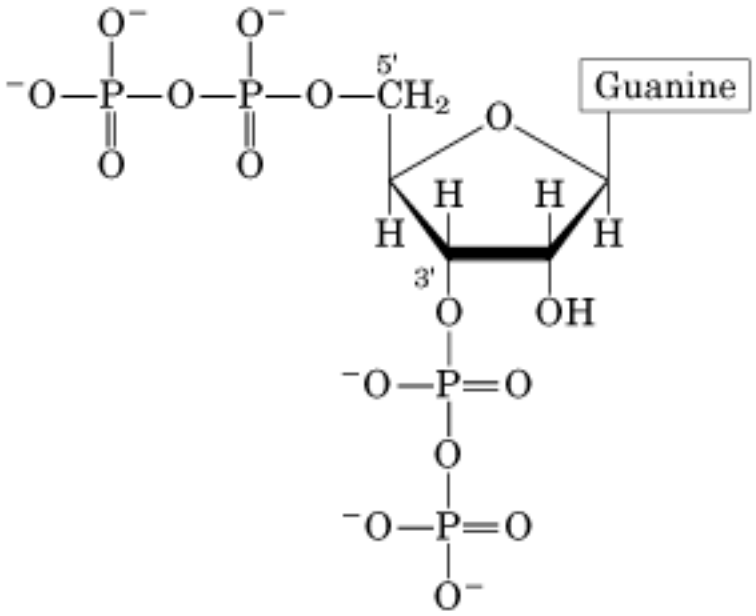




Adenosine 3',5'-cyclic monophosphate
(cyclic AMP; cAMP)



Guanosine 3',5'-cyclic monophosphate
(cyclic GMP; cGMP)



Guanosine 3'-diphosphate,5'-diphosphate
(guanosine tetraphosphate)
(ppGpp)

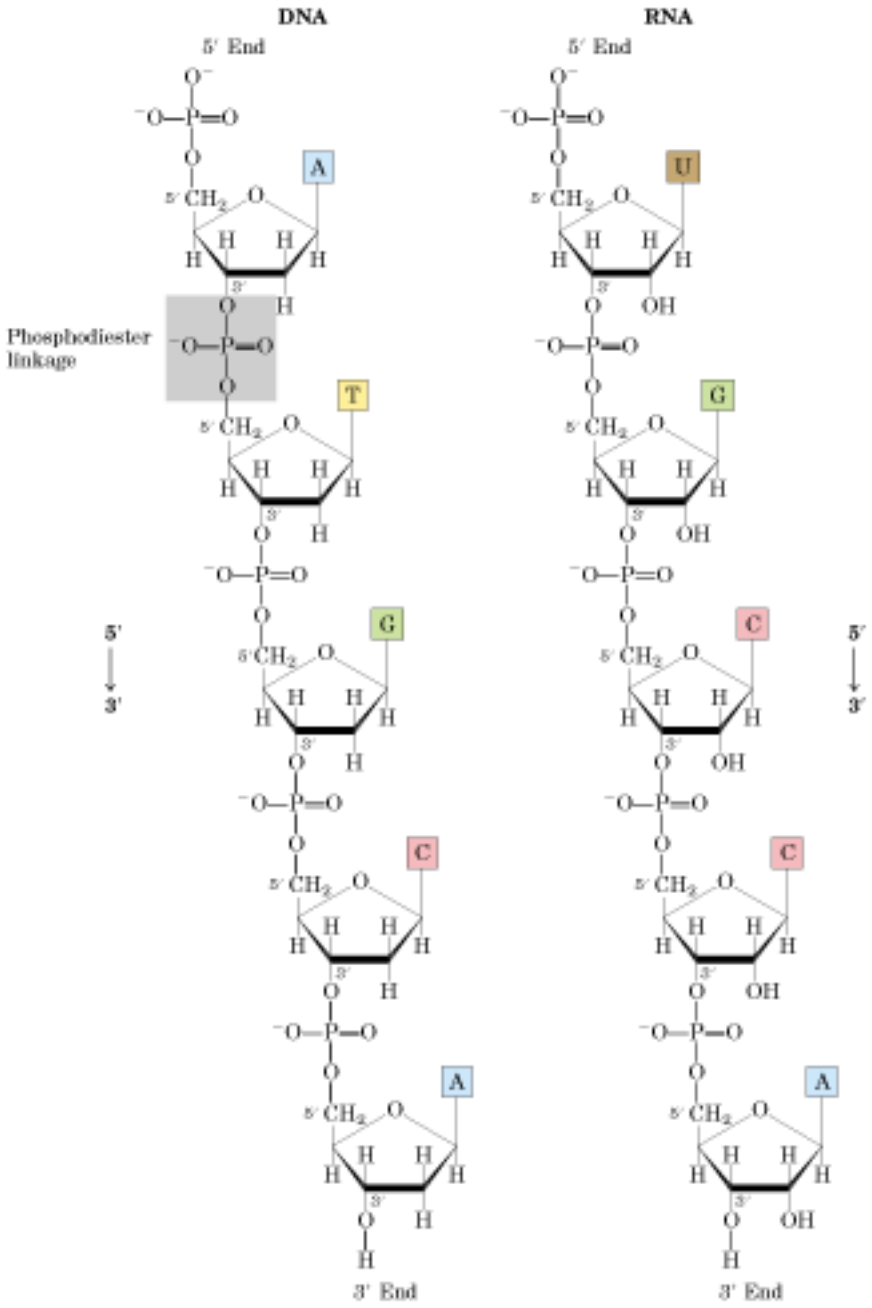
2. Phosphodiester bonds link successive nucleotides in nucleic acids (in both DNA and RNA)

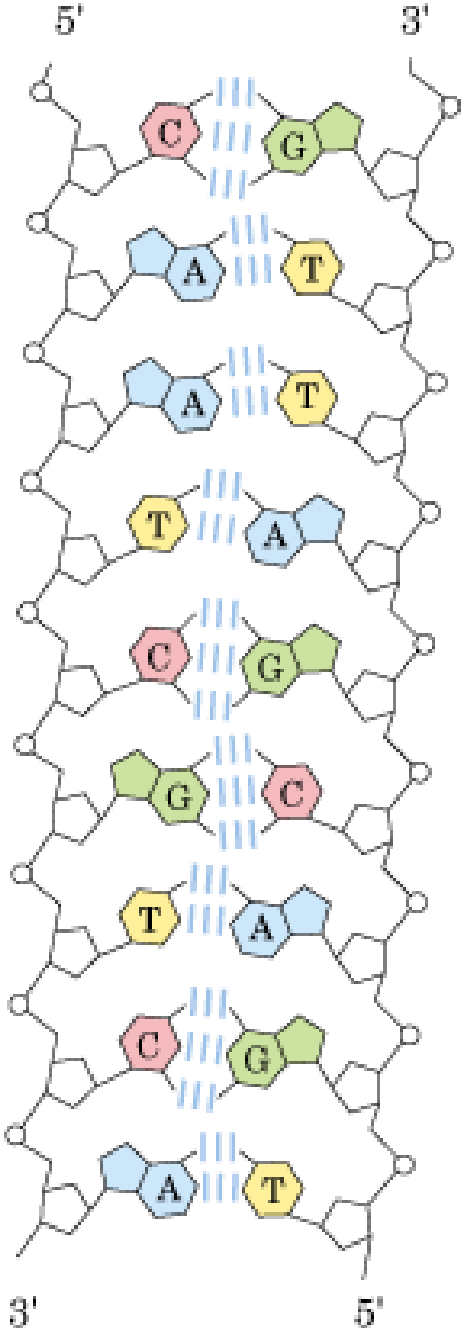
2.1 The 3'-hydroxyl group of one nucleotide is joined to the 5'-phosphate group of the next nucleotide by a phosphodiester bridge.

2.1.1 The covalent backbones of nucleic acids consist of alternating phosphate and pentose (β-D-deoxyribose in DNA, β-D-ribose in RNA) residues.



phosphodiester




5'
3'
C
G
A
T
A
T
T
A
C
G
G
C
T
A
C
G
A
T
3'
5'

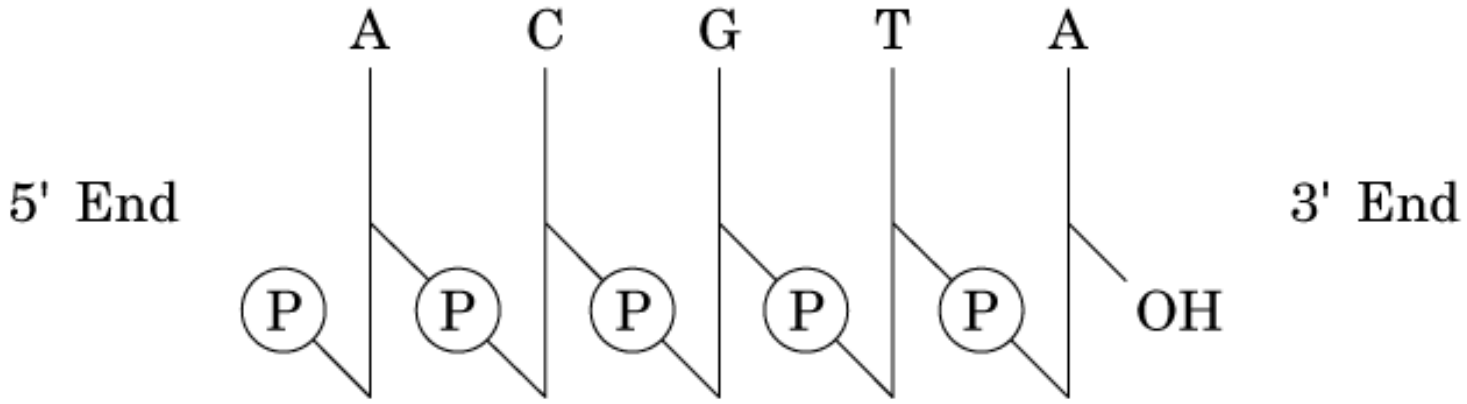
2.1.2 The characteristic bases can be regarded as side groups attaching to the backbone at regular intervals (similar to the R groups on a peptide chains).

2.1.3 Each DNA and RNA strands have a specific polarity with a distinct 5' end (the end lacking a nucleotide at the 5' position) and a 3' end (the end lacking a nucleotide at the 3' position). 5'-pCpGpT-3'-OH

2.1.4 The base sequence of a DNA or RNA molecule is always written with the 5' end on the left and 3' end on the right by convention.



2.1.5 The nucleotide sequences of short segment of nucleic acids can be represented in different ways.



$pA\text{-}C\text{-}G\text{-}T\text{-}A_{OH}$

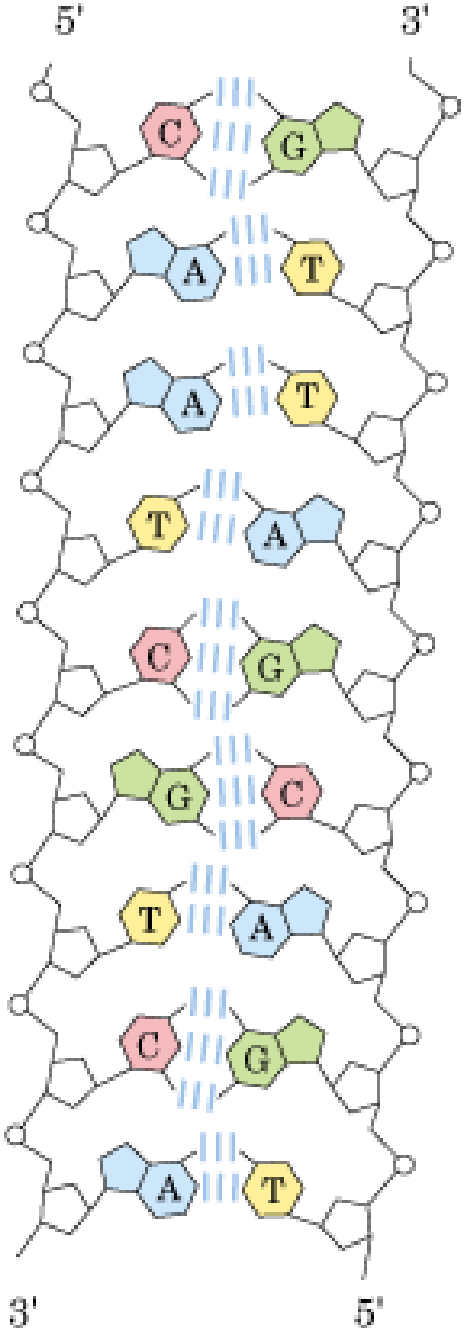
pApCpGpTpA

ACGTA

2.1.6 An oligonucleotide refers to nucleic acids shorter than about 50 nucleotides.

2.1.7 The backbones of both DNA and RNA are hydrophilic, having negative charges at physiological pH, that are generally neutralized by positively charged proteins, metal ions, and polyamines in cells.



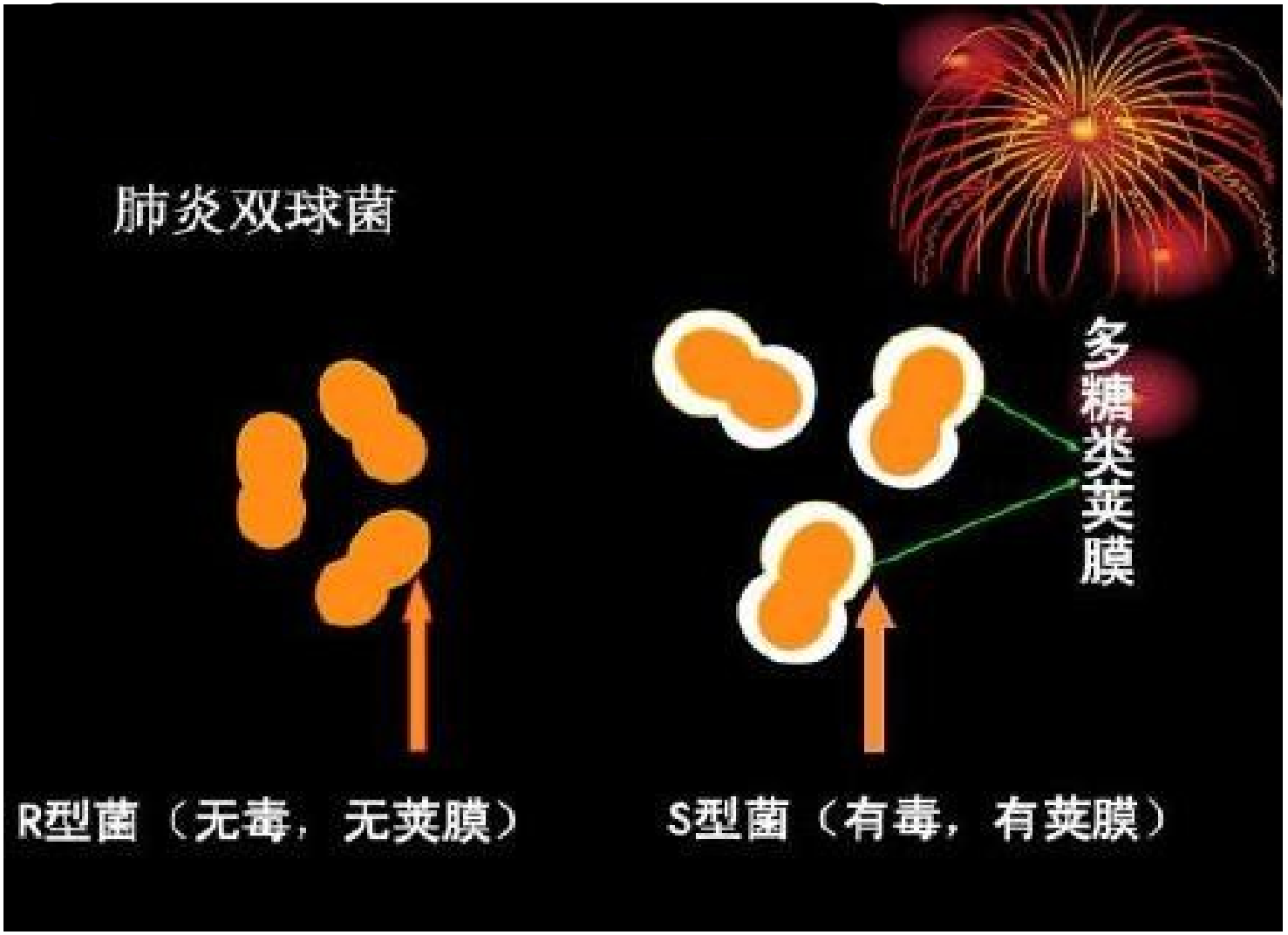

5'
3'
C
G
A
T
A
T
T
A
C
G
G
C
T
A
C
G
A
T
3'
5'

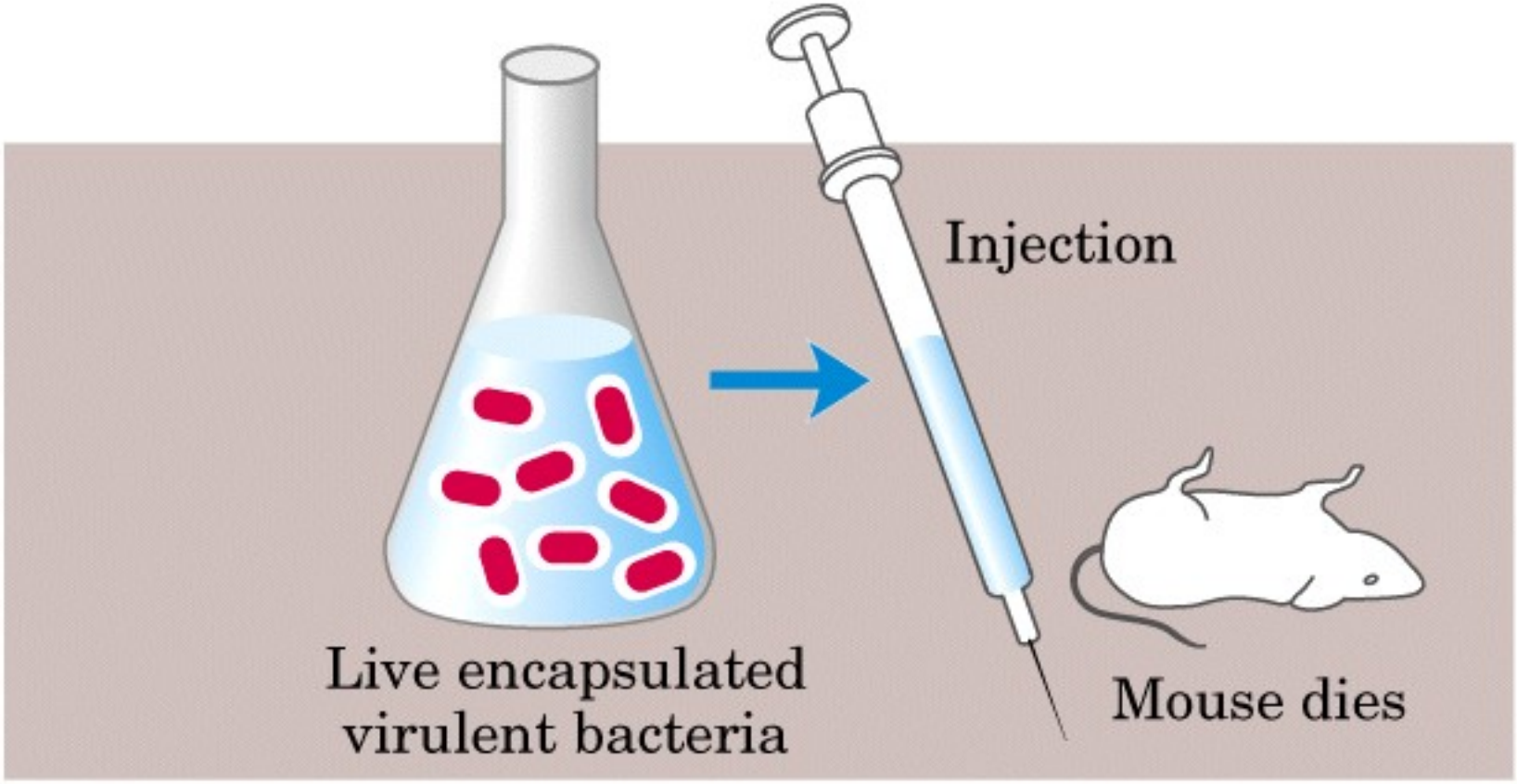
3. DNA was found to be the molecule storing the genetic information.

The biochemical investigation of DNA began with Friedrich Miescher in 1868.

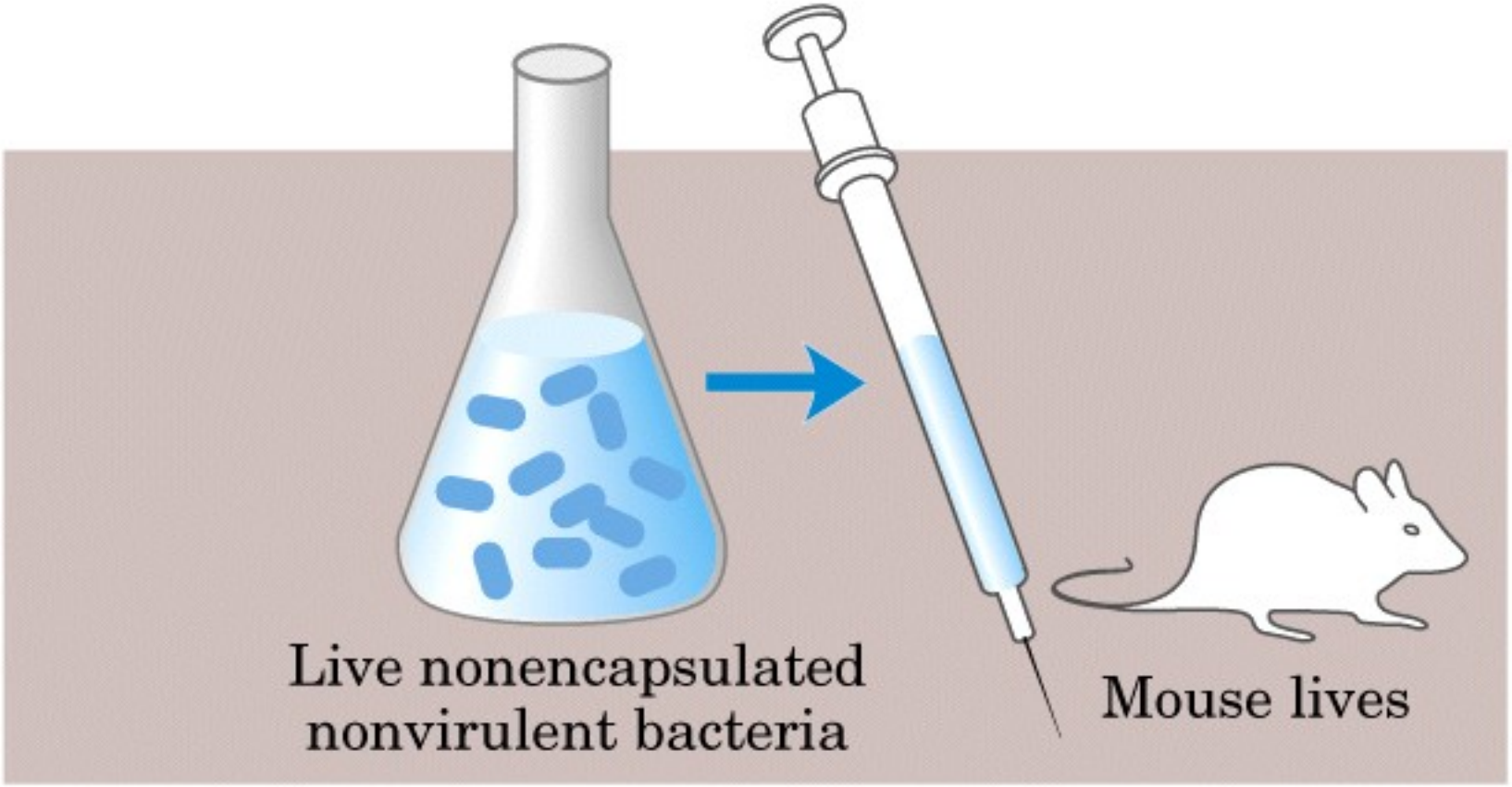
The first direct evidence that DNA is the bearer of genetic information came in 1944 through a discovery made by Oswald T. Avery et al.



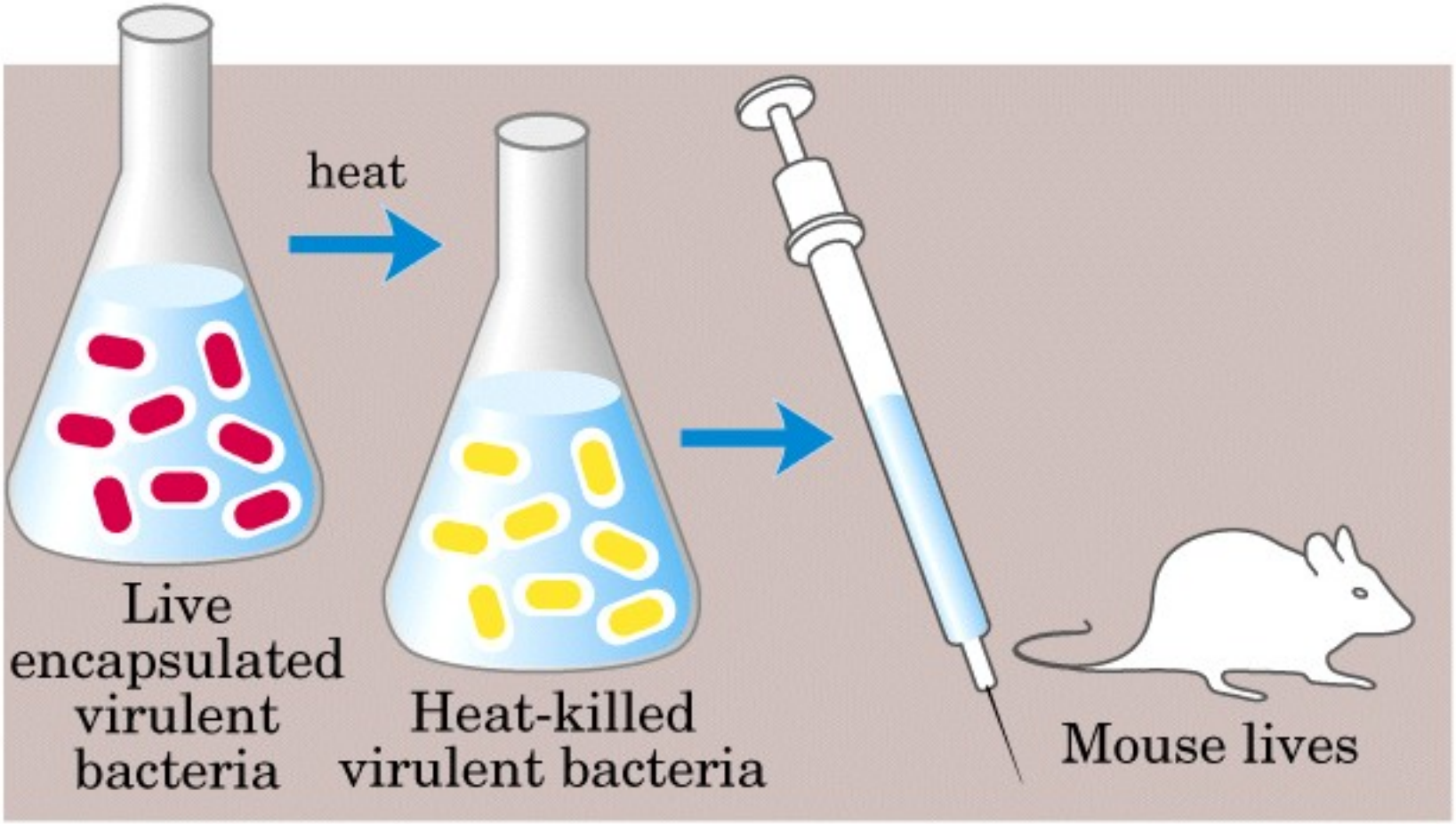

肺炎双球菌
多糖类荚膜
R型菌（无毒，无荚膜）
S型菌（有毒，有荚膜）


Injection
Live encapsulated
virulent bacteria
Mouse dies

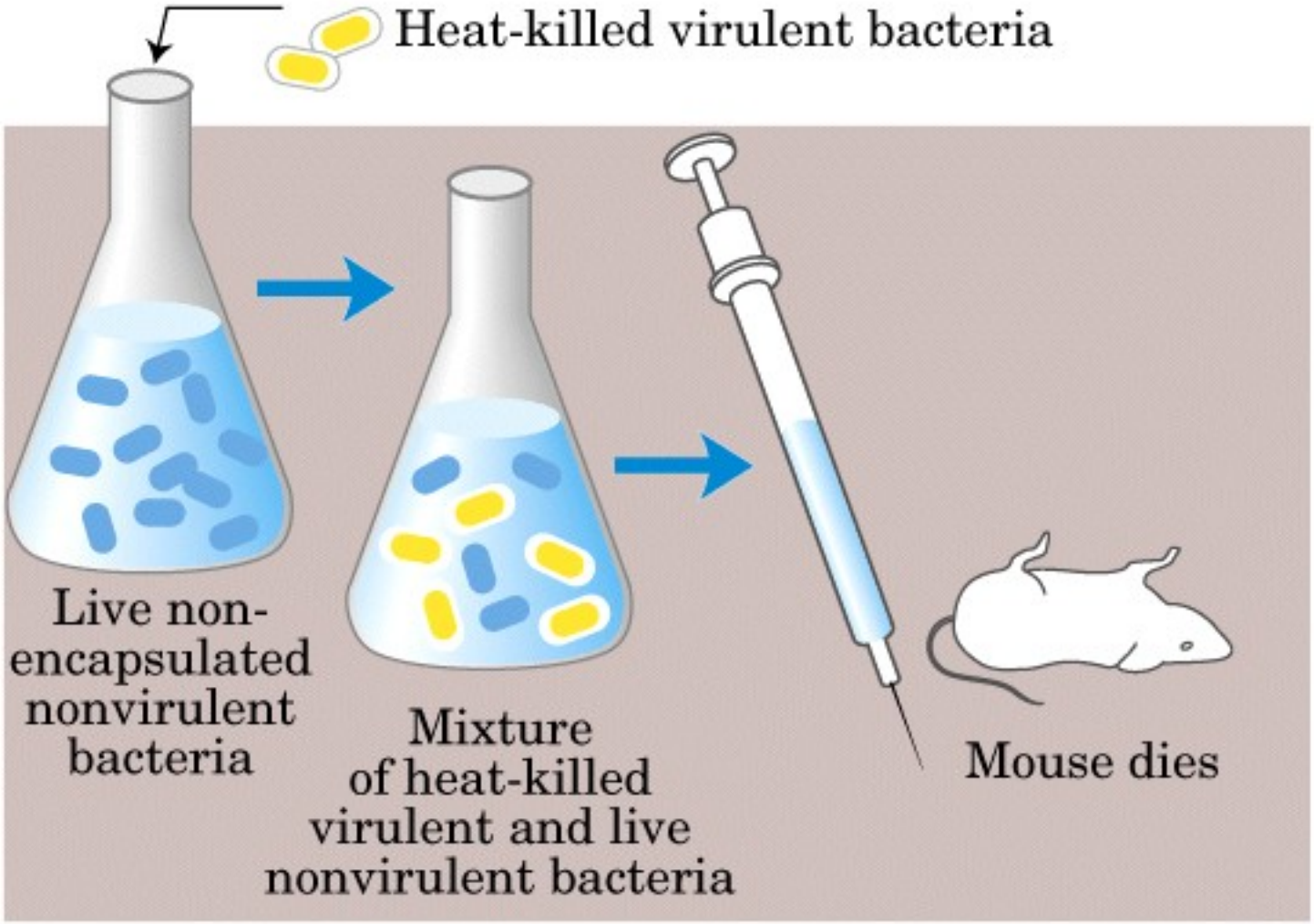
(a)



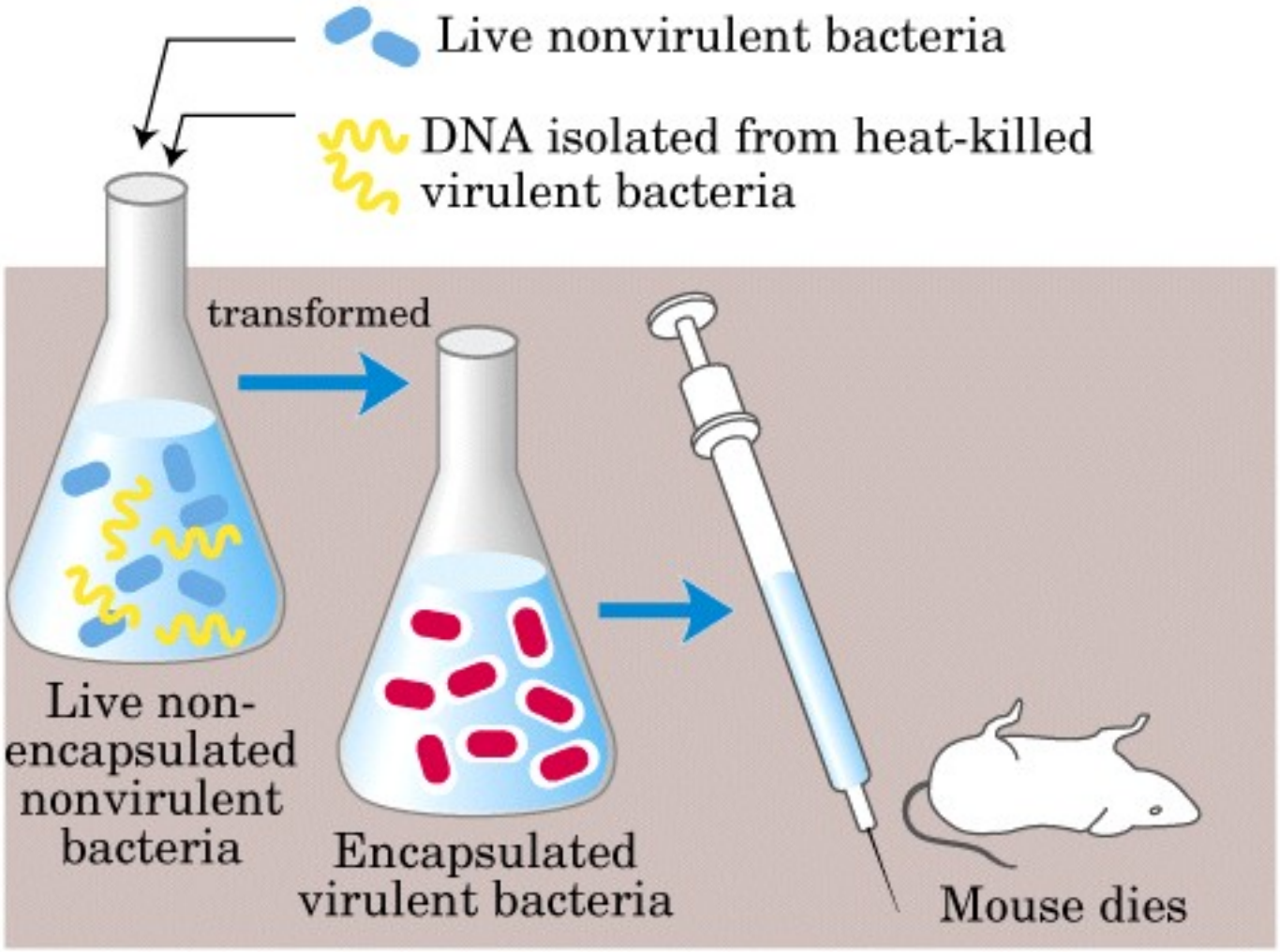
(b)


heat
Live encapsulated virulent bacteria
Heat-killed virulent bacteria
Mouse lives

(c)


Heat-killed virulent bacteria
Live non-encapsulated nonvirulent bacteria
Mixture of heat-killed virulent and live nonvirulent bacteria
Mouse dies

(d)


Live nonvirulent bacteria
DNA isolated from heat-killed virulent bacteria
transformed
Live non-encapsulated nonvirulent bacteria
Encapsulated virulent bacteria
Mouse dies

(e)

3.1 Avery and his colleagues discovered that a nonvirulent R form of pneumococcus bacterium (with rough colonies) can be transformed into the virulent S form (of smooth colonies).



3.2 DNA was found to carry the genetic information for virulence in the pneumococci transformation experiment.

3.2.1 Addition of DNA extracted from the heat-killed S form pneumococci (with protein removed as completely as possible) into live nonvirulent R form bacteria transformed the R form into a virulent S form permanently.

3.2.2 Treatment with proteolytic enzymes (trypsin, chymotrypsin) did not have any effect on the transformation activity.



3.2.3 Treatment with ribonuclease (known to digest RNA) had no effect on the transformation activity.

3.2.4 Treatment with deoxyribonuclease (known to digest DNA) destroyed the transformation activity.

3.2.5 Chromosomal proteins were assumed to carry the genetic information (with DNA playing a secondary role) until Avery, MacLeod, McCarty performed these experiments in 1944.



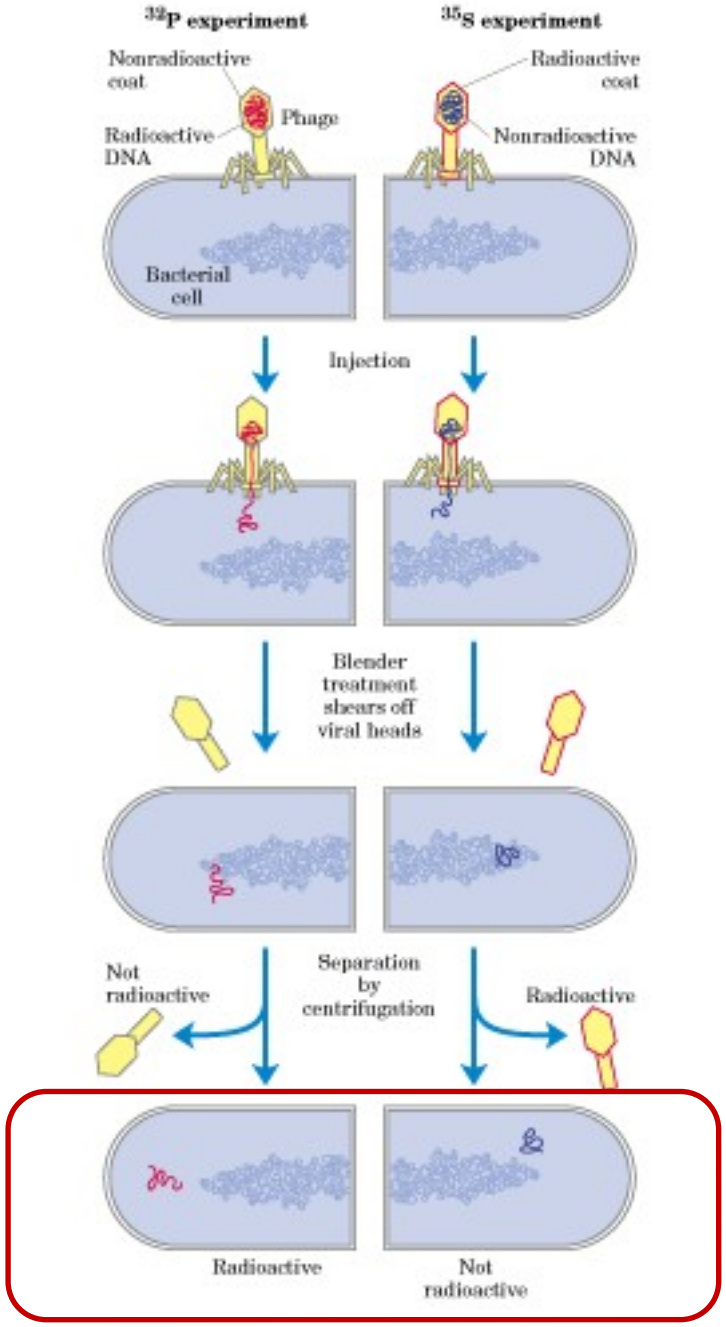
3.3 Further support for the genetic role of DNA came from the studies of T2 bacteriophage (a bacterial virus) that infects E.coli.

3.3.1 The T2 bacteriophage consists of a core of DNA surrounded by a protein coat.

3.3.2 Alfred Hershey and Martha Chase demonstrated that at infection only DNA (labeled with radioisotope ^{32}P) entered E.coli cells, proteins (labeled with ^{35}S) did not enter the host cells (1952).

3.3.3 DNA provided the genetic information for bacteriophage replication within the E.coli cells.




^{32}P experiment
^{35}S experiment
Nonradioactive coat
Radioactive DNA
Phage
Radioactive coat
Nonradioactive DNA
Bacterial cell
Injection
Blender treatment shears off viral heads
Not radioactive
Separation by centrifugation
Radioactive
Radioactive
Not radioactive

4. DNA molecules are double helices.

The Nobel Prize in Physiology or Medicine 1962



Photo from the Nobel Foundation archive.

Francis Harry Compton Crick

Prize share: 1/3



Photo from the Nobel Foundation archive.

James Dewey Watson

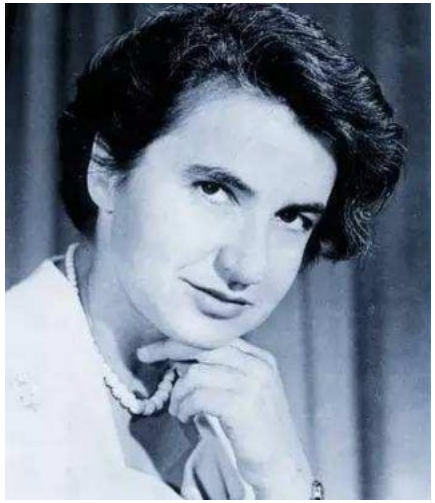
Prize share: 1/3



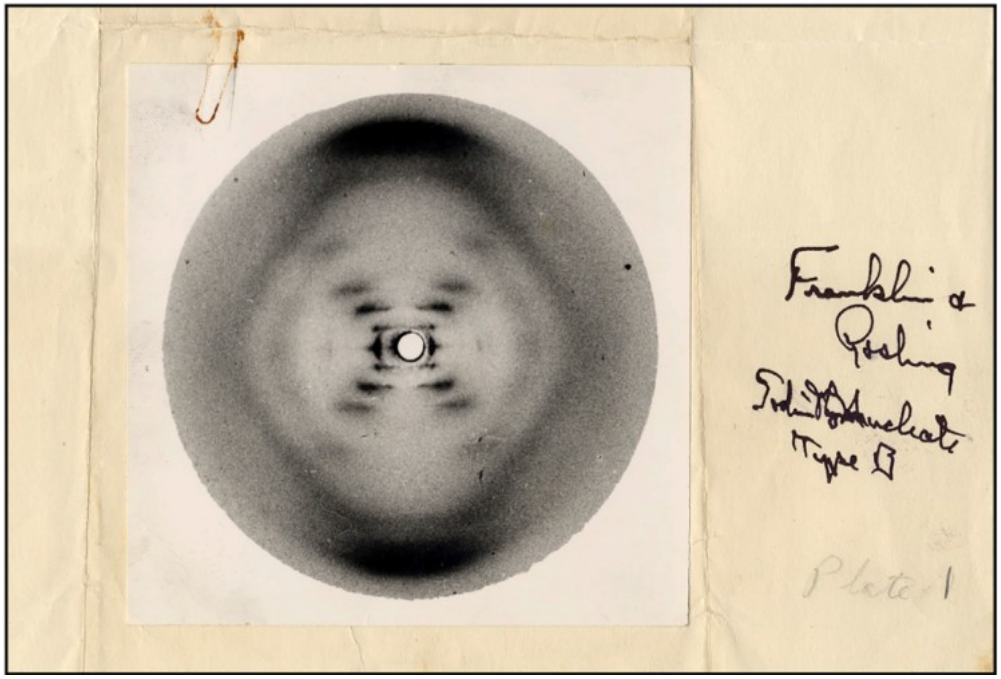
Photo from the Nobel Foundation archive.

Maurice Hugh Frederick Wilkins

Prize share: 1/3



Rosalind Franklin


Franklin & Gosling
Type B
Plate 1



米歇尔



Erwin Chargaff

Watson and Crick proposed a double helix model based mainly on three main pieces of evidence.

A) the fact that the DNA molecule is composed of bases, deoxyriboses, and phosphate groups linked together as a polydeoxyribonucleotide.

B) X-ray diffraction pattern of DNA fibers, suggesting a helical structure with two distinctive regularities of 3.4 and 34 Angstroms along the axis of the molecule.

C) Chargaff's discovery on the quantitative relationships between the bases (A=T, G=C).



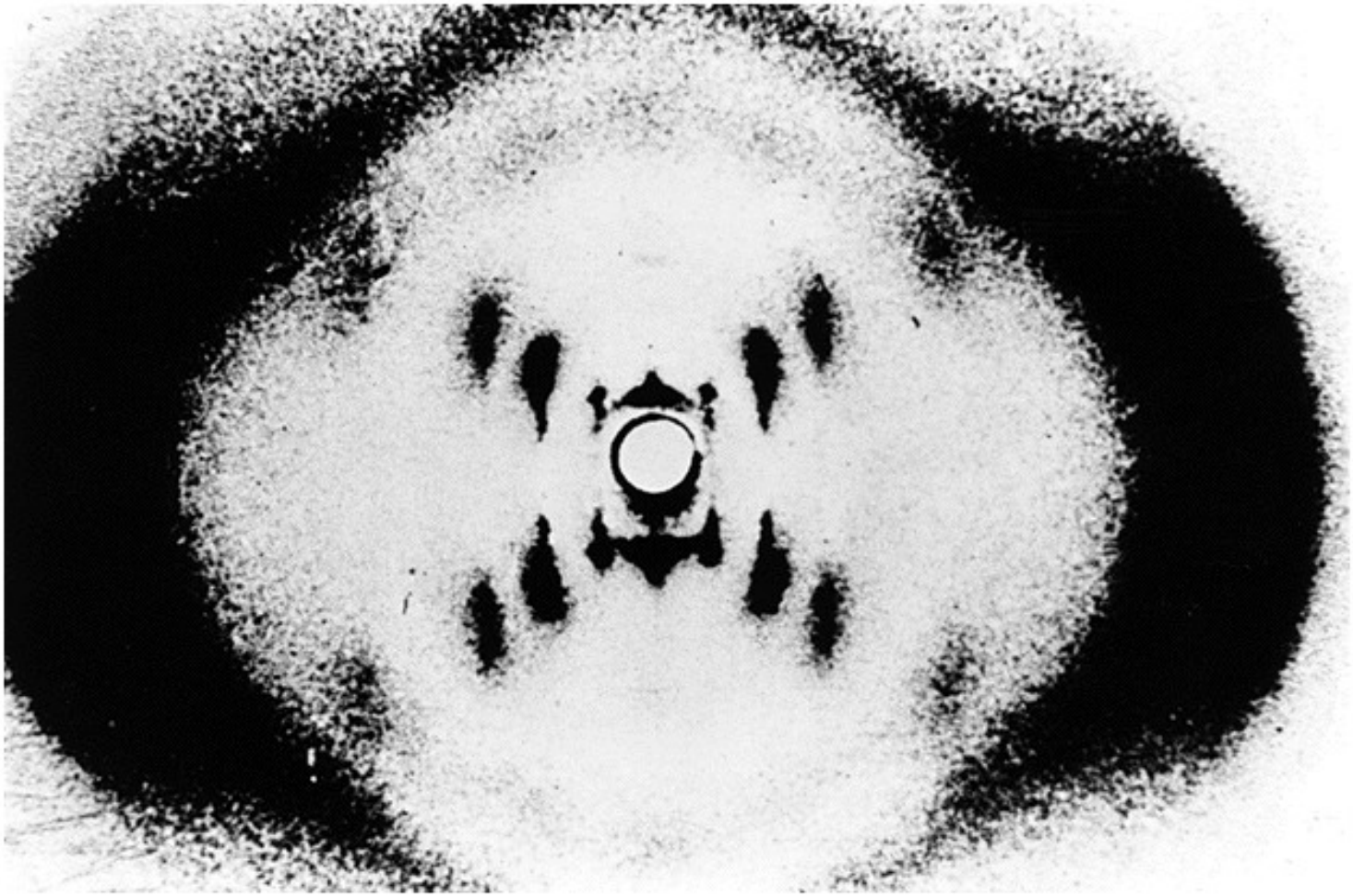
DNA 的碱基组成

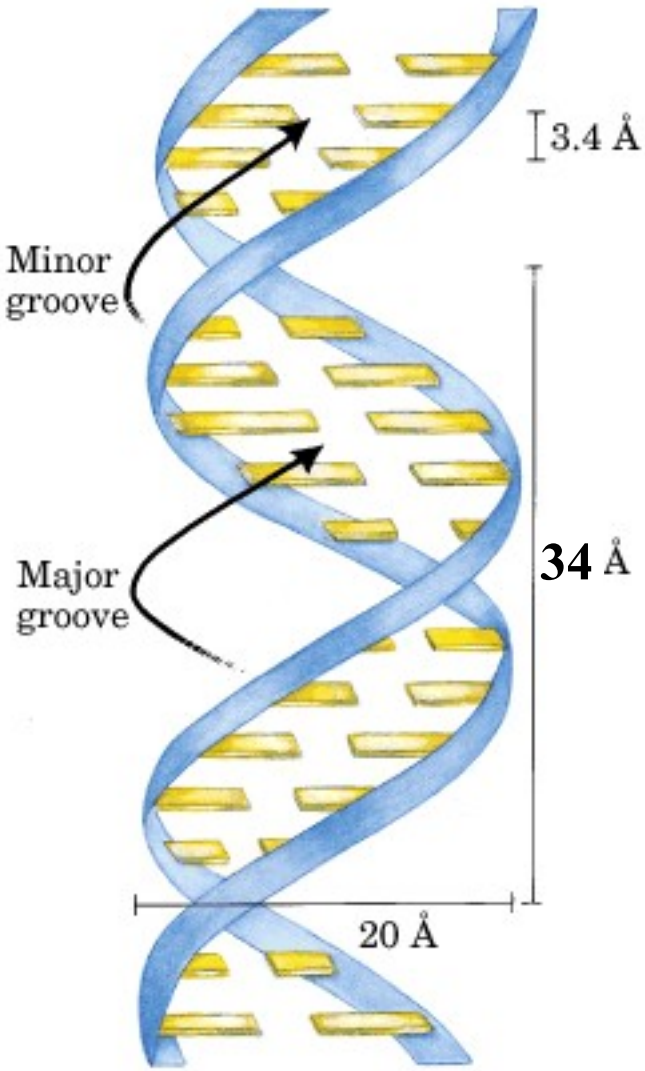
Chargaff 规则：

- 一种生物的所以体细胞的 DNA 碱基组成均相同
- 不同生物碱基组成差异大，可用不对称比率表示

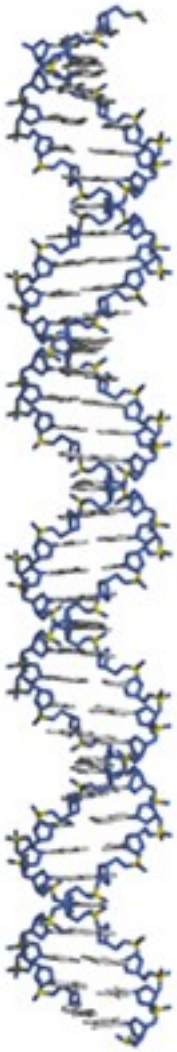
  （A + T） /(G+C)
- 亲缘近的生物，其碱基组成相似
- 同一生物内， DNA 分子组成 A=T G=C
- 同一生物的 DNA 分子 Pu=Py



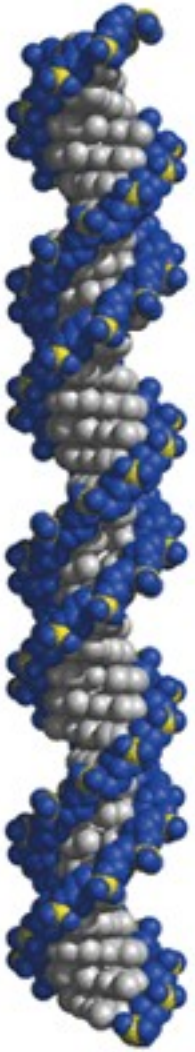




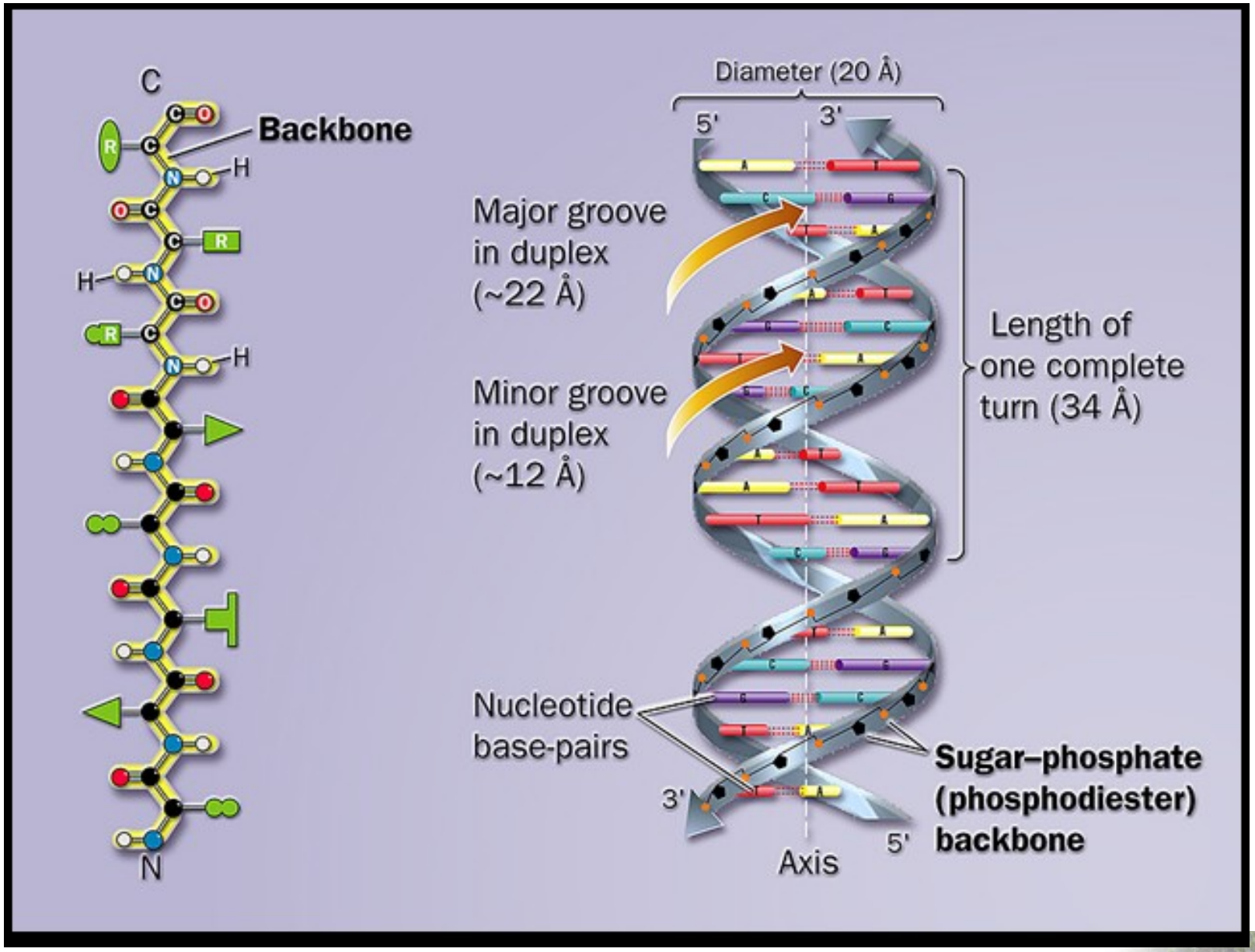
(a)

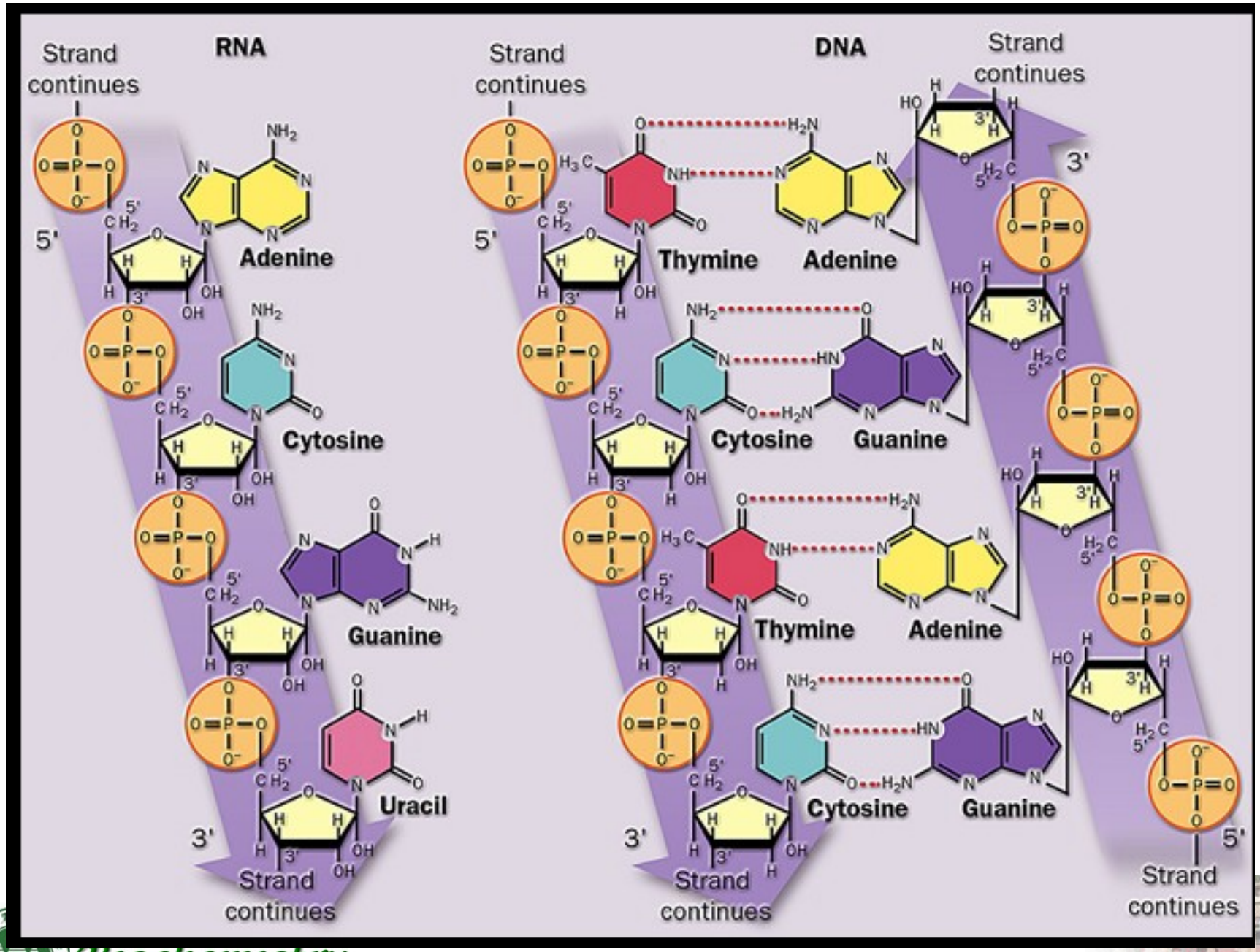


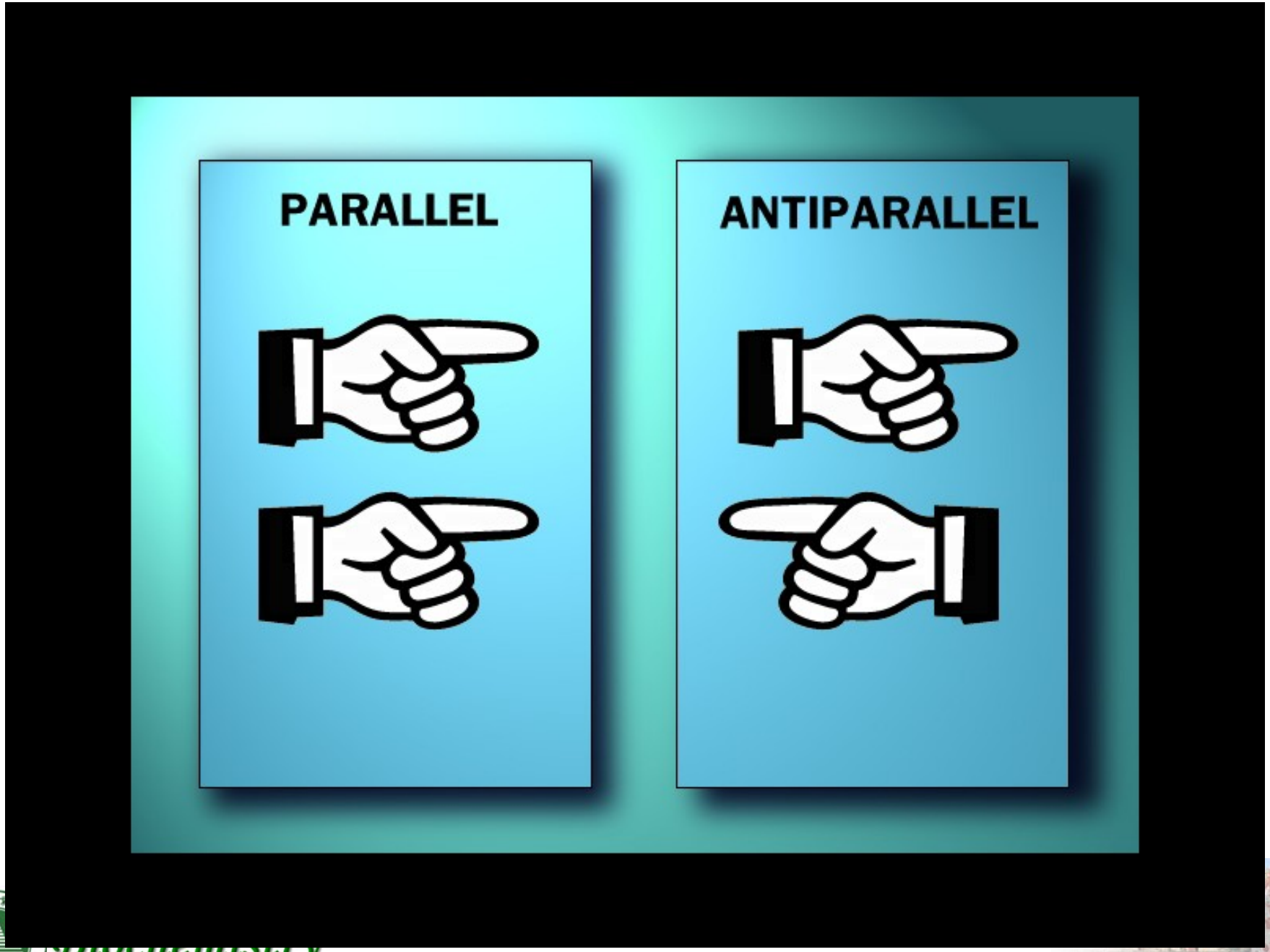
(b)

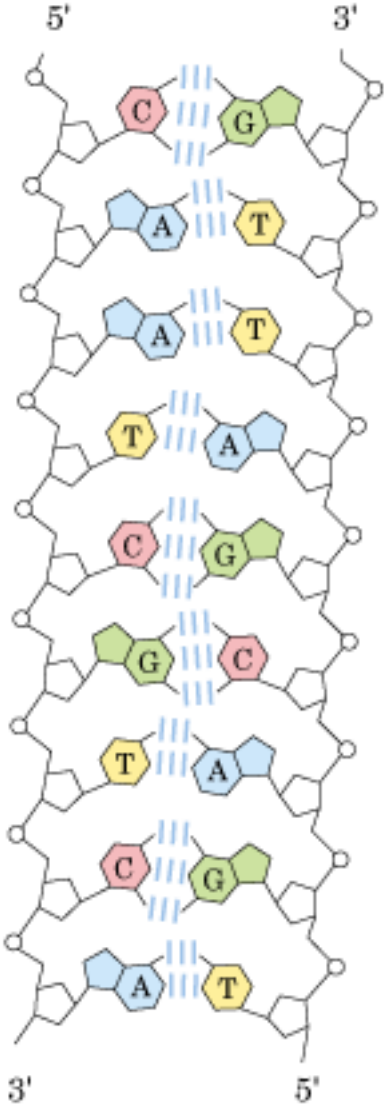


(c)


C
Backbone
H
H
H
N
Diameter (20 Å)
5'
3'
Major groove
in duplex
(~22 Å)
Minor groove
in duplex
(~12 Å)
Length of
one complete
turn (34 Å)
Nucleotide
base-pairs
Sugar–phosphate
(phosphodiester)
backbone
3'
5'
Axis


RNA
Strand continues
Adenine
Cytosine
Guanine
Uracil
5'
3'
Strand continues
DNA
Strand continues
Thymine
Adenine
Cytosine
Guanine
Thymine
Adenine
Cytosine
Guanine
3'
5'
Strand continues
Strand continues


PARALLEL
ANTIPARALLEL


5'
3'
C
G
A
T
A
T
T
A
C
G
G
C
T
A
C
G
A
T
3'
5'

4.1 DNA double helix

4.1.1 The DNA molecule is a right-handed double helix containing two antiparallel strands.

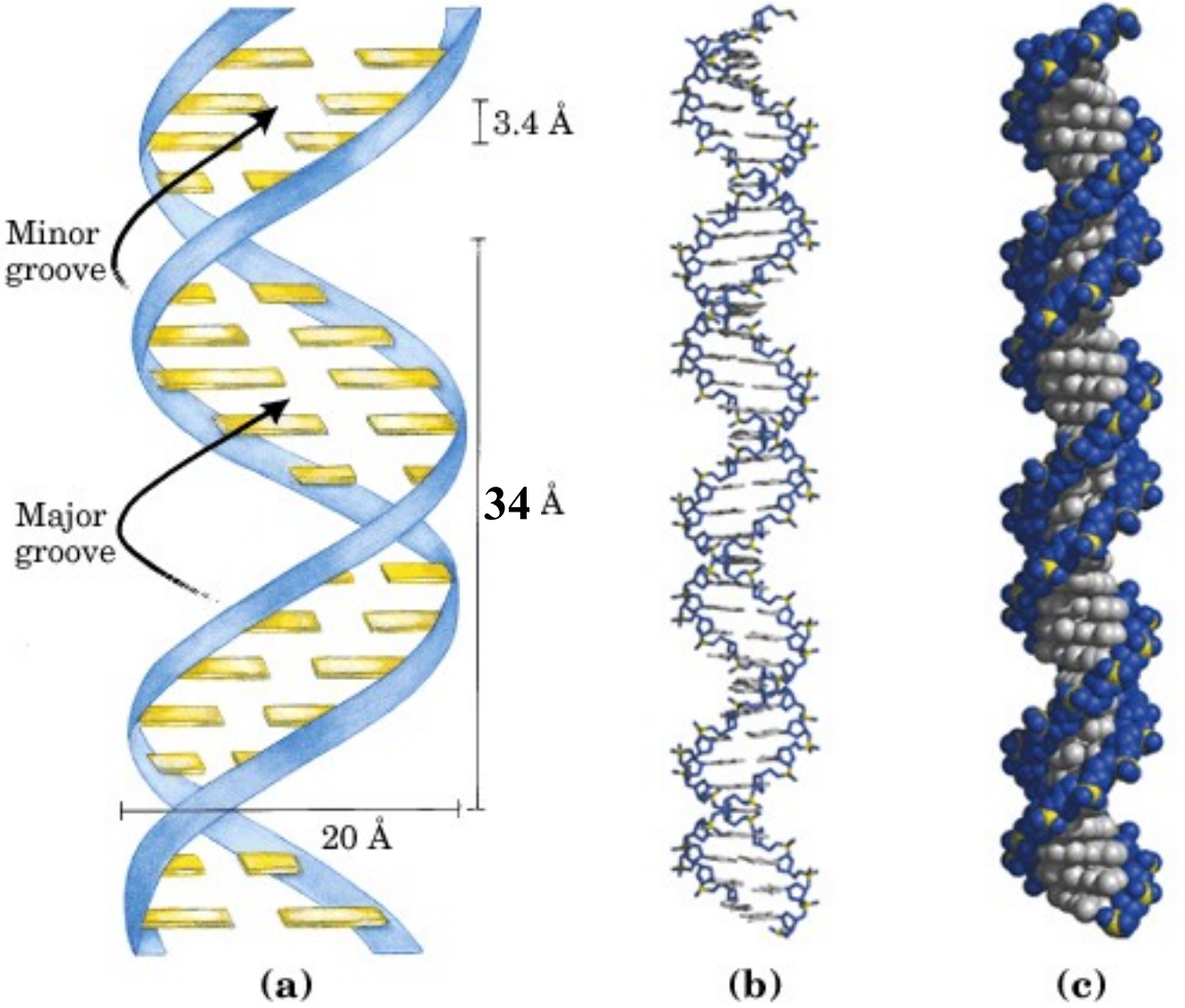
4.1.2 The phosphate-deoxyribose backbones are on the outside of the helix, whereas the purine and pyrimidine bases are stacked inside.

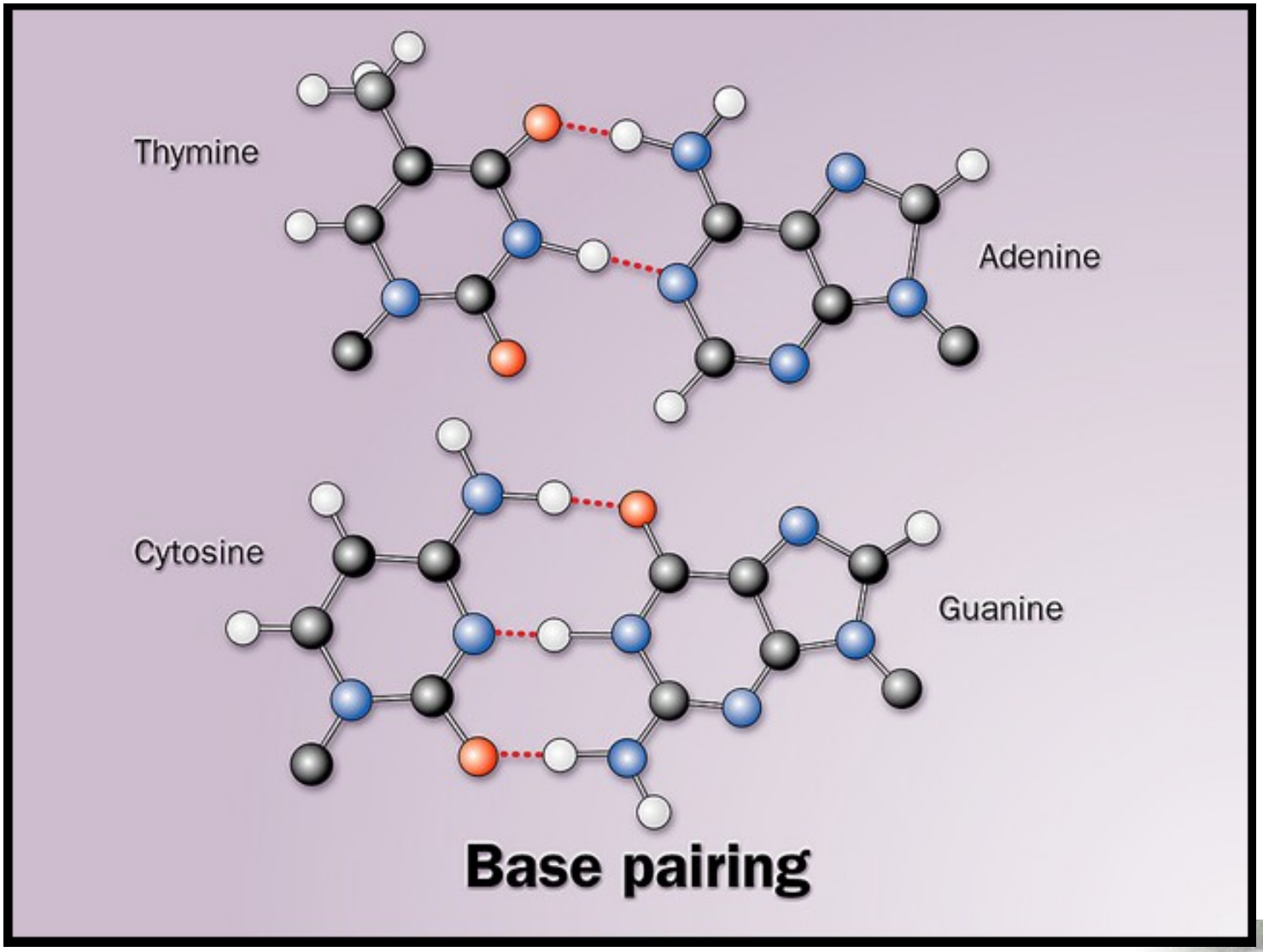
4.1.3 The two antiparallel chains are complementary to each other through hydrogen bonds between pairs of bases （ A=T, G≡C ） .

4.1.4 The base-stacking interactions make a major nonspecific contribution to the stability of the duplex.

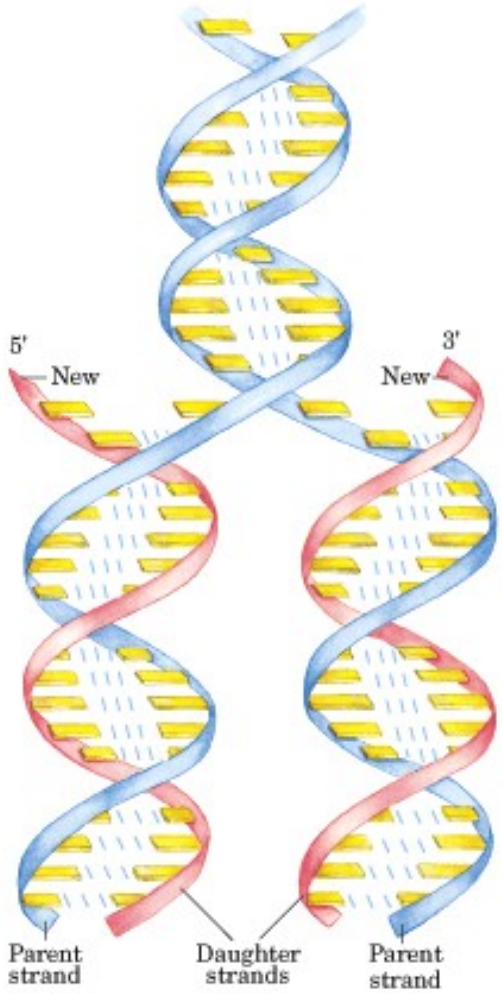


Detailed information of DNA double helix




Thymine
Adenine
Cytosine
Guanine
Base pairing

4.2 The double-helical model of DNA immediately suggested a mechanism for the replication of DNA.



4.3 DNA can occur in different structural forms.

4.3.1 DNA is remarkably flexible molecule with many rotatable bonds (thermal fluctuations producing bending, stretching, and unpairing).

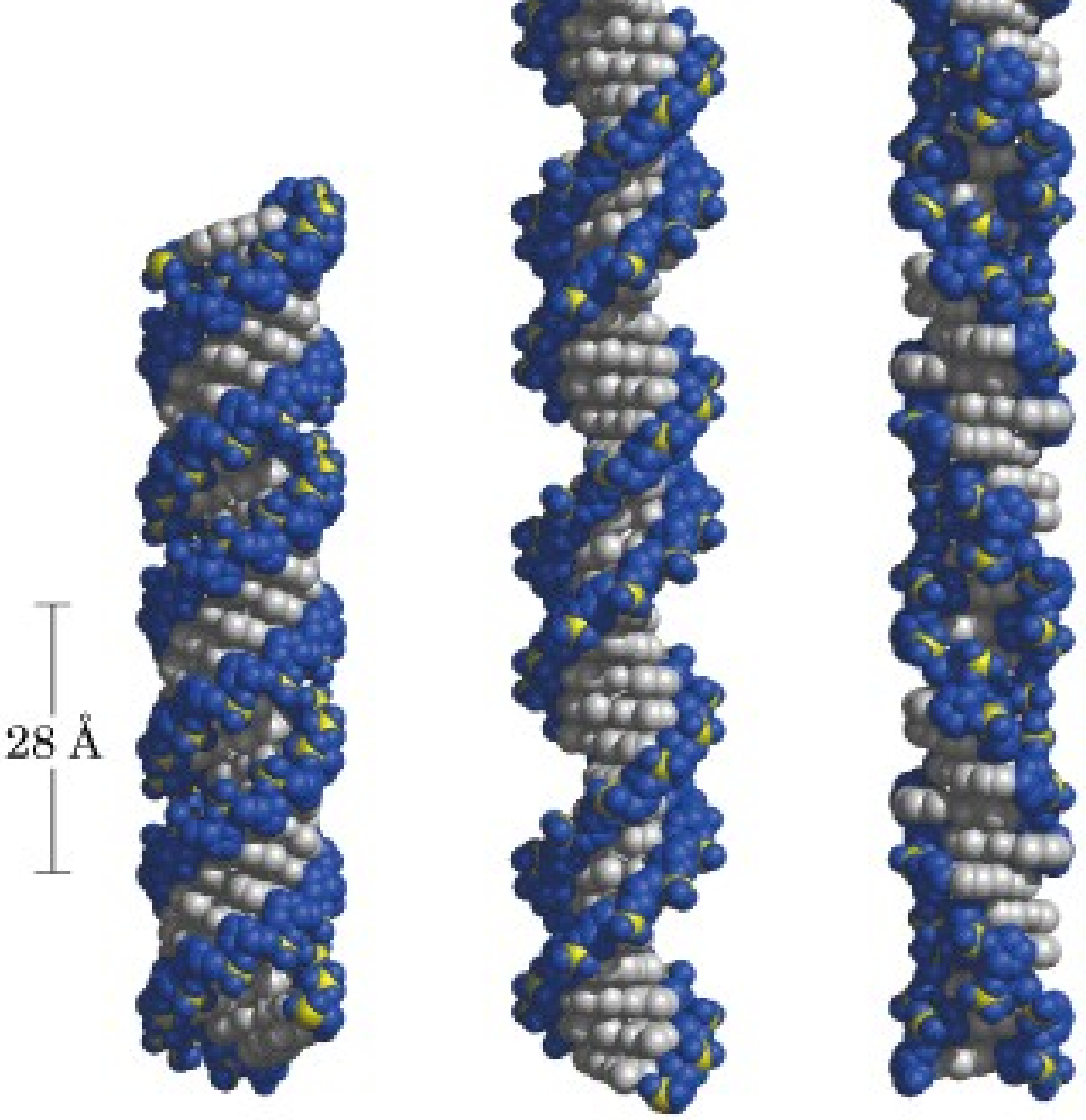
4.3.2 The duplex structure proposed by Watson and Crick is referred as the B-form DNA, and is found to be the most stable structure for a random nucleotide sequence under physiological conditions. Thus it is the standard structure for DNA molecules.



4.3.3 At reduced humidity the DNA molecule will take the A-form: it is still a right-handed duplex made up of antiparallel strands held together by Watson-Crick base pairing.

4.3.4 The A-form helix is wider and shorter than the B-form helix. The plane of the base pairs in A-DNA is tilted about 20° with respect to the helix axis.




28 Å
A form
B form
Z form

4.3.5 The Z-form DNA is a left-handed double helix in which backbone phosphates zigzag.

4.3.6 The biological roles of Z-DNA is uncertain (may play roles in gene expression and genetic recombination).



5. RNA molecules do not form simple, regular secondary structure but many form complex and unique three dimensional structures. (G1)

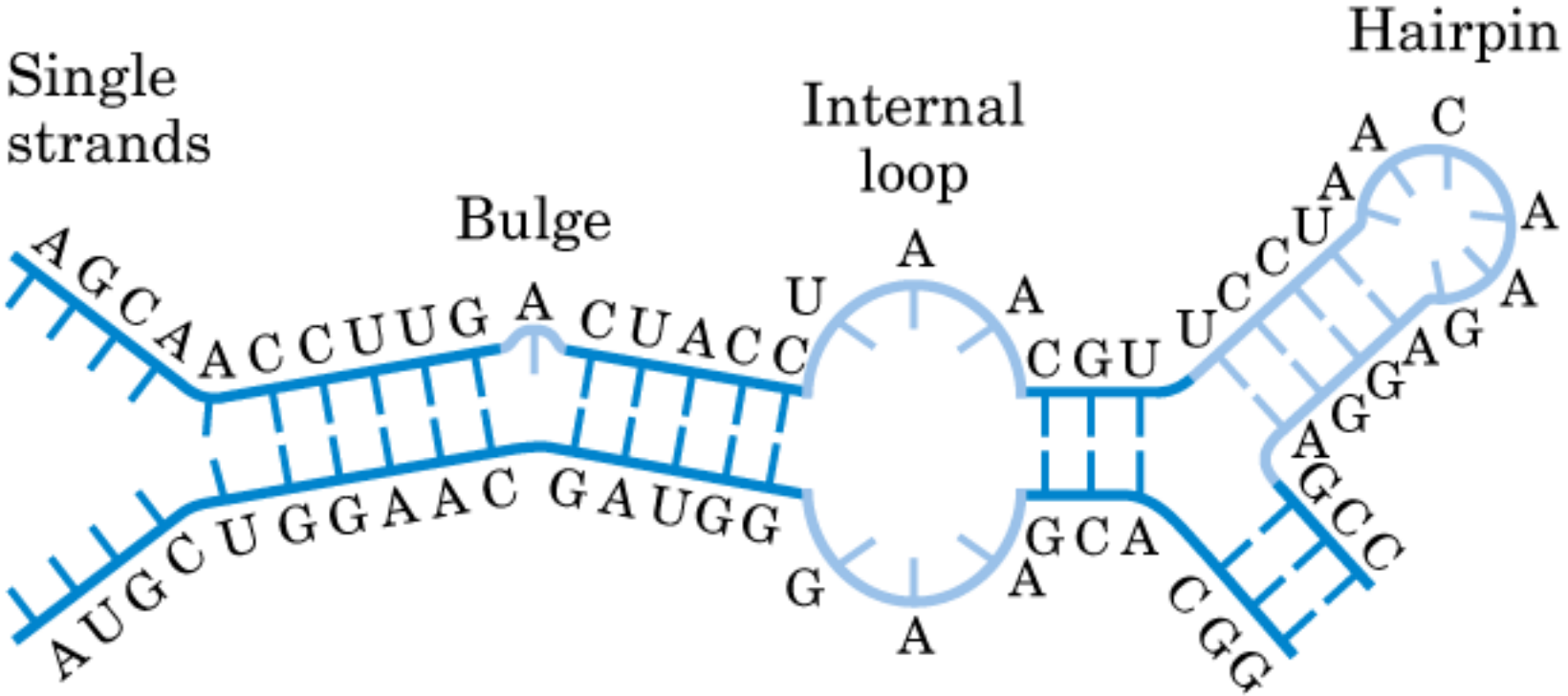
5.1 Single strand RNA tends to take right-handed helical conformation.

5. 2 Self-complementary sequences on a RNA molecule lead to specific structures.

5. 3 The standard base-pairing rules are followed (i.e., A with U, G with C).

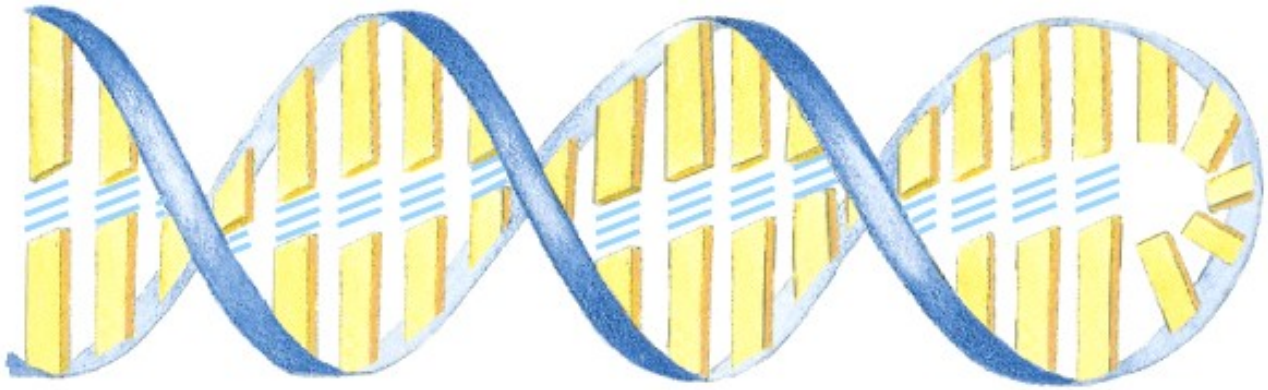
5.4 Intrastrand base pairing makes structures including internal loops, and hairpins (helical).




Single strands
Bulge
Internal loop
Hairpin
AGCAACCUUG
A
CUACC
U
A
CGU
UCCU
AC
A
A
G
AG
G
AGCC
CGG
AUGCUGGAAC
GAUGG
G
A
GCA
A

(a)





Hairpin double helix
(b)

6. Duplex DNA and RNA molecules can be denatured and renatured.

6.1 Denaturation

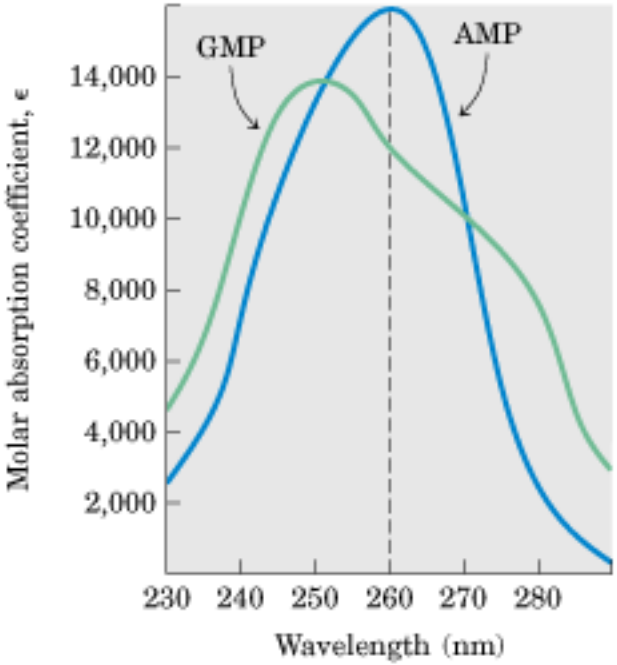
6.1.1 Viscosity decreases sharply.

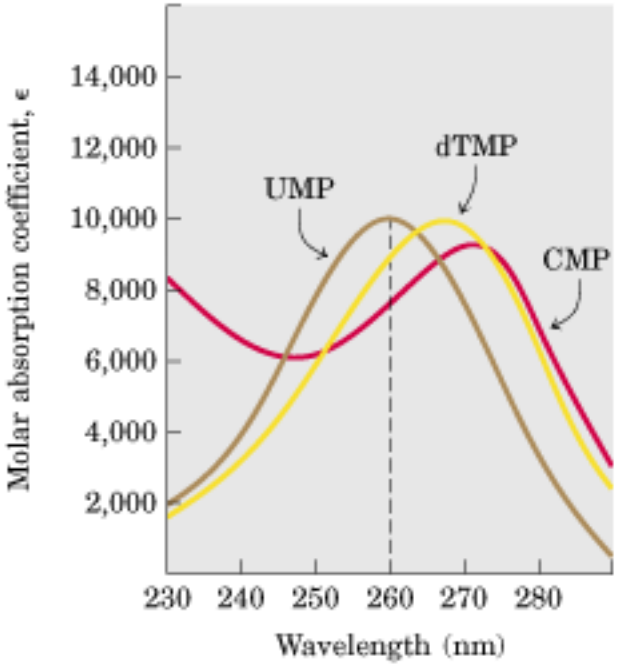
6.1.2 UV absorption at 260 nm increases significantly, an effect called hyperchromism or hyperchromic effect (增色效应).

DNA 复性——减色效应（ hypochromic effect ）

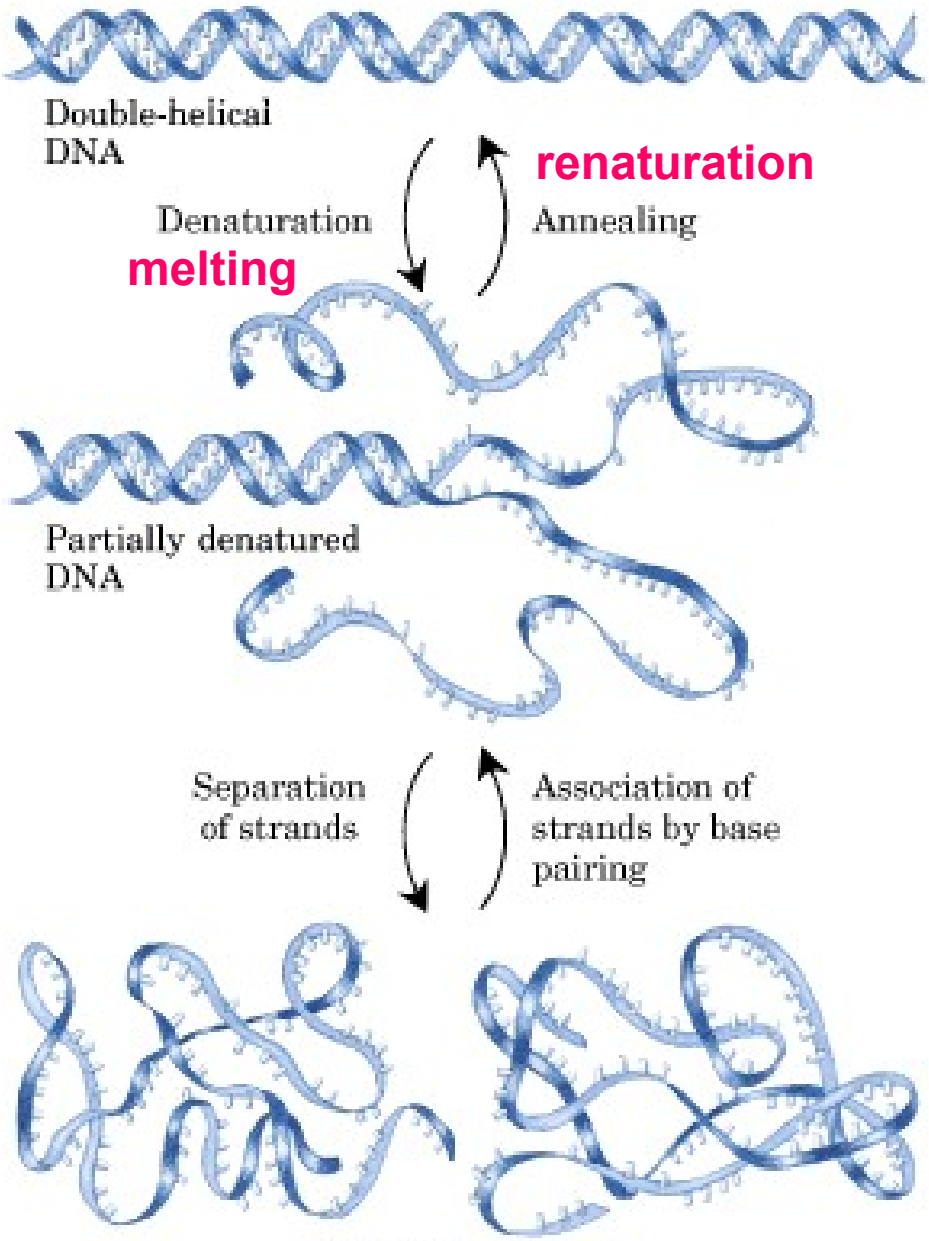


核酸、核苷酸的紫外吸收性质





Molar absorption coefficient at 260 nm, ϵ_{260} ($M^{-1}cm^{-1}$)	
AMP	15,400
GMP	11,700
CMP	7,500
UMP	9,900
dTMP	9,200


Double-helical DNA
renaturation
Denaturation
melting
Annealing
Partially denatured DNA
Separation of strands
Association of strands by base pairing
Separated strands of DNA in random coils

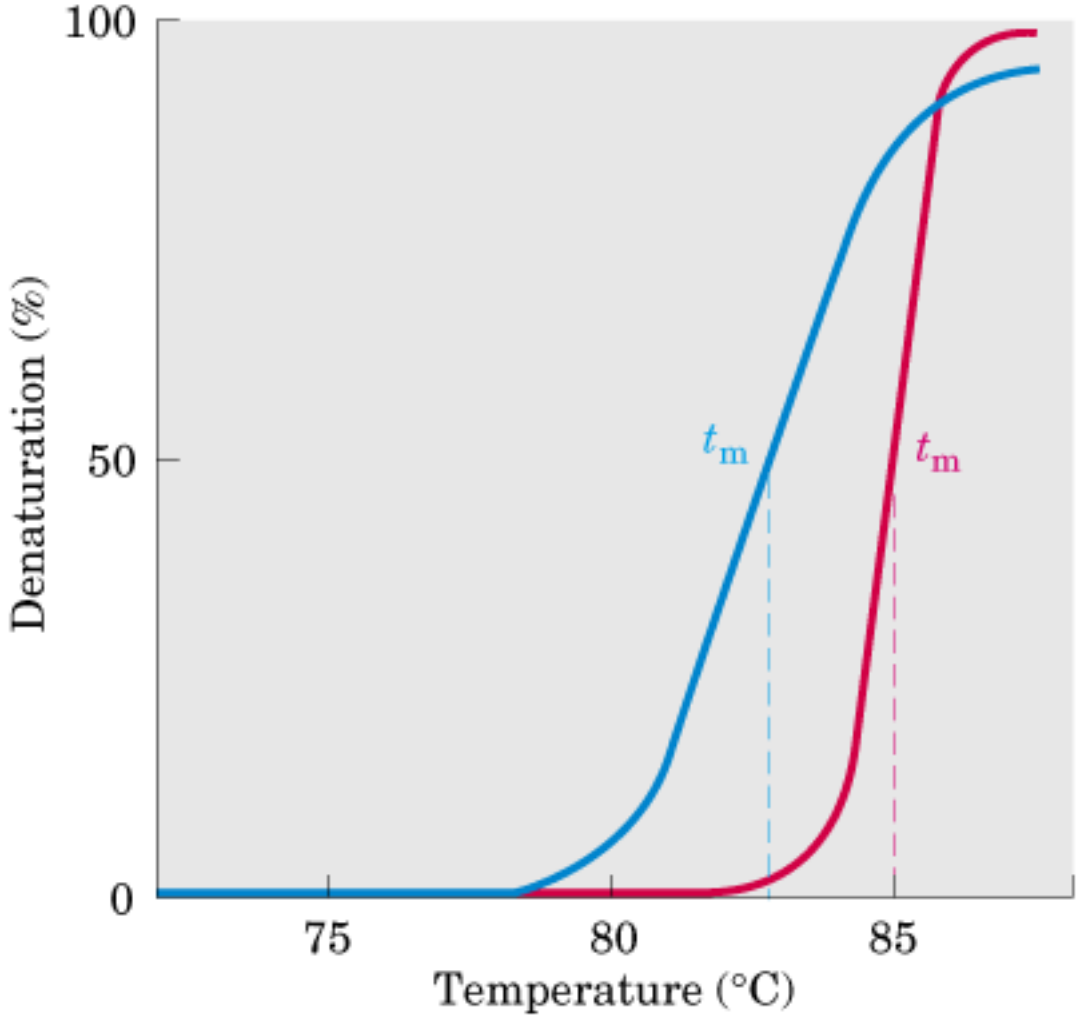
6.1.3 The unwinding (i.e., denaturation) of the double helix is called melting.

6.1.4 Each species of DNA has a characteristic melting temperature (Tm or tm) at which half of the duplex chain is separated. (Τηε μιδποιντ οφ τηε τεμπερατυρε ρανγε οωερ ωηιχη δενατυρατιον οχχυρσ ισ χαλλεδ τηε μελτινγ τεμπερατυρε, ορ *T*μ.)

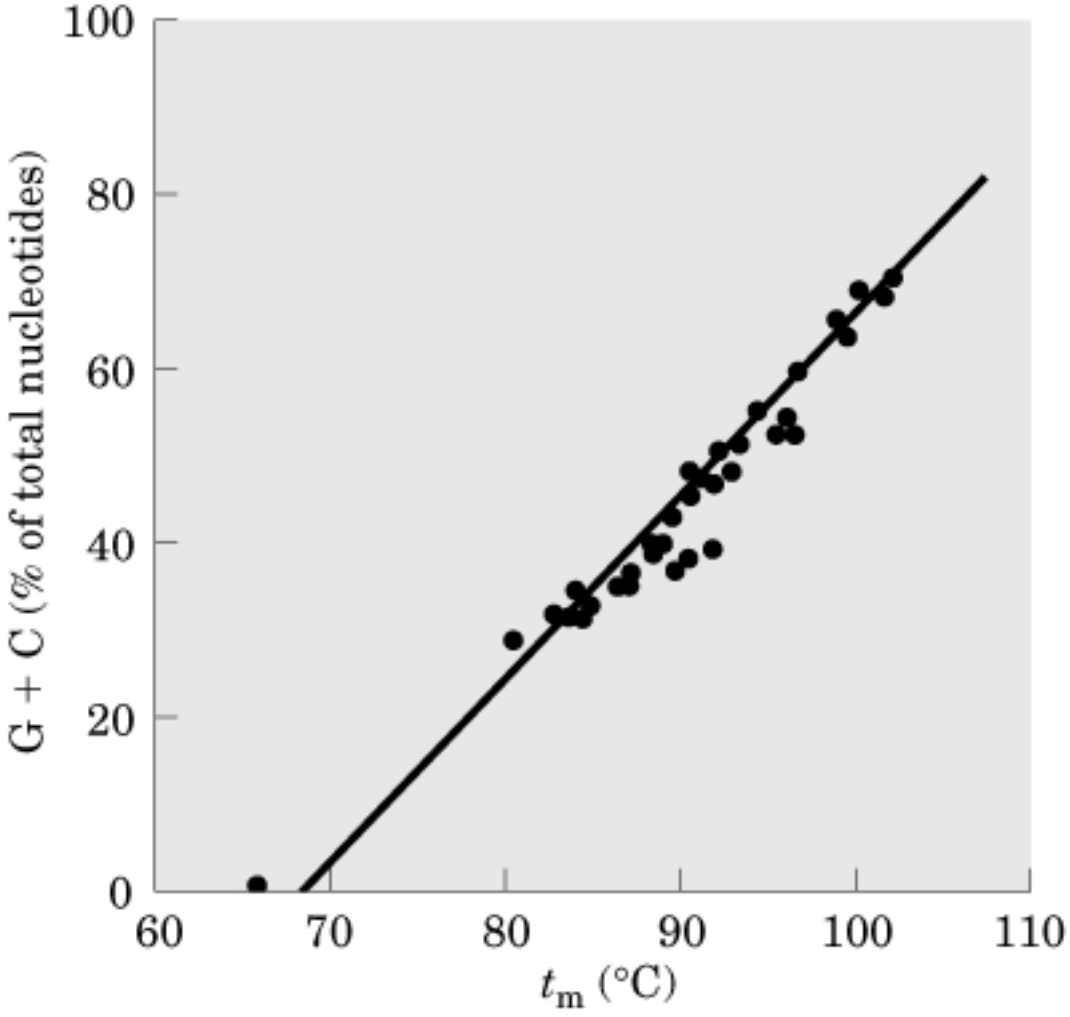
6.1.5 Tm has an approximate linear relationship with (G+C)%).

6.1.6 DNA segments rich in AT base pairs are melted first (at lower temperatures).

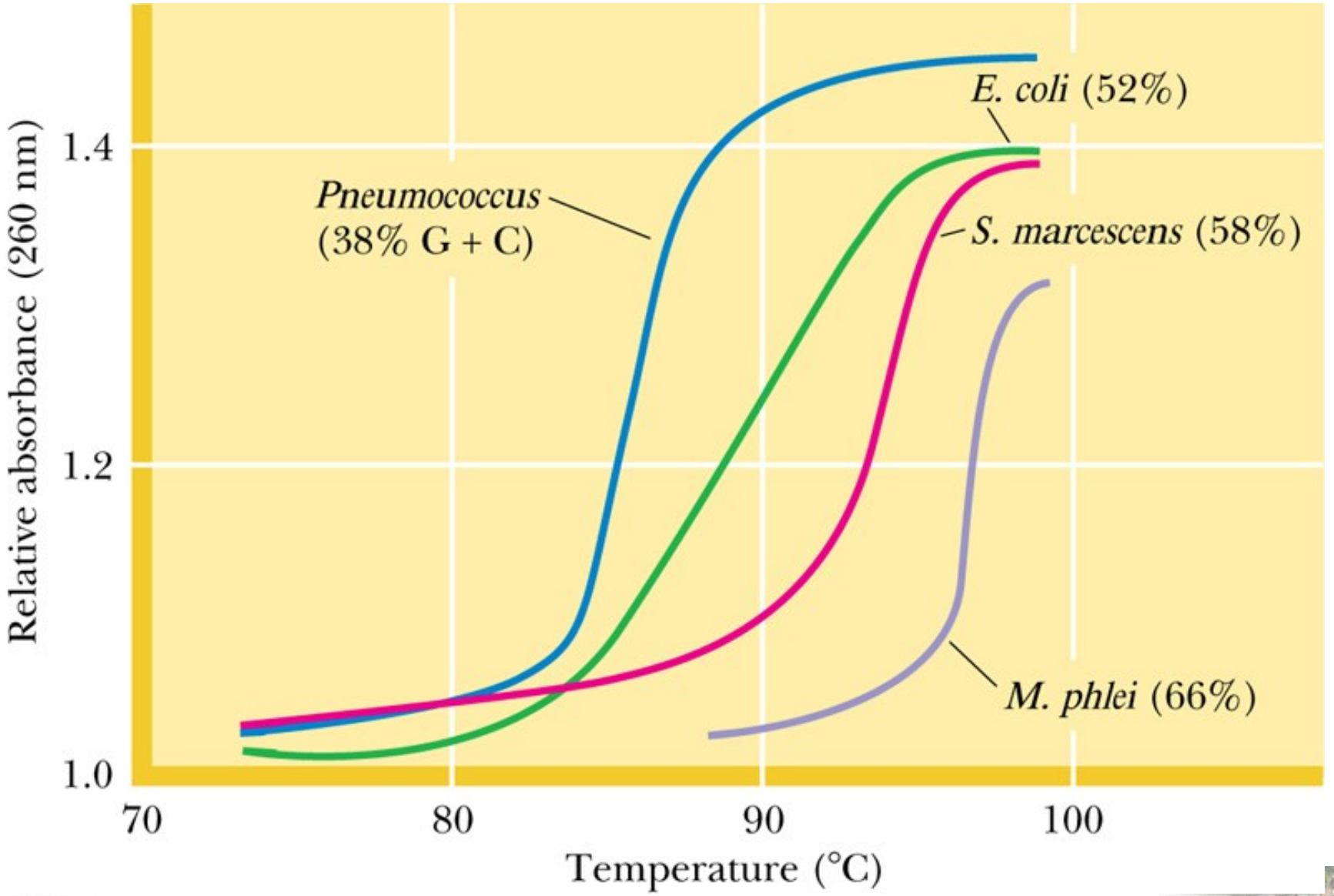


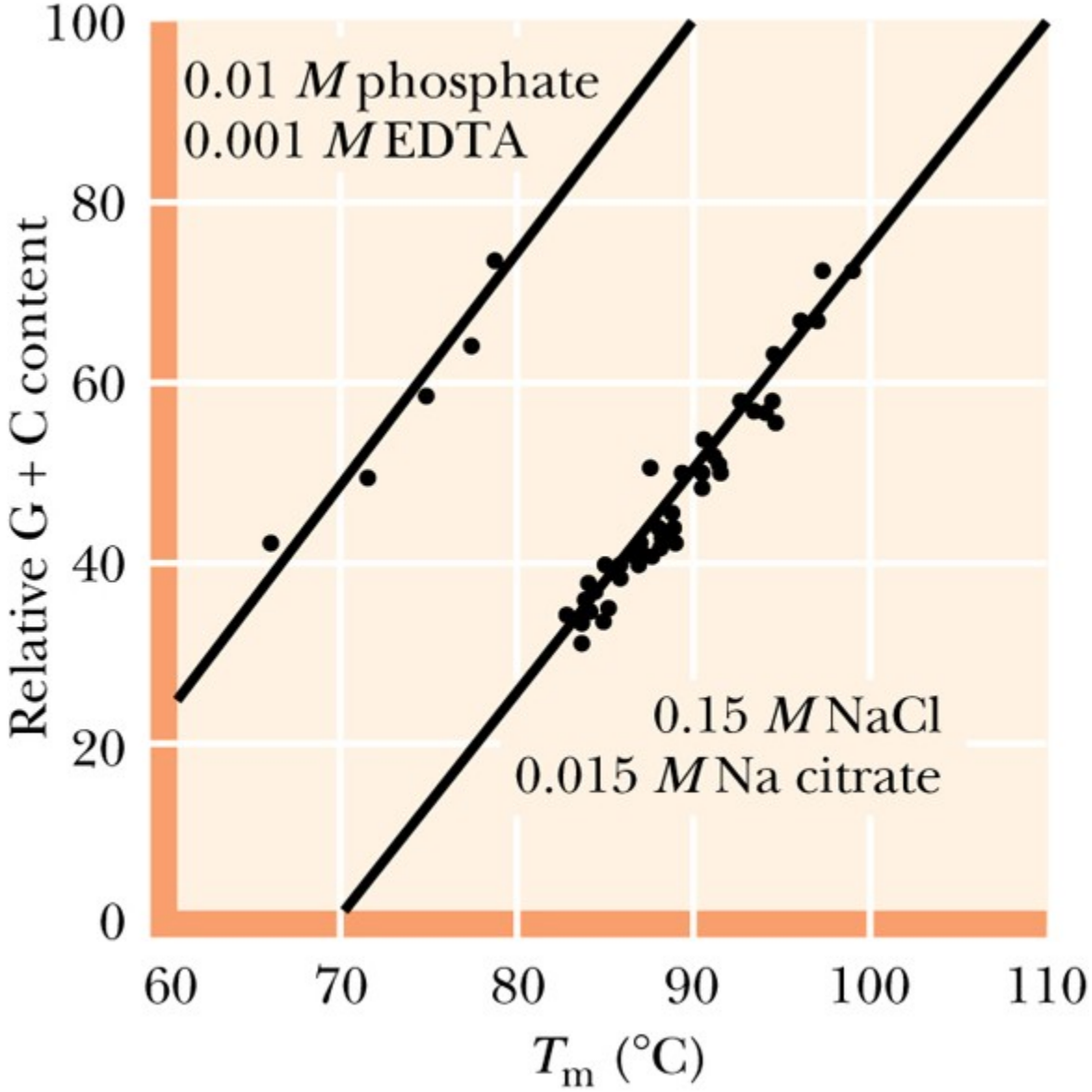


(a)



(b)


Relative absorbance (260 nm)
1.4
1.2
1.0
70
80
90
100
Temperature (°C)
E. coli (52%)
Pneumococcus
(38% G + C)
S. marcescens (58%)
M. phlei (66%)


100
80
60
40
20
0
Relative G + C content
0.01 M phosphate
0.001 M EDTA
0.15 M NaCl
0.015 M Na citrate
60
70
80
90
100
110
T_m (°C)

DNA 的 Tm 值与下列因素有关

DNA 的均一性
G-C 含量
介质的离子强度



6.2 When the temperature or pH is returned to the biological range, the two separated complementary strands will spontaneously rewind (renature) to form a duplex structure.

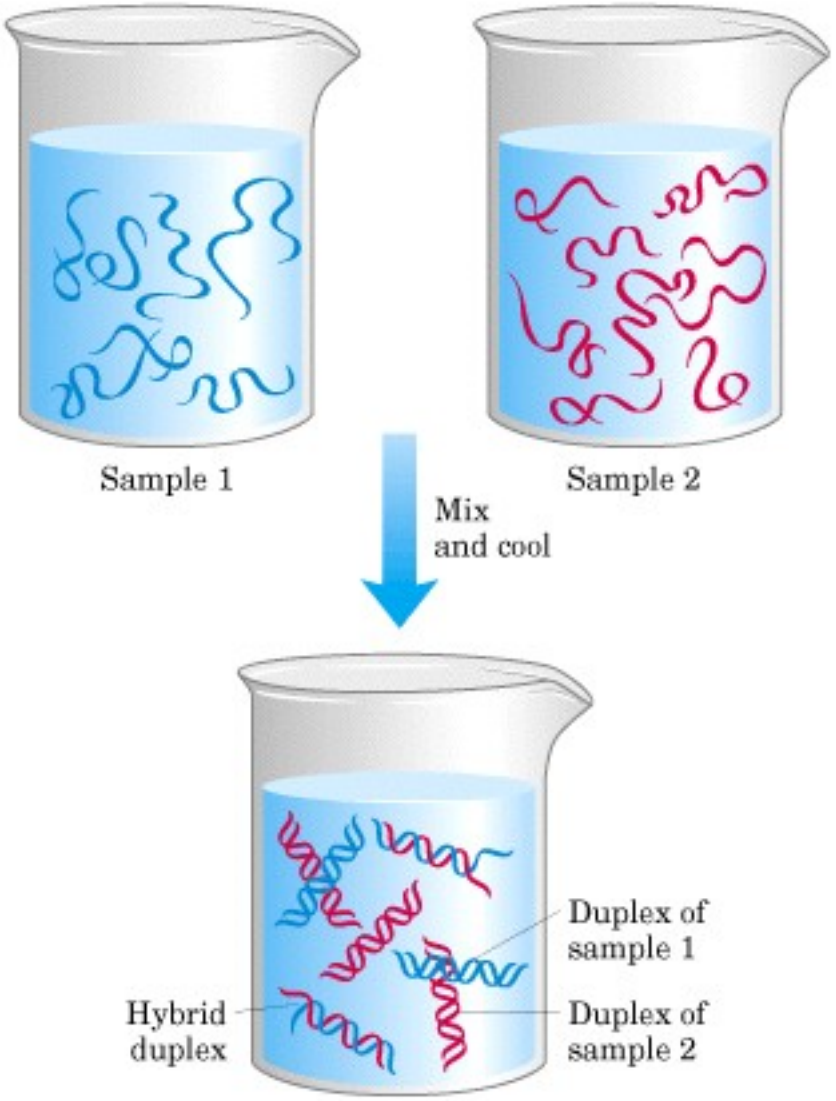
6.2.1 This renaturation process is called annealing.

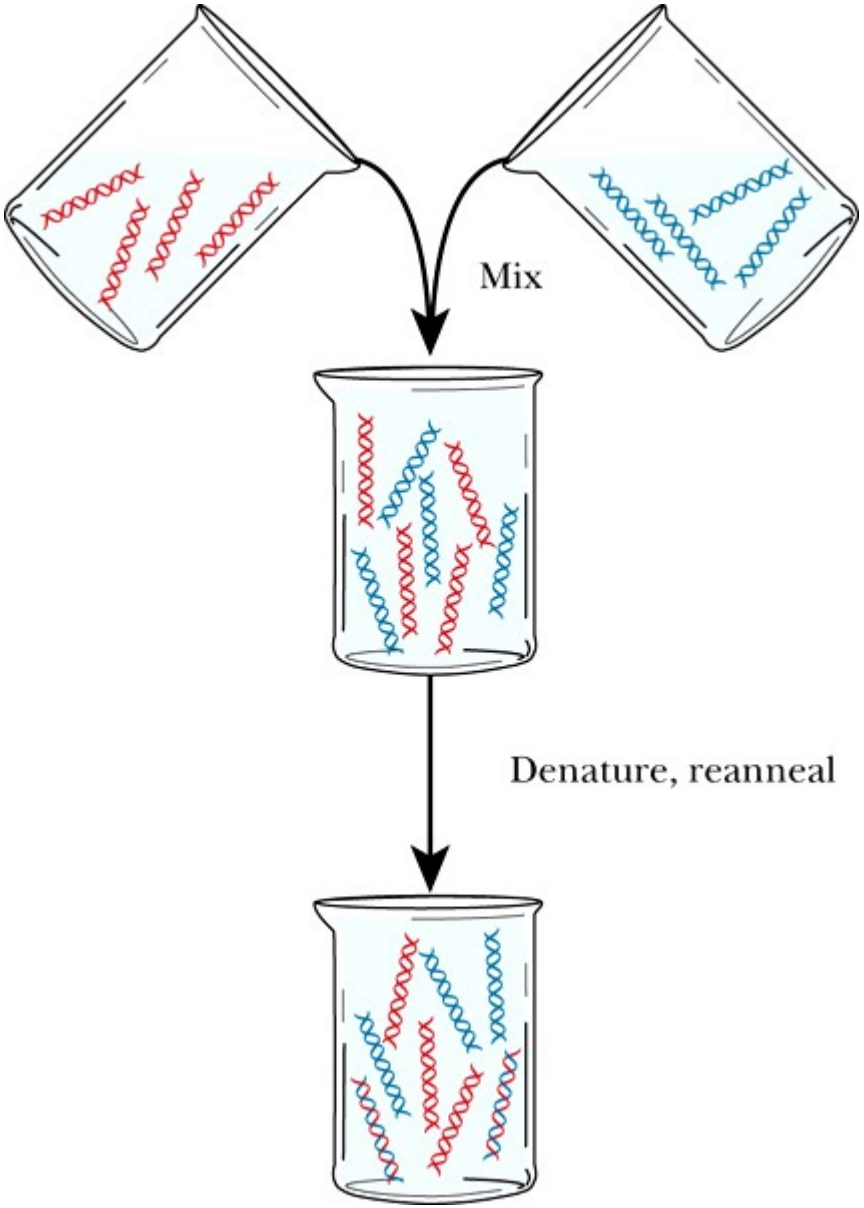
6.2.2 Annealing can occur between complementary DNA, RNA, or DNA-RNA hybrids.

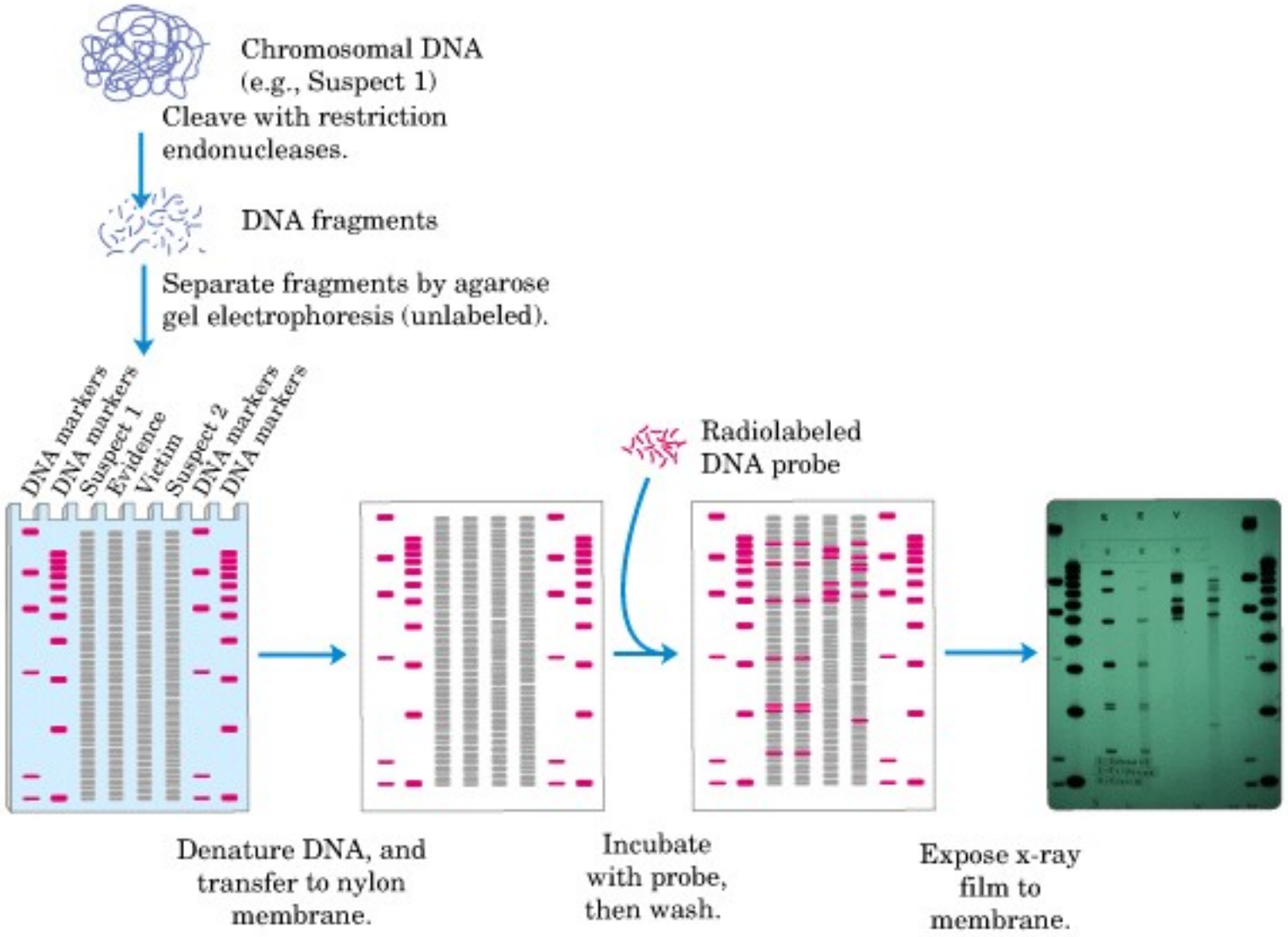
6.2.3 This hybridization principle is widely used in detecting existence of specific DNA or RNA species in cells.



分子杂交


Sample 1
Sample 2
Mix
and cool
Duplex of
sample 1
Hybrid
duplex
Duplex of
sample 2


Mix
Denature, reanneal


Chromosomal DNA
(e.g., Suspect 1)
Cleave with restriction endonucleases.
DNA fragments
Separate fragments by agarose gel electrophoresis (unlabeled).
DNA markers
DNA markers
Suspect 1
Evidence
Victim
Suspect 2
DNA markers
DNA markers
Radiolabeled DNA probe
Denature DNA, and transfer to nylon membrane.
Incubate with probe, then wash.
Expose x-ray film to membrane.

Southern blot

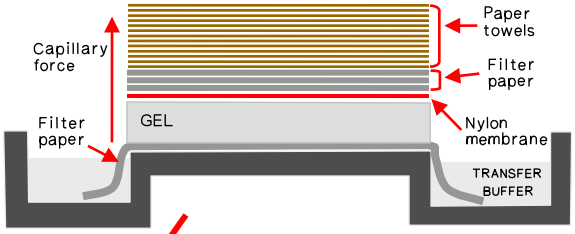
1. Digest DNA with restriction enzymes



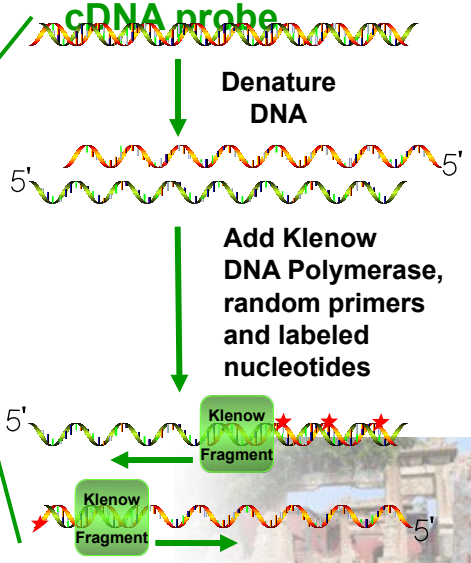
2. Separate fragments on agarose gel



3a. Transfer separated DNA onto membrane for further analysis



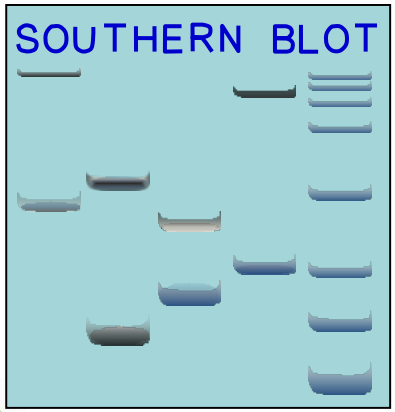
3b. Synthesize labeled cDNA probe



4. Hybridize denatured probe with the membrane



5. Wash the membrane and expose with the film

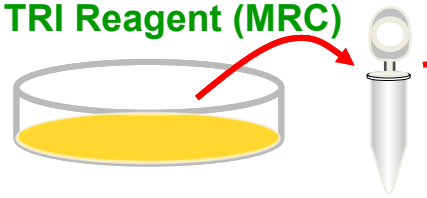


Biochemistry

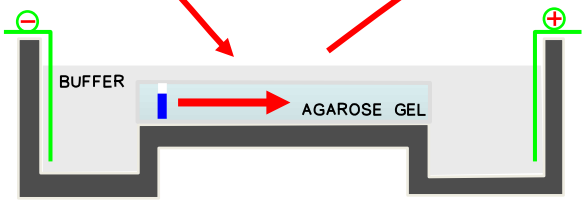
Northern blot

1. Collect cells or tissue

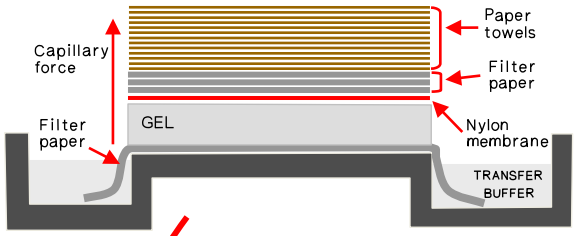
and extract RNA with TRI Reagent (MRC)



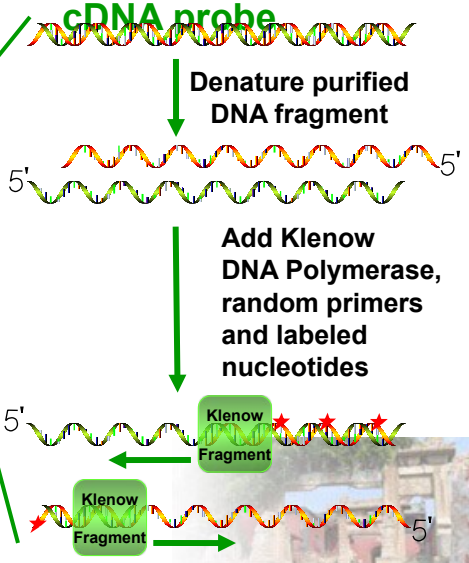
2. Separate RNA on agarose gel



3a. Transfer separated RNA onto membrane



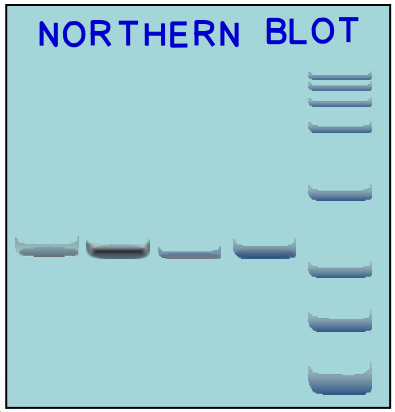
3b. Synthesize labeled cDNA probe



4. Hybridize denatured probe with the membrane



5. Wash the membrane and expose with the film



Biochemistry

F2 DNA organization in chromosomes

Every cell of a multicellular organism generally contains the same genetic material.

DNA molecules are the largest macromolecules in the cell and are commonly packaged into structures called chromosomes.

Most bacteria and viruses have a single chromosome, whereas eukaryotic cells usually contain many.



DNA Molecules Are Much Longer than the Cellular or Viral Packages That Contain Them

Virus	Size of viral DNA (bp)	Length of viral DNA (nm)	Long dimension of viral particles (nm)
ϕX174	5,386	1,939	25
T7	39,936	14,377	78
λ (lambda)	48,502	17,460	190
T4	168,889	60,800	210

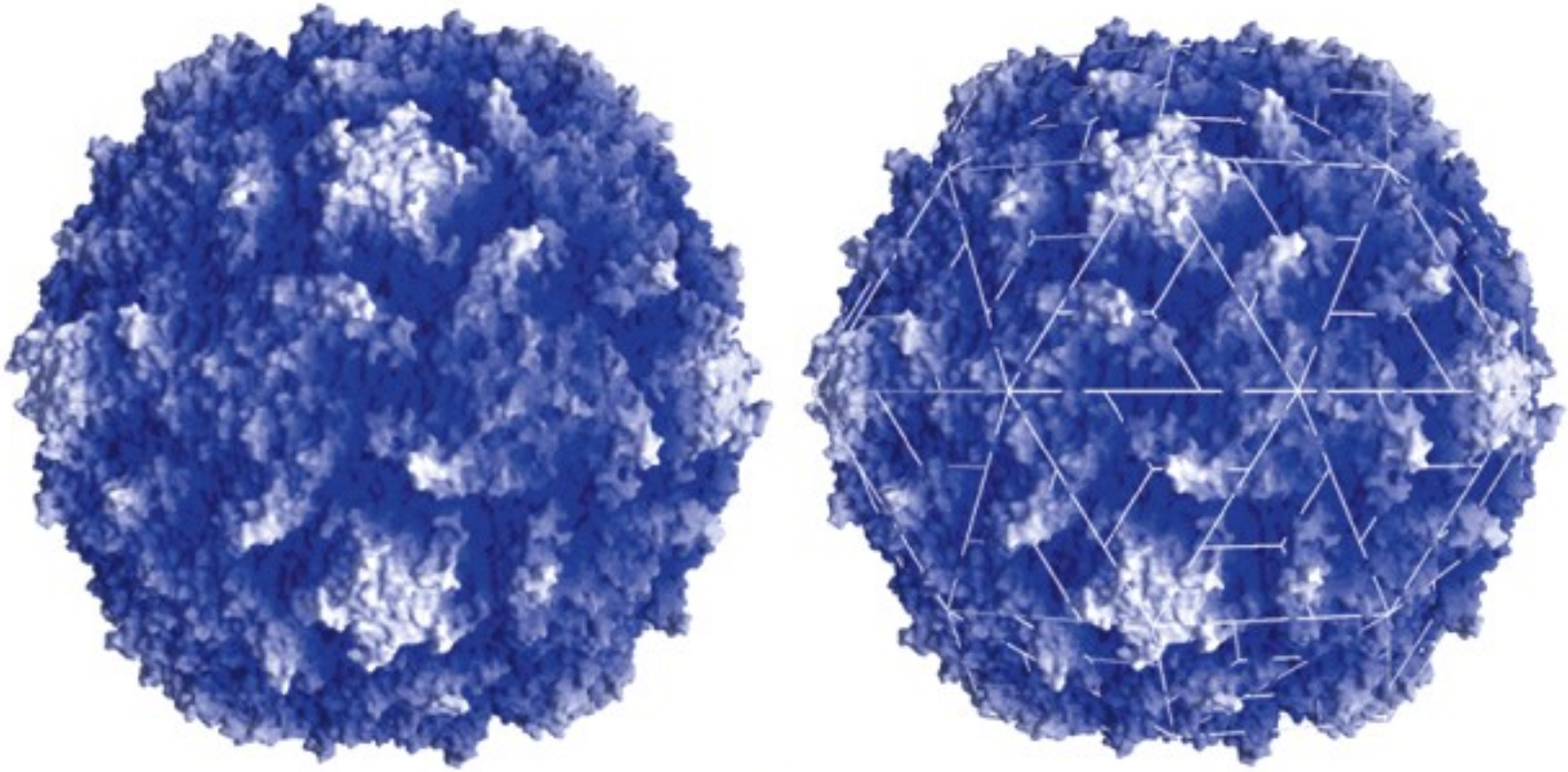
Note: Data on size of DNA are for the replicative form (double-stranded). The contour length is calculated assuming that each base pair occupies a length of 3.4 Å (see Fig. 8-13).

1.Viral DNA molecules are relatively small

1.1 Viruses generally contain considerably less genetic information than cells.

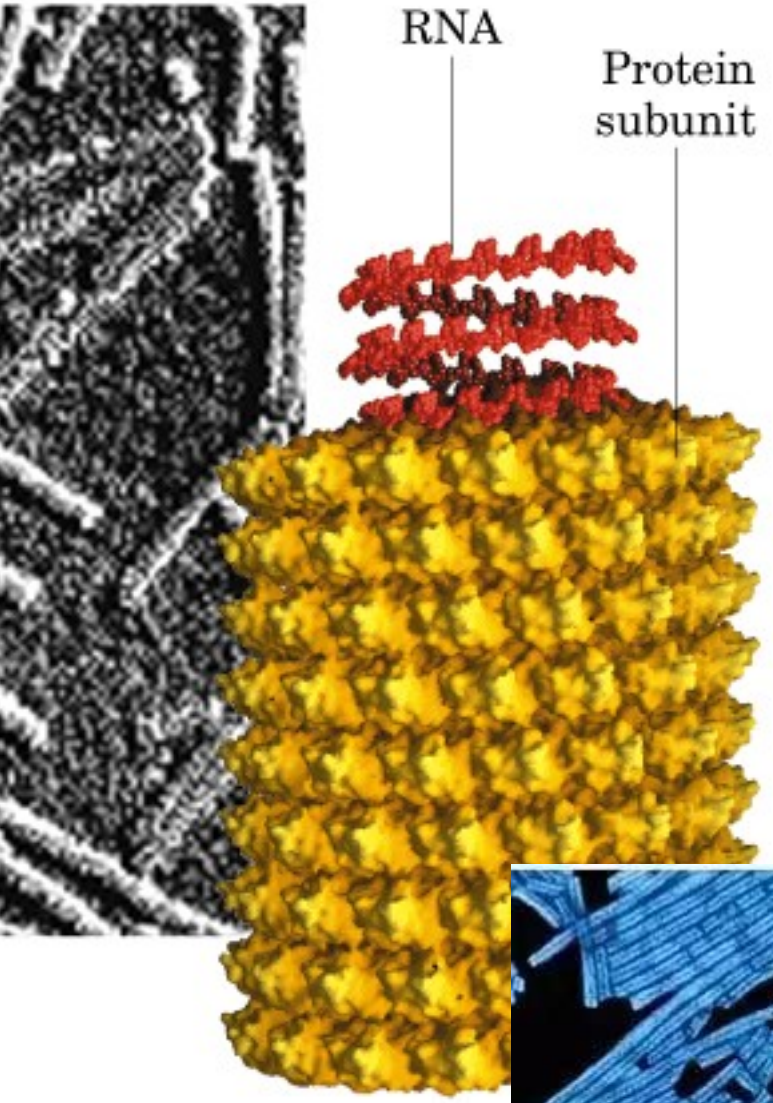
An infectious virus particle often consists a single RNA or DNA molecule surrounded by a protein coat.



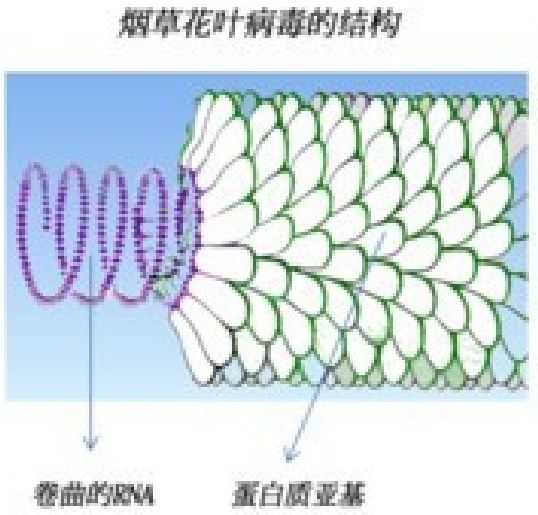


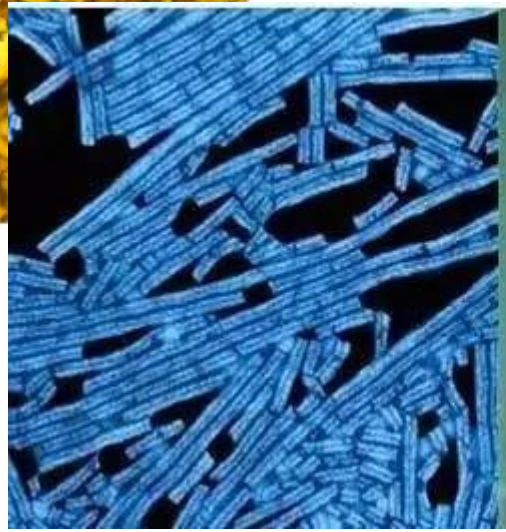
(a)





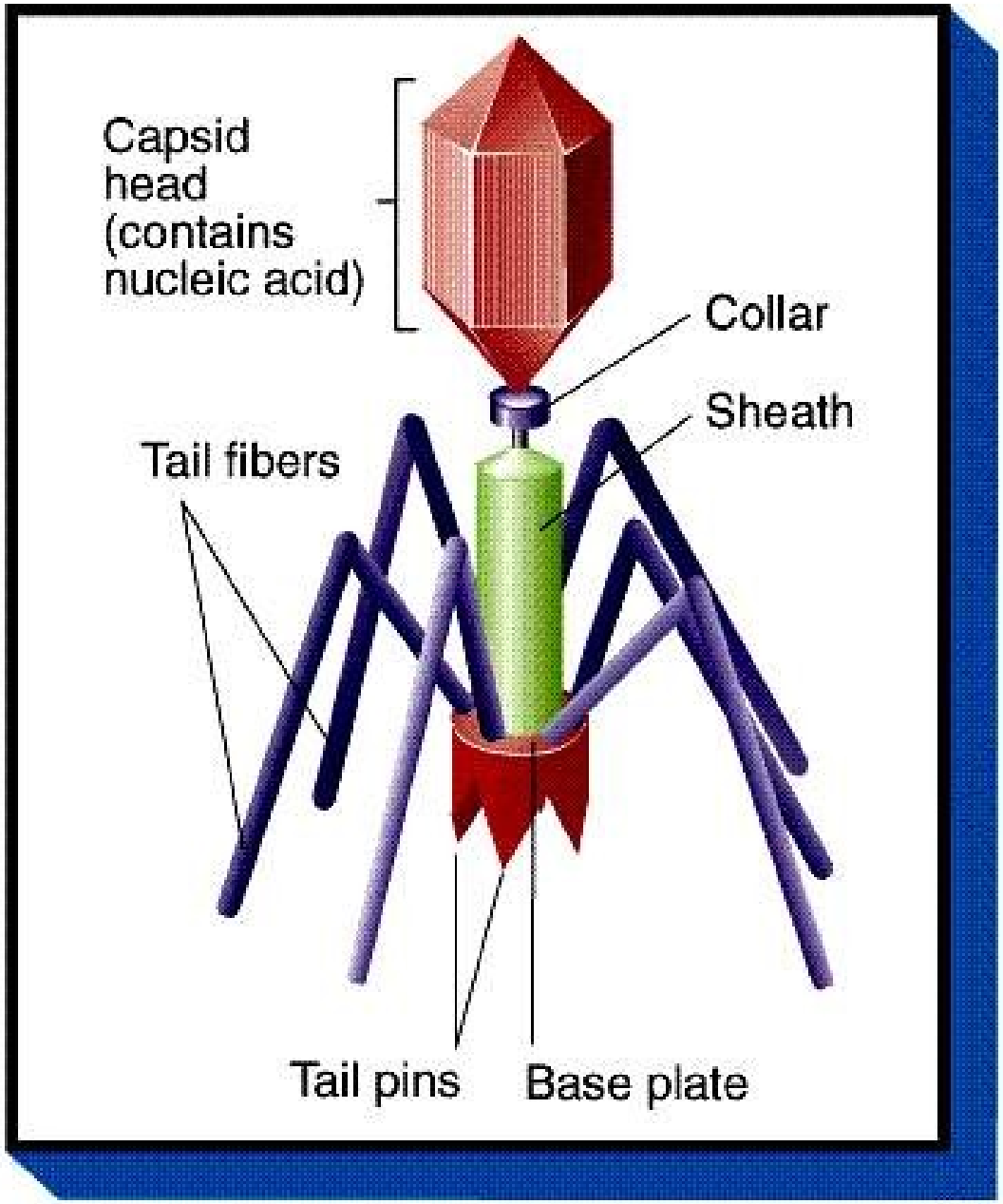
(b)

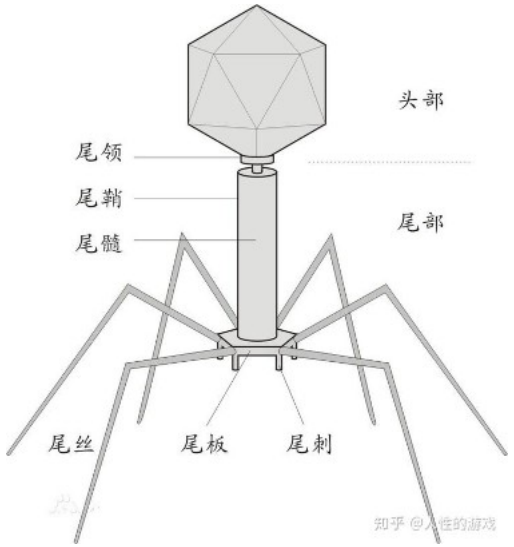


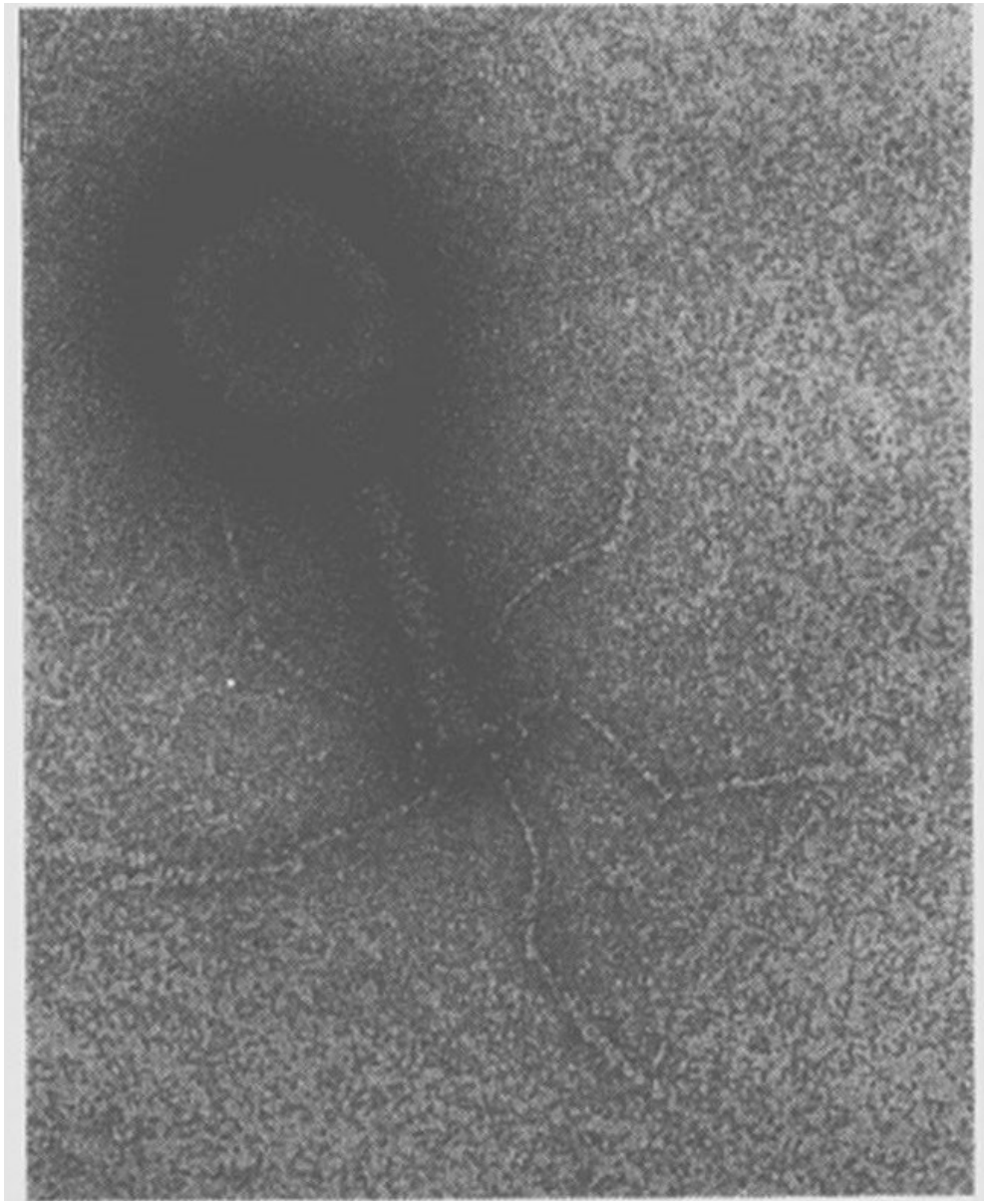


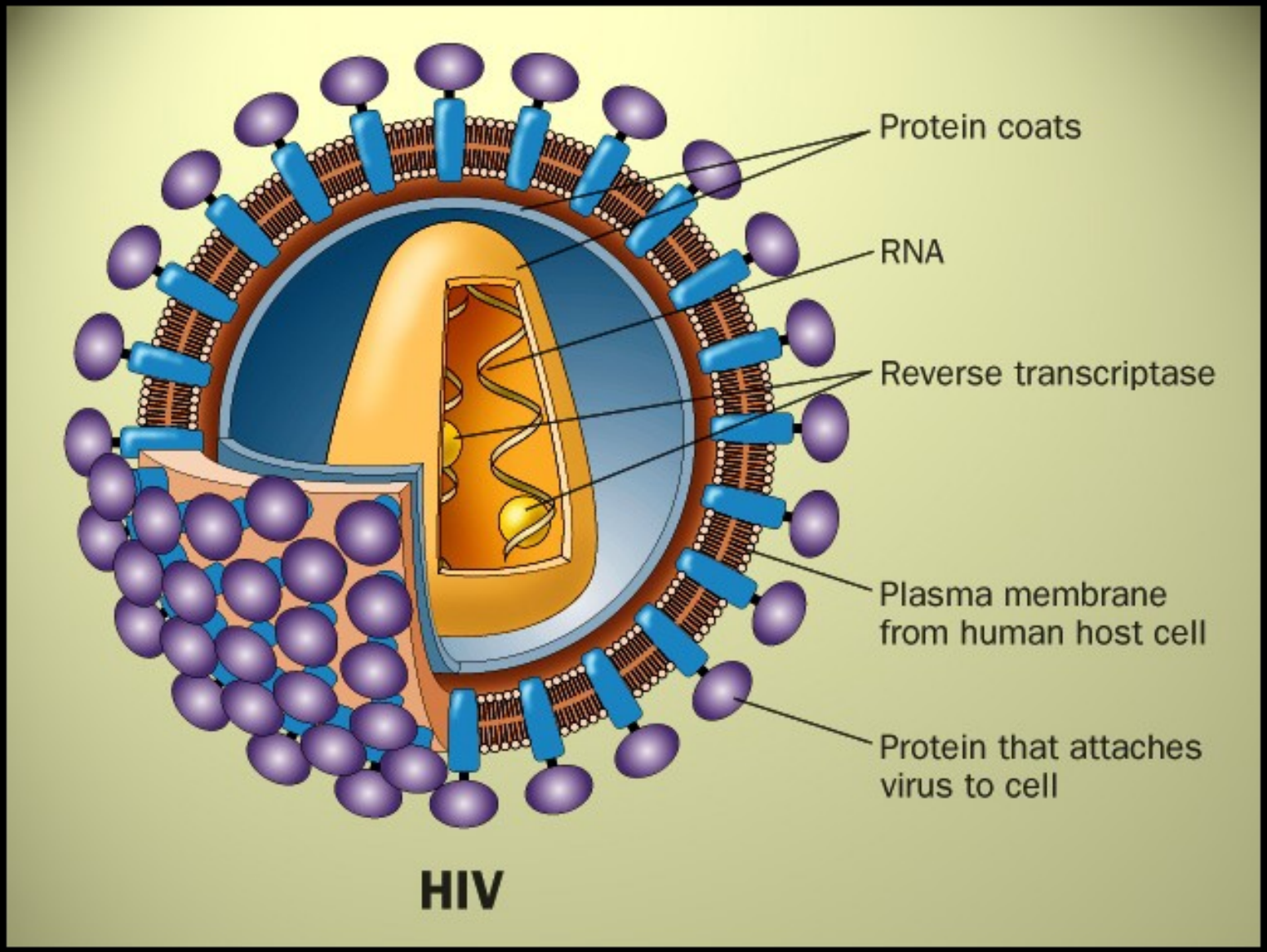


Biochemistry

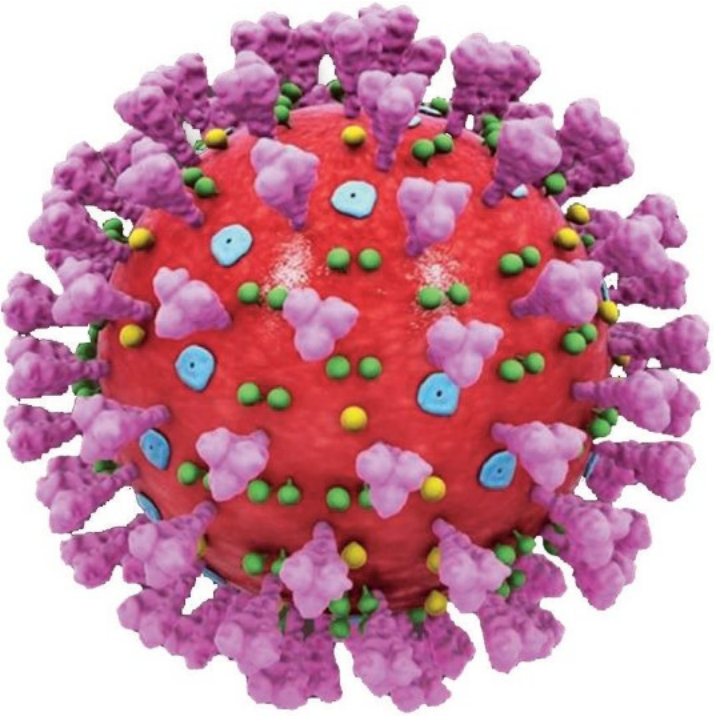

Capsid
head
(contains
nucleic acid)
Collar
Sheath
Tail fibers
Tail pins
Base plate

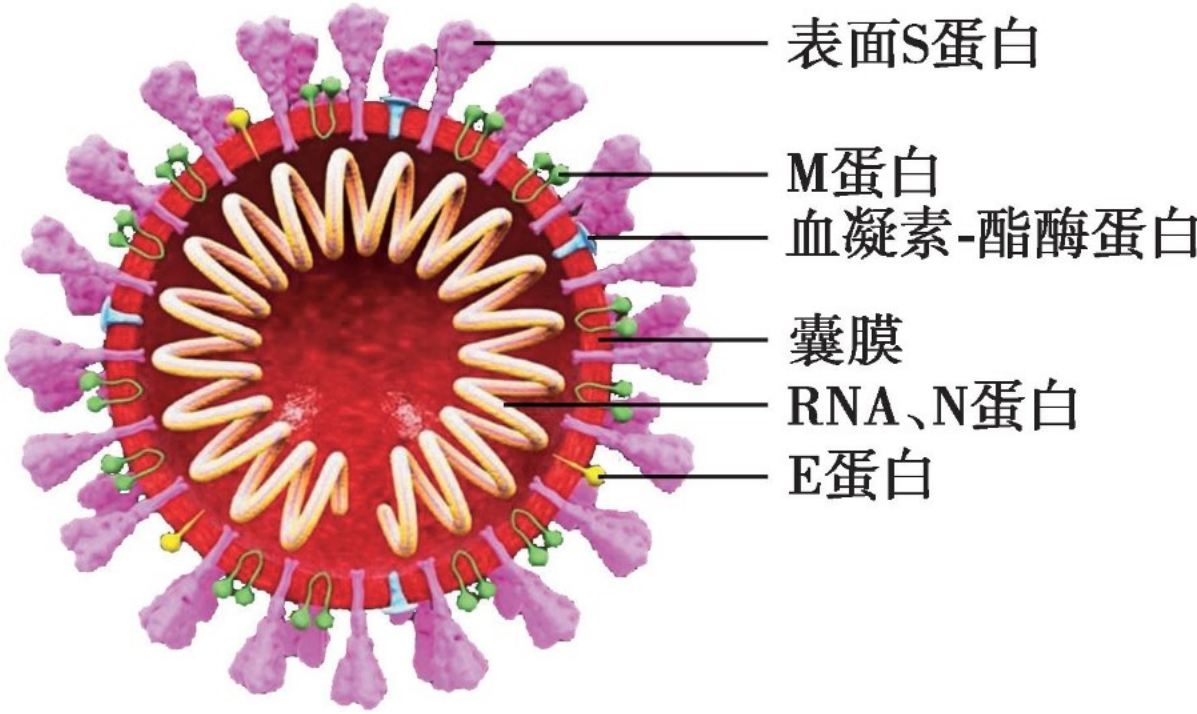

头部
尾领
尾鞘
尾髓
尾部
尾丝
尾板
尾刺




Protein coats
RNA
Reverse transcriptase
Plasma membrane from human host cell
Protein that attaches virus to cell
HIV

新型冠状病毒



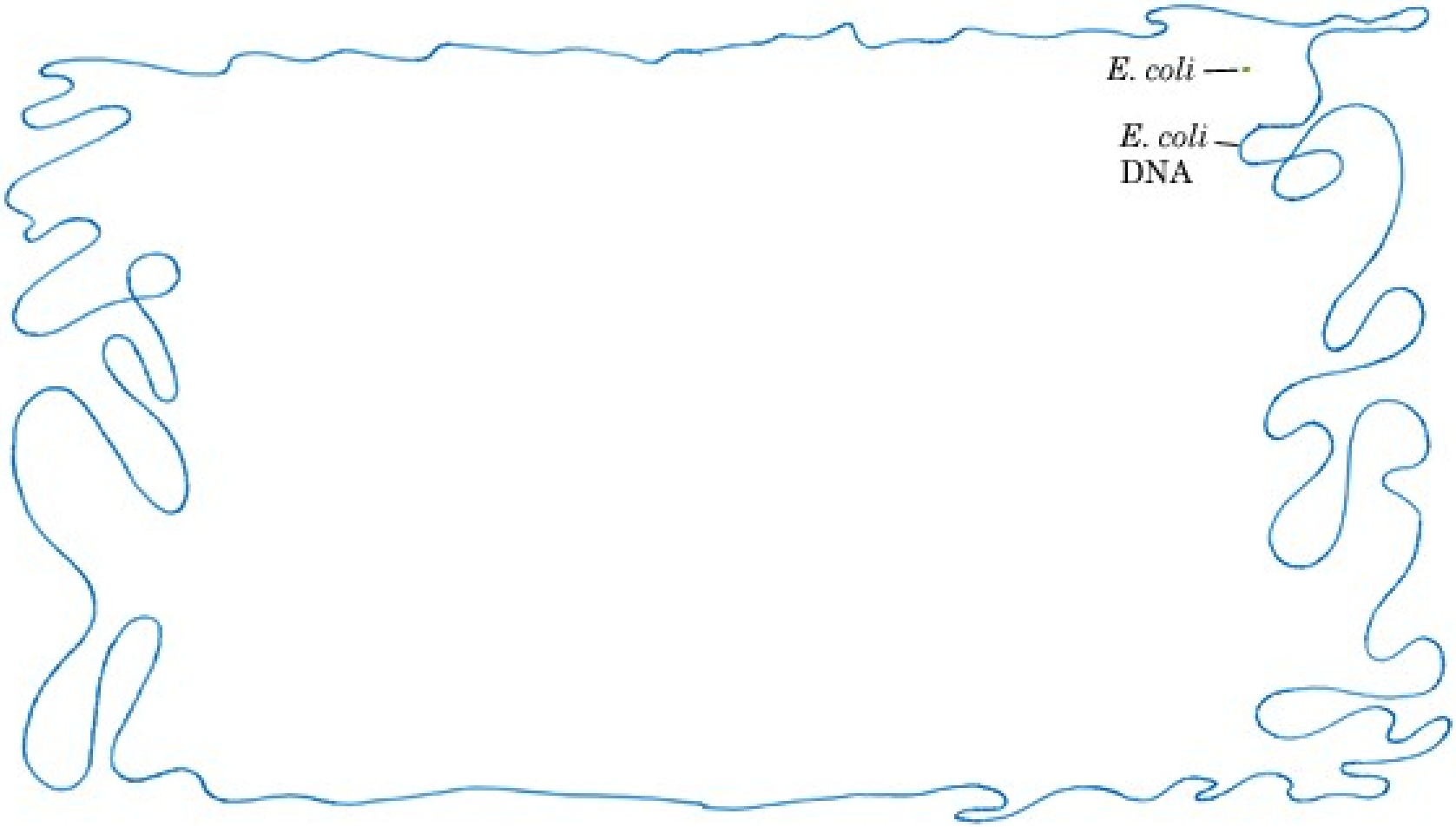


2 Bacteria contain chromosomes and extrachromosomal DNA

2.1 Chromosomal DNA

The chromosome of an *E. coli* cell is a single double-stranded circular DNA molecule. Its 4,639,221 base pairs have a contour length of about 1.7 mm, some **850 times** the length of the E. coli cell. Bacterial DNA must have an even more tightly compacted tertiary structure than the viral DNA.





The length of the E. coli chromosome (1.7 mm) is depicted in linear form relative to the length of a typical E. coli cell (2 μm)

蛋白质骨架

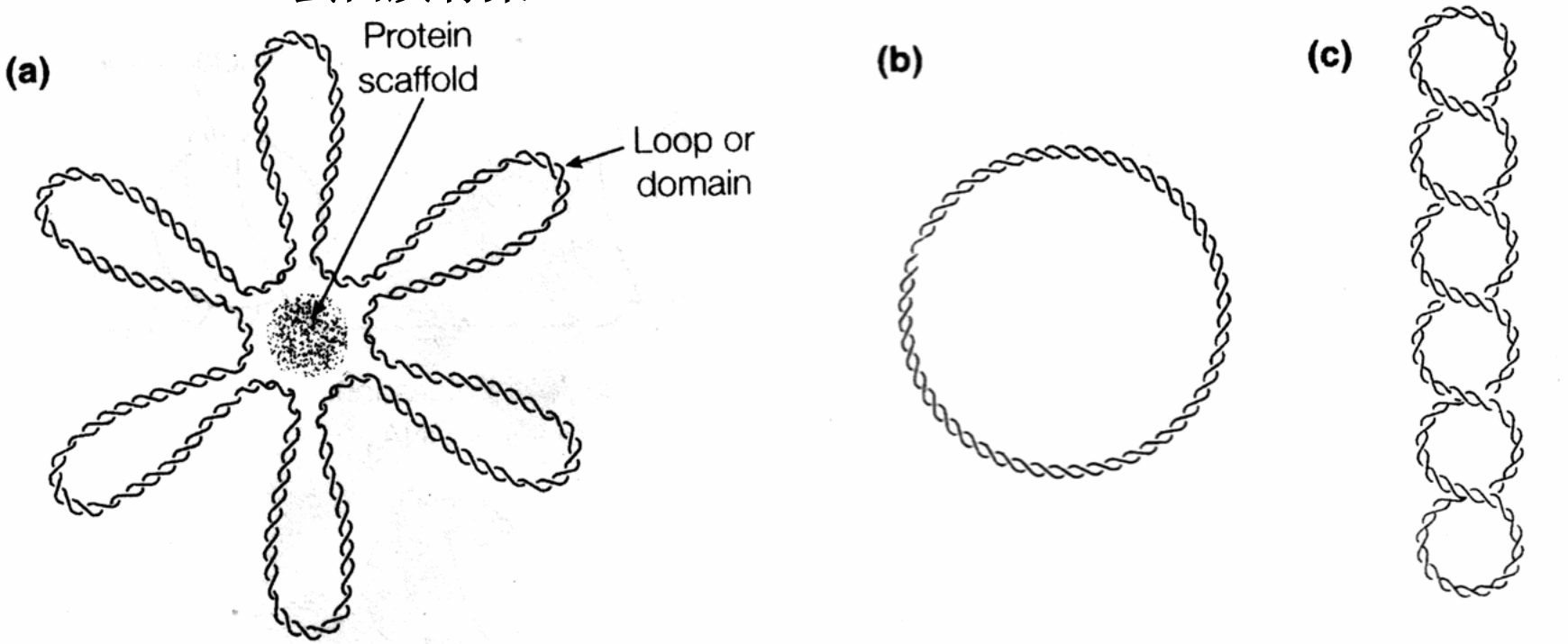
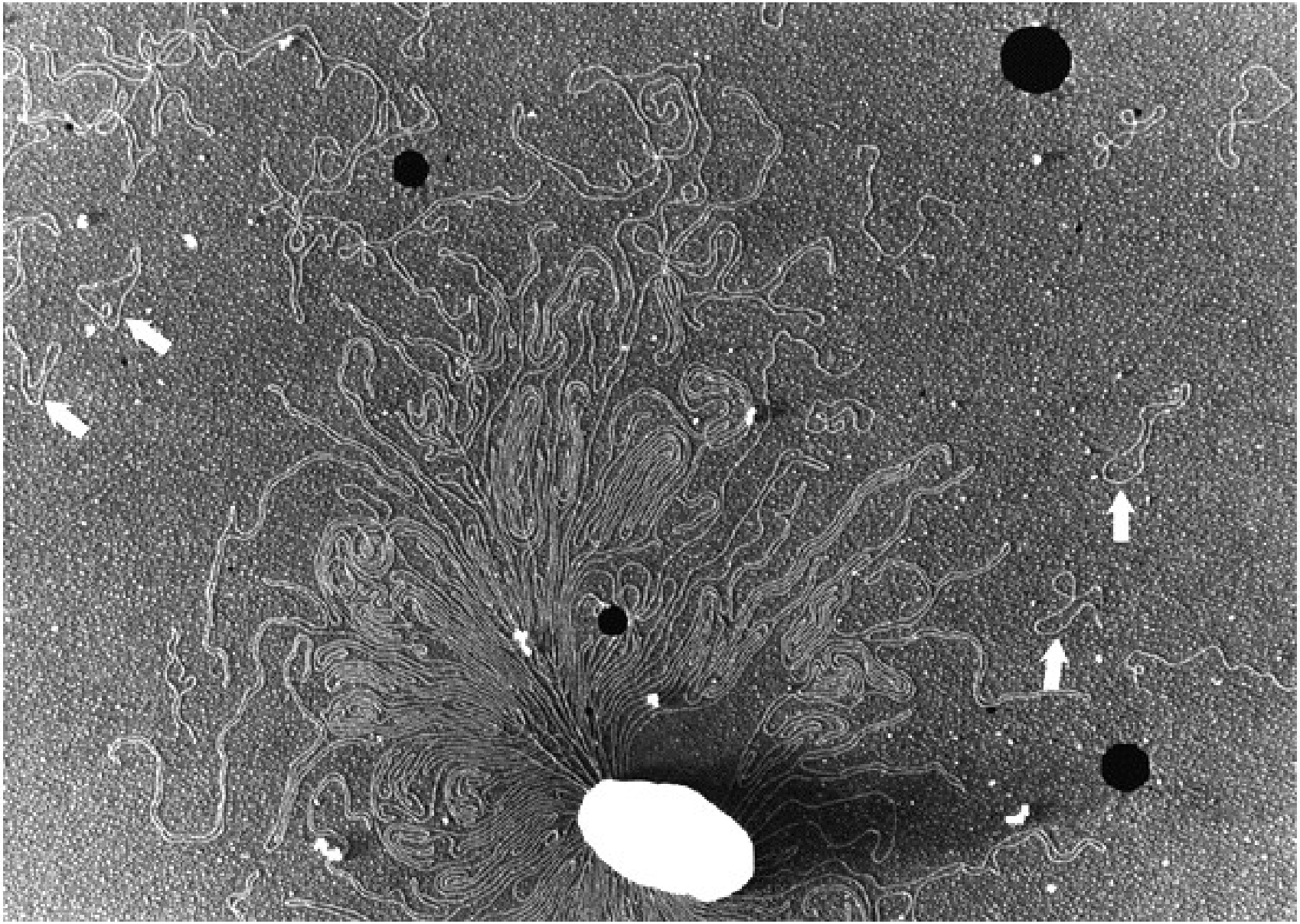
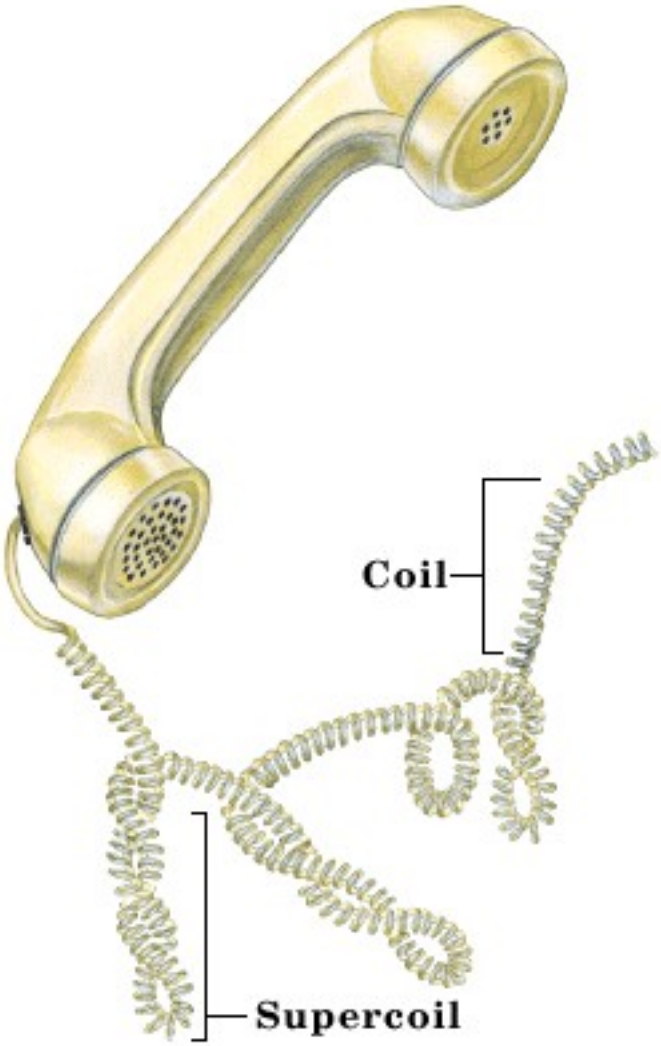
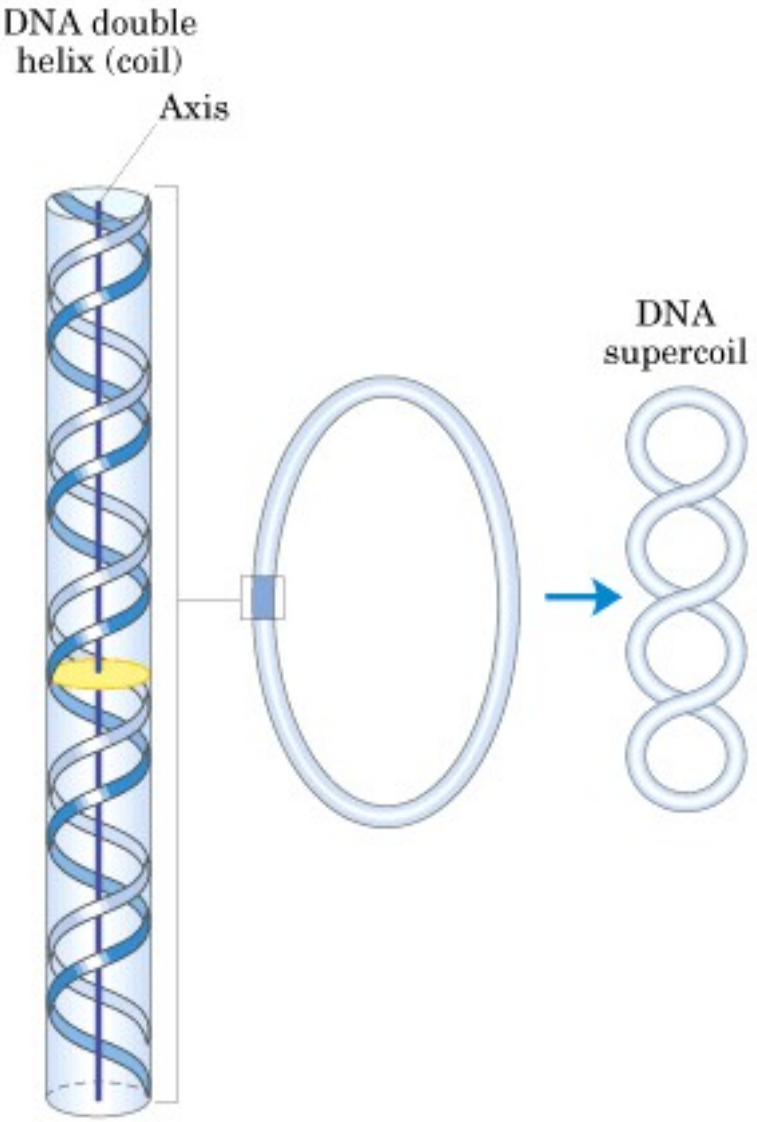


Fig. 1. (a) The association of circular bacterial DNA with a protein scaffold; (b) a circular double-stranded DNA molecule; (c) supercoiled DNA.

The DNA is negatively supercoiled, complexed to several histone-like proteins and organize into about 50 domains bound to a protein scaffold.

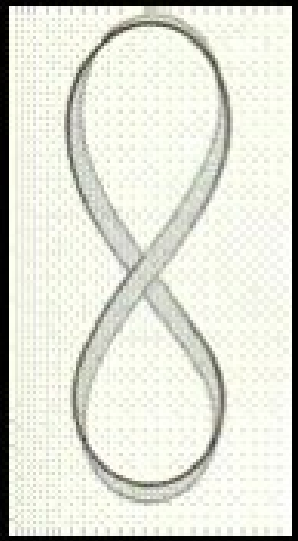


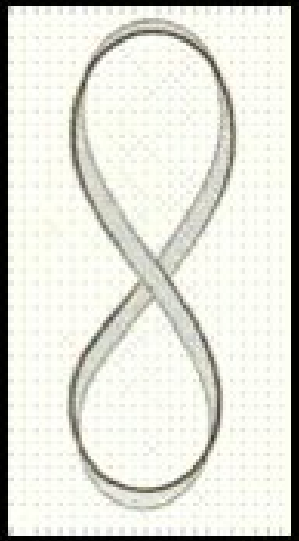

Coil
Supercoil


DNA double helix (coil)
Axis
DNA supercoil

负超螺旋

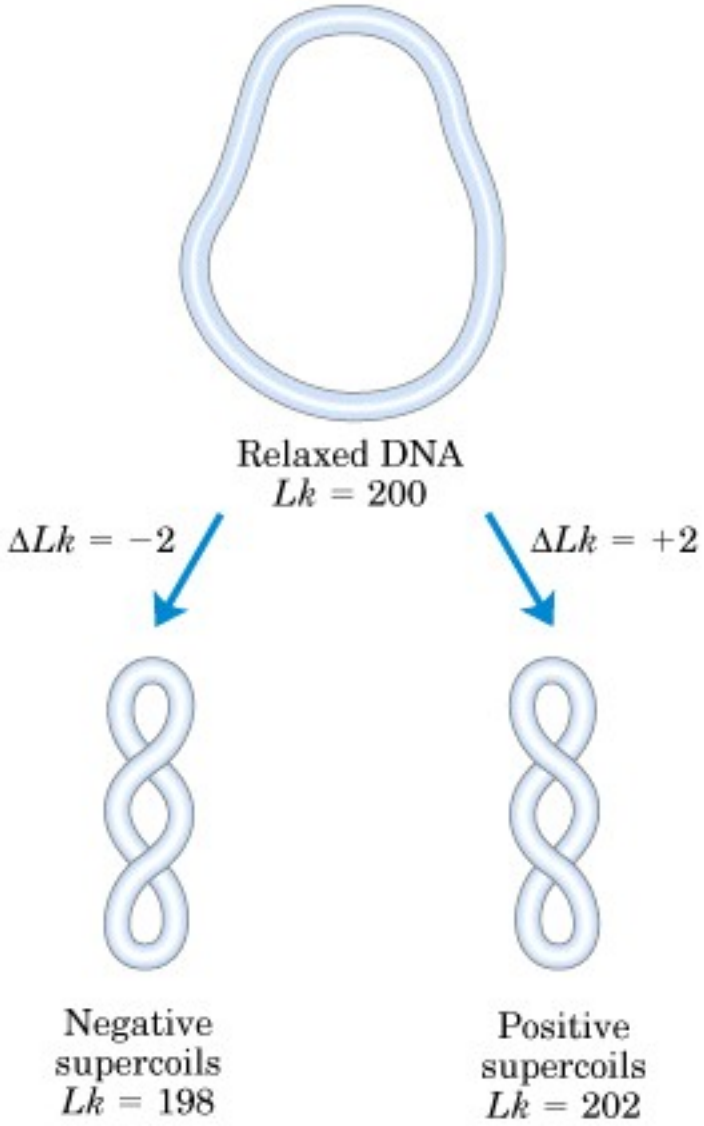
右旋



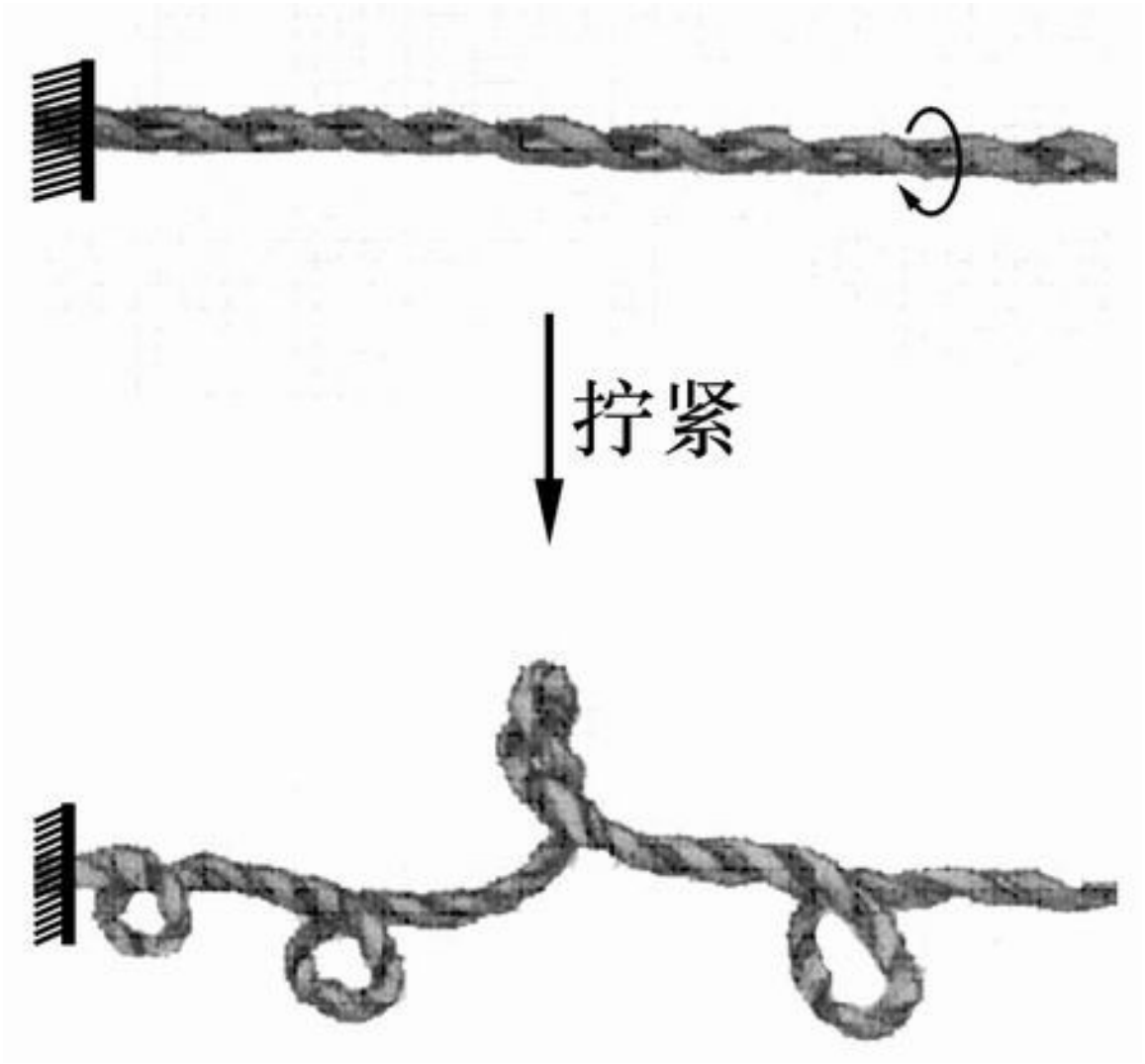


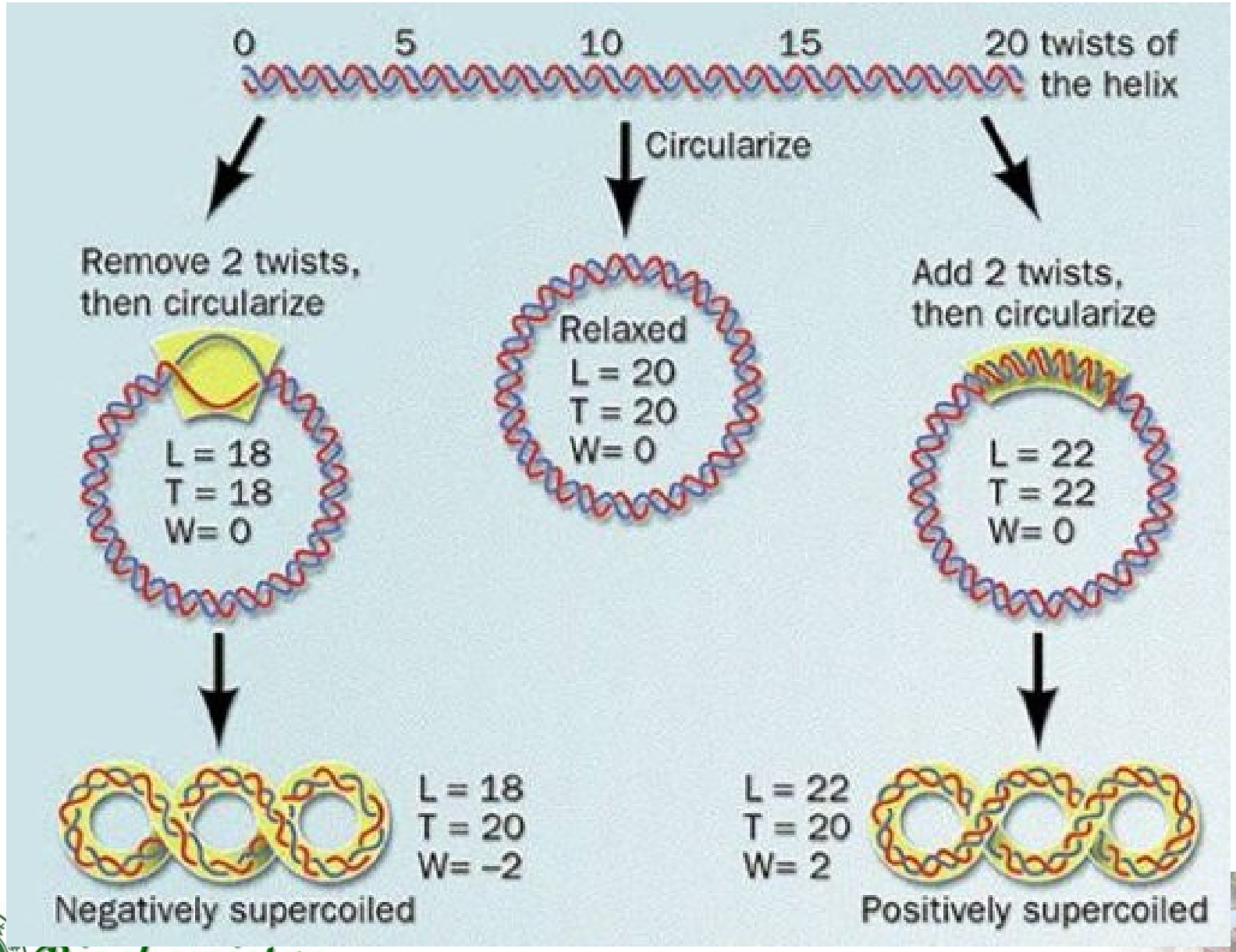
正超螺旋

左旋


Relaxed DNA
$Lk = 200$
$\Delta Lk = -2$
$\Delta Lk = +2$
Negative supercoils
$Lk = 198$
Positive supercoils
$Lk = 202$




拧紧


0
5
10
15
20 twists of the helix
Circularize
Remove 2 twists, then circularize
Relaxed
L = 20
T = 20
W= 0
Add 2 twists, then circularize
L = 18
T = 18
W= 0
L = 22
T = 22
W= 0
L = 18
T = 20
W= −2
L = 22
T = 20
W= 2
Negatively supercoiled
Positively supercoiled

DNA 超螺旋结构的拓扑学性质

DNA 不同的空间分子构象又称为拓扑异构体 (topoisomer)，其拓扑学特性可以用 3 个参数来描述。

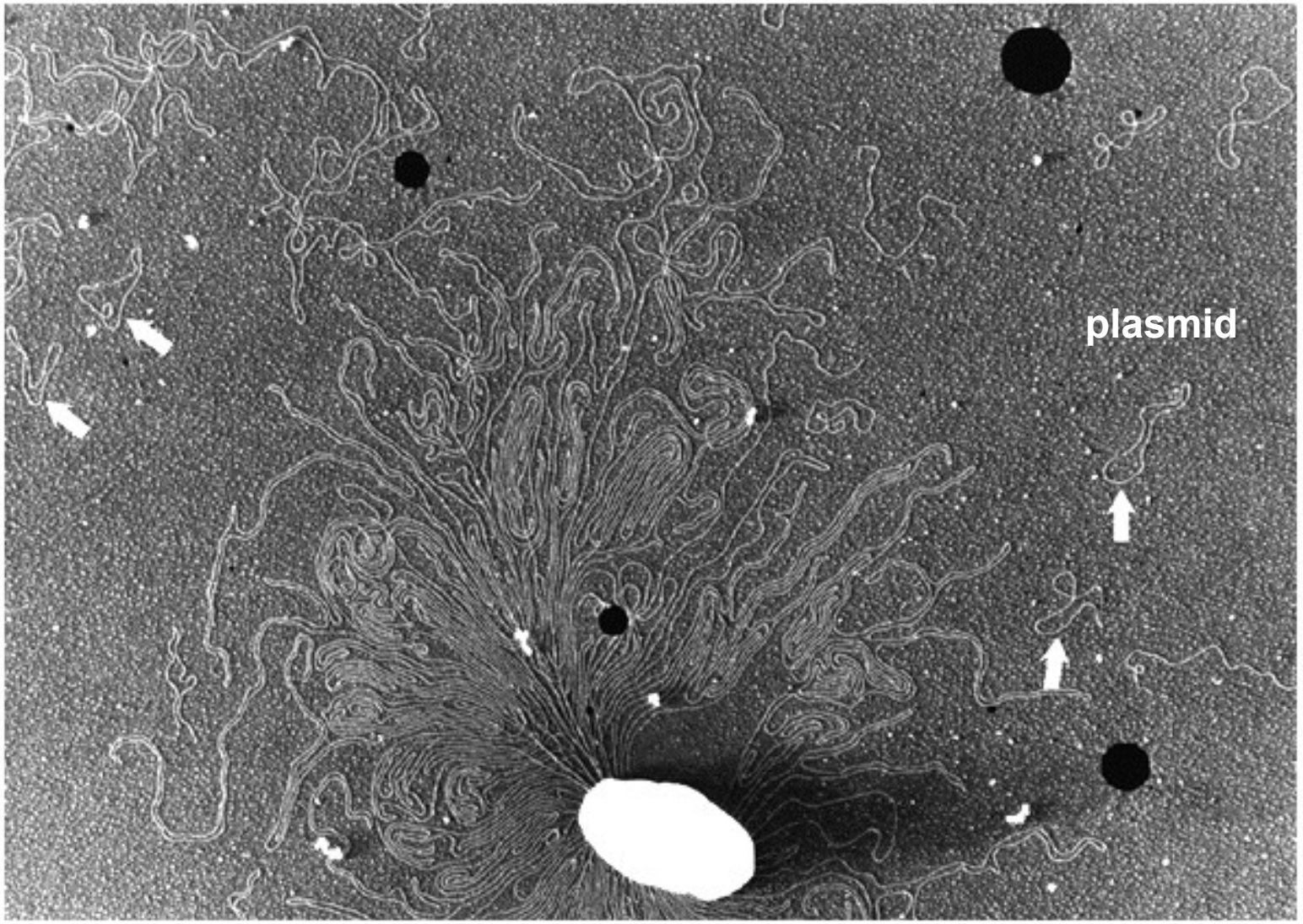
连环数（*L*, linking number）、
盘绕数（*T*, twsting number）、
超螺旋周数（*W*, writhing number）

$$L = T + W$$

2.2 Extrachromosomal DNA

In addition to the very large, circular DNA chromosome in the nucleoid, many bacteria contain one or more small, circular DNA molecules that are free in the cytosol. These extrachromosomal elements are called **plasmids**.





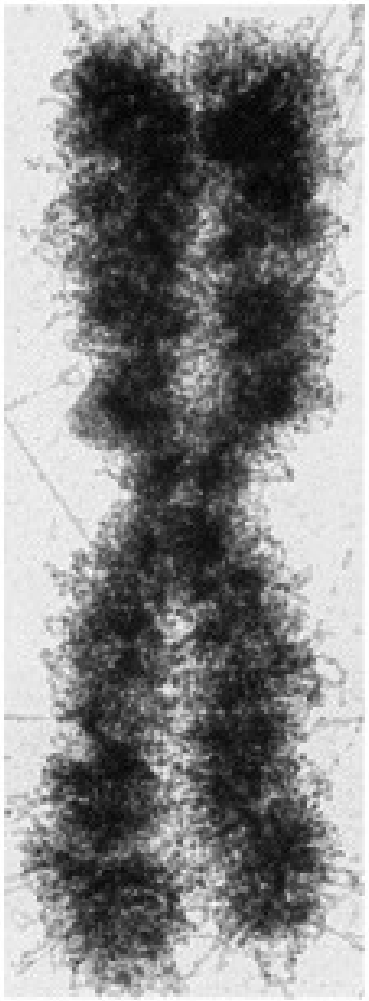
DNA from a lysed E. coli cell

3. Eukaryotic cells contain more DNA than prokaryotes

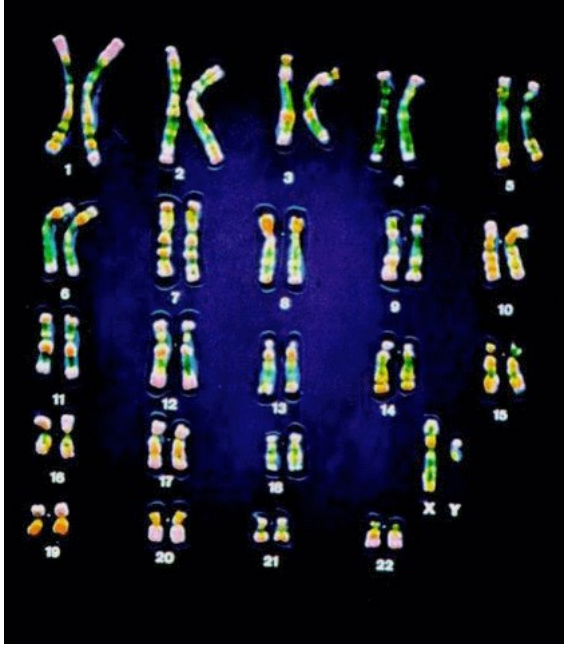
Human cells and many other mammalian cells have about 600 times as much DNA as E. coli. The plants contain an even greater amount.

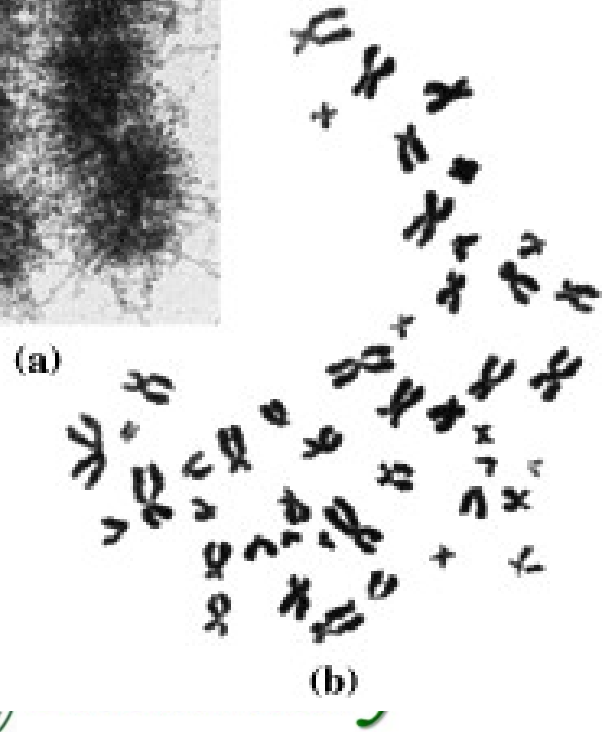
The total contour length of all the DNA in a single human cell is about 2 m.





A single human chromosomes.





Complete set of chromosomes from a leukocyte. There are 46 chromosomes in each normal human somatic cell.

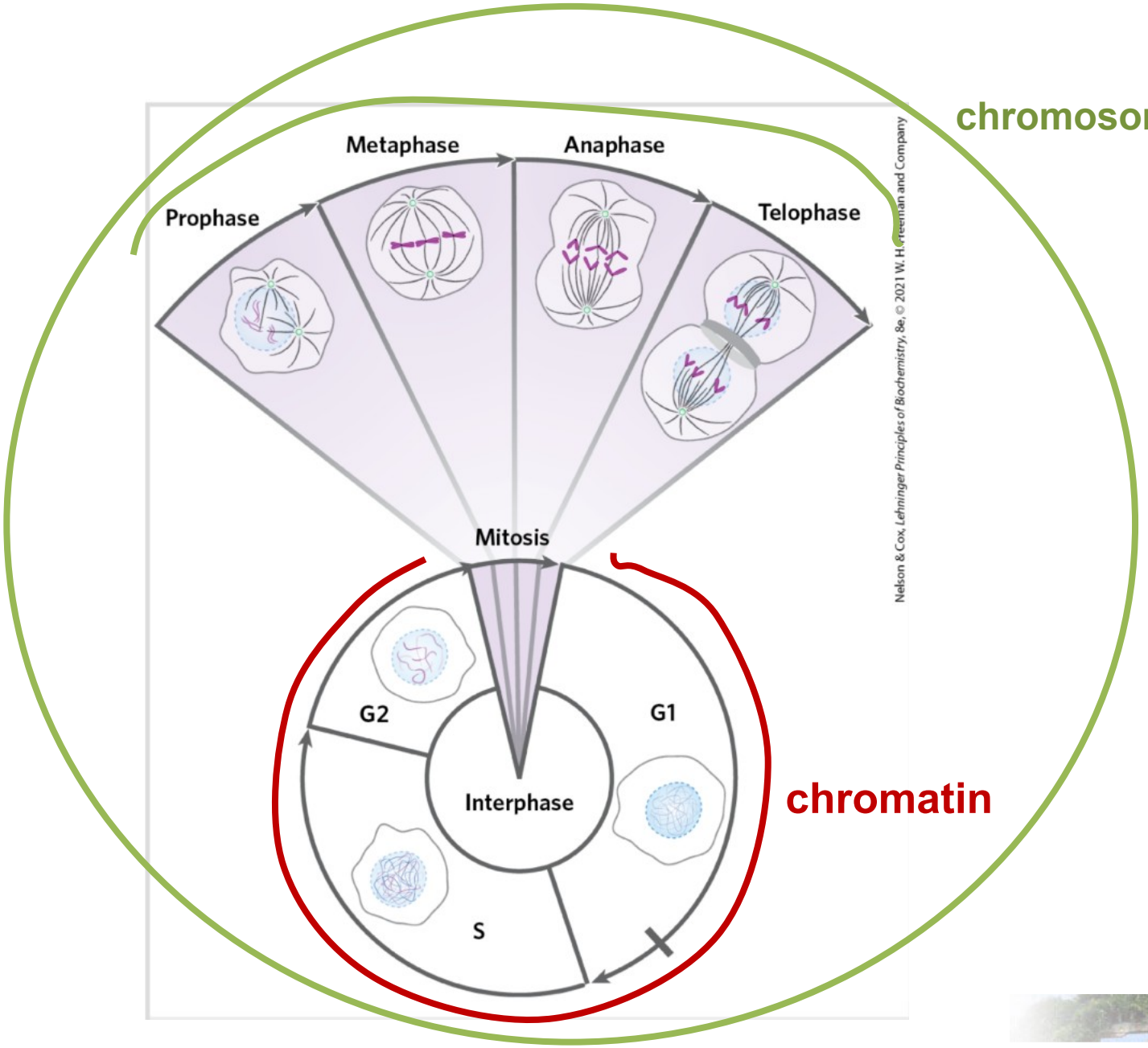


The eukaryotic cell cycle produces remarkable changes in the structure of chromosomes (Fig. 24-22). In nondividing eukaryotic cells (in the G0 phase) and those in interphase (G1, S, and G2), the chromosomal material, **chromatin**, is amorphous. In the S phase of interphase, the DNA in this amorphous state replicates, each chromosome producing two **sister chromosomes** (called sister **chromatid**s) that remain associated with each other after replication is complete. The chromosomes become much more condensed during prophase of mitosis, taking the form of a species-specific number of well-defined pairs of sister chromatids

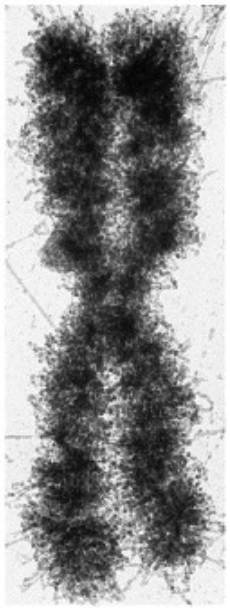


In the eukaryotic cell cycle, interphase chromosomes appear, at first viewing, to be dispersed and amorphous. Nevertheless, several forms of chromatin can be found along these chromosomes. About 10% of the chromatin in a typical eukaryotic cell is in a more condensed form than the rest of the chromatin. This form, **heterochromatin**, is transcriptionally inactive. Heterochromatin is generally associated with particular chromosome structures — the centromeres, for example. The remaining, less condensed chromatin is called **euchromatin**.

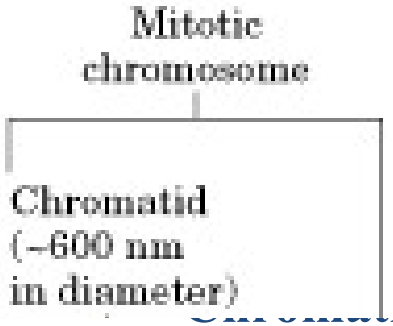




Changes in chromosome structure during the eukaryotic cell cycle.





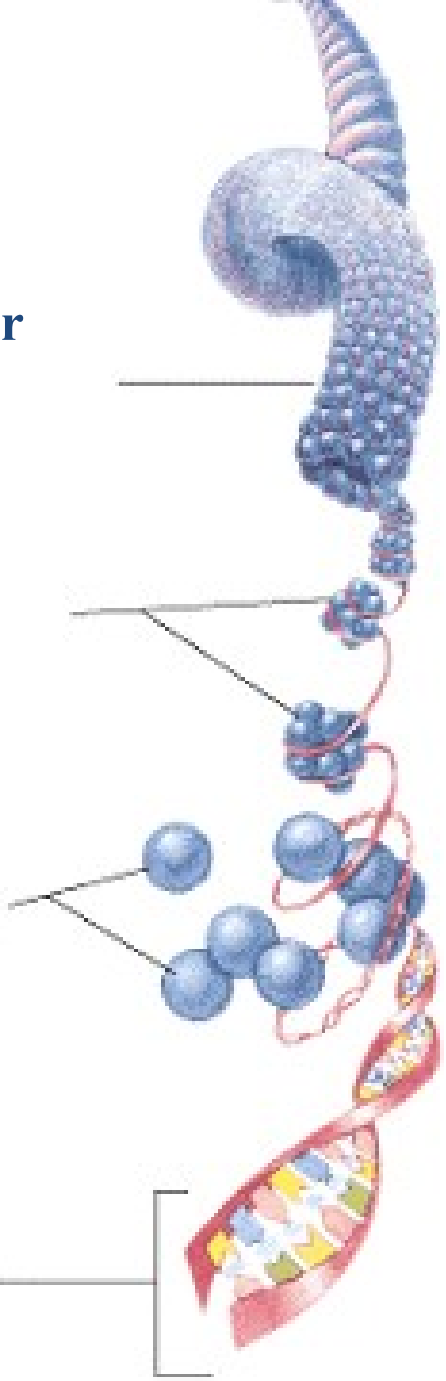


Chromatin fiber

(30 nm in diameter)

Nucleosomes

(10 nm in diameter)



Histones

DNA





Chromosomes contain both DNA and protein. Most of the protein are histones, but there are also nonhistone proteins.

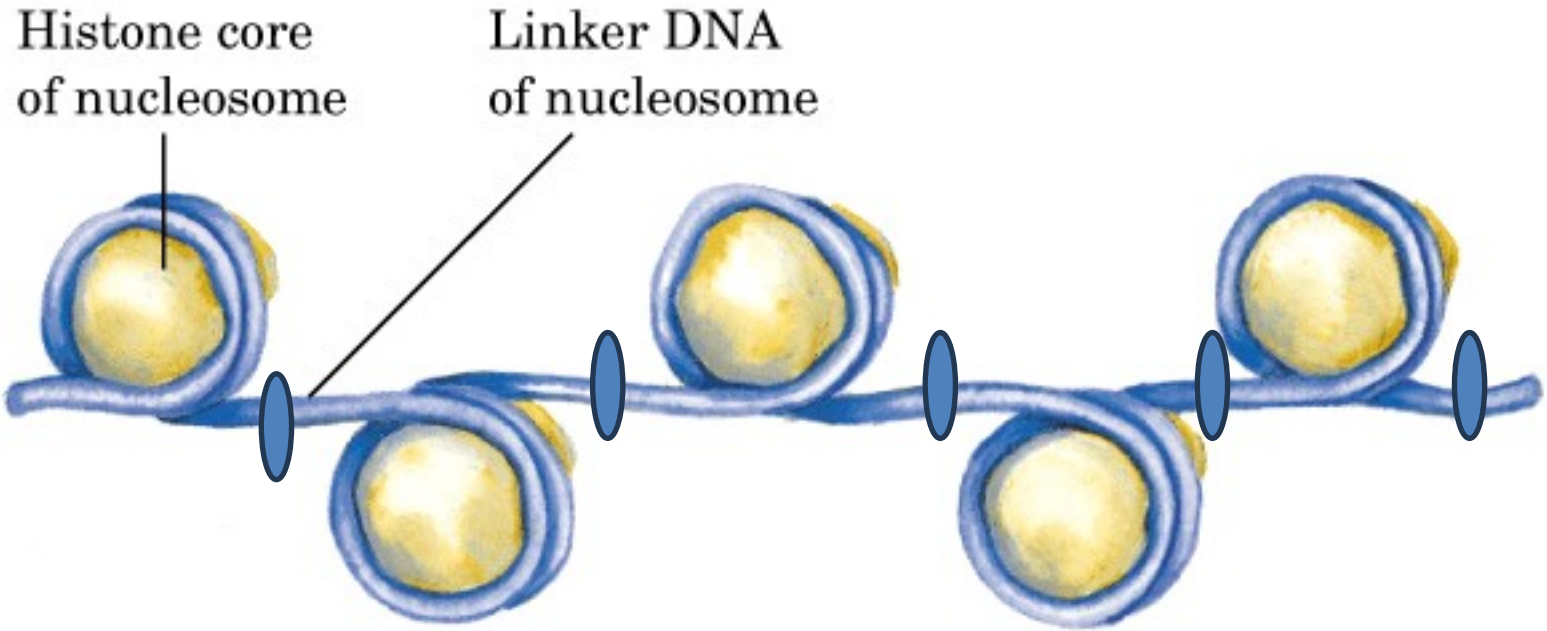
In the nucleus, each chromosome contains a single linear double-stranded DNA molecule.



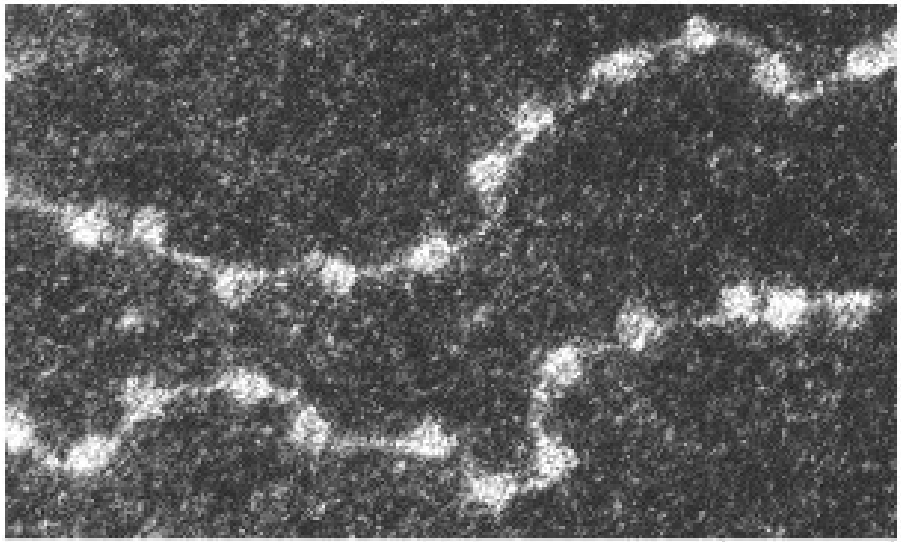
4. Nucleosomes

When chromosomes are gently "condensed", they have the appearance under the electron microscope of "beads on a string". The "beads" are called nucleosomes and consist of DNA complexed with histones.

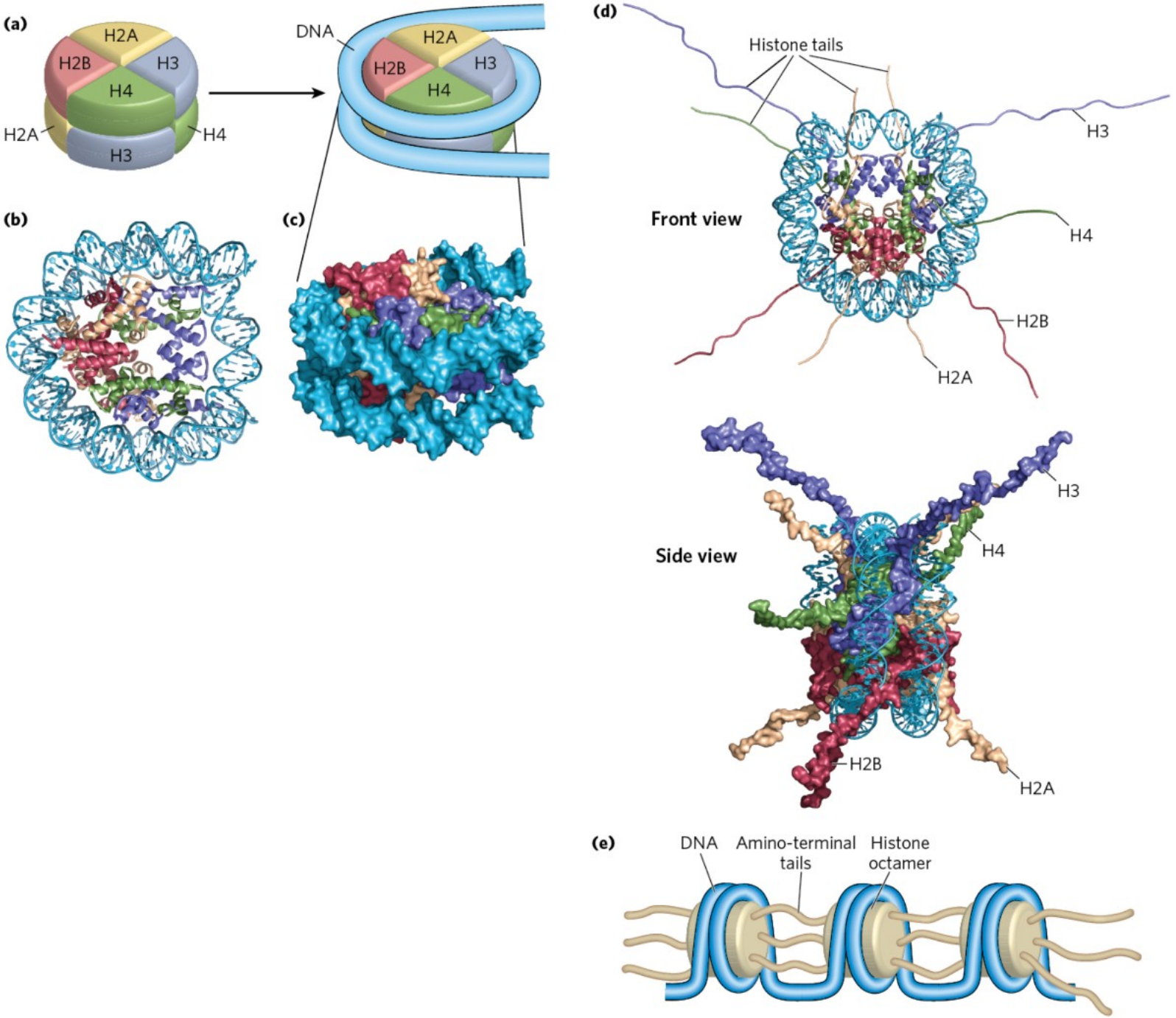




(a)



(b)



Nelson & Cox, *Lehninger Principles of Biochemistry*, 8e, © 2021 W. H. Freeman and Company

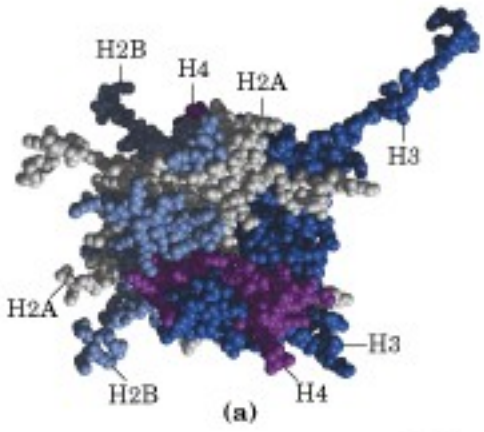
table 24–3

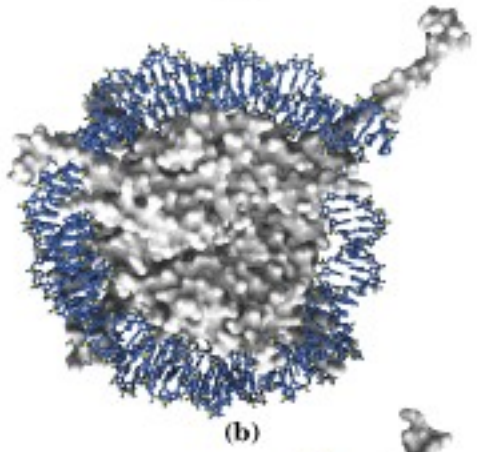
Types and Properties of Histones

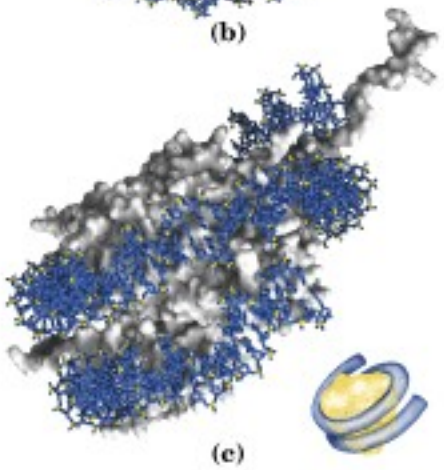
Histone	Molecular weight	Number of amino acid residues	Content of basic amino acids (% of total)	
			Lys	Arg
H1*	21,130	223	29.5	1.3
H2A*	13,960	129	10.9	9.3
H2B*	13,774	125	16.0	6.4
H3	15,273	135	9.6	13.3
H4	11,236	102	10.8	13.7

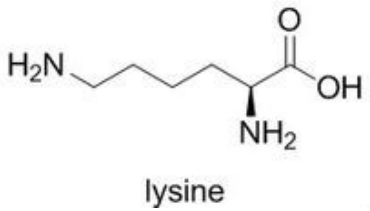
*The sizes of these histones vary somewhat from species to species. The numbers given here are for bovine histones.

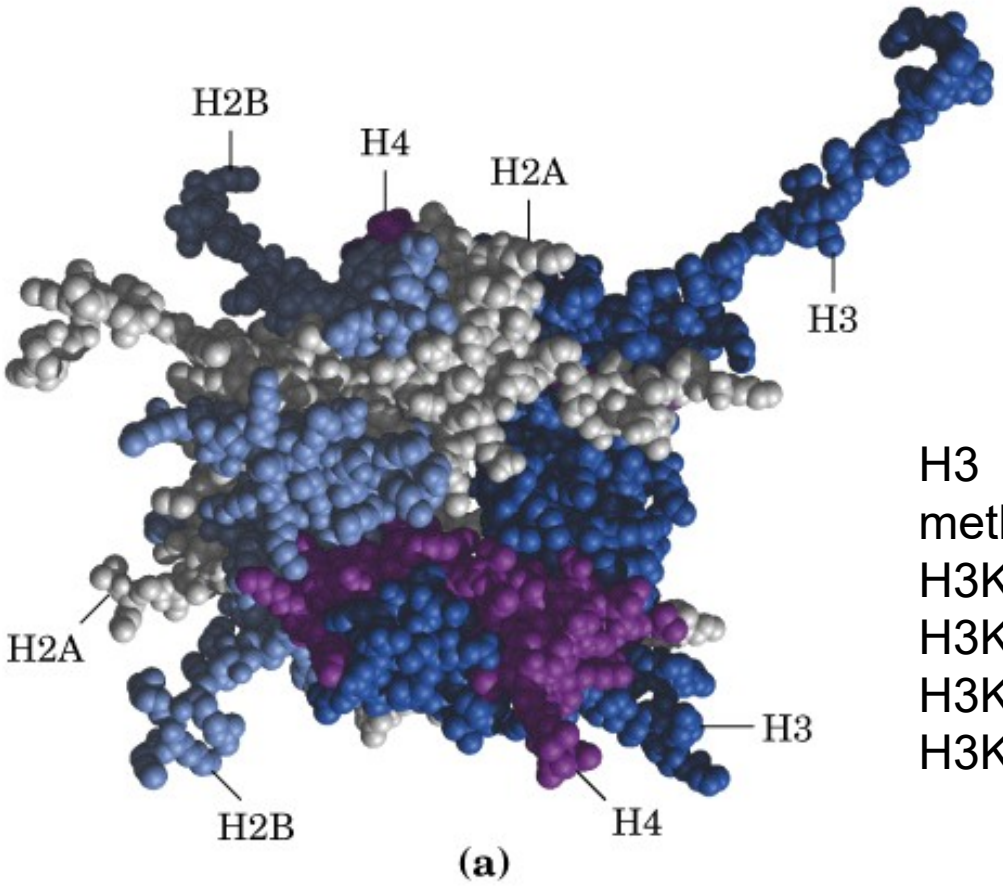






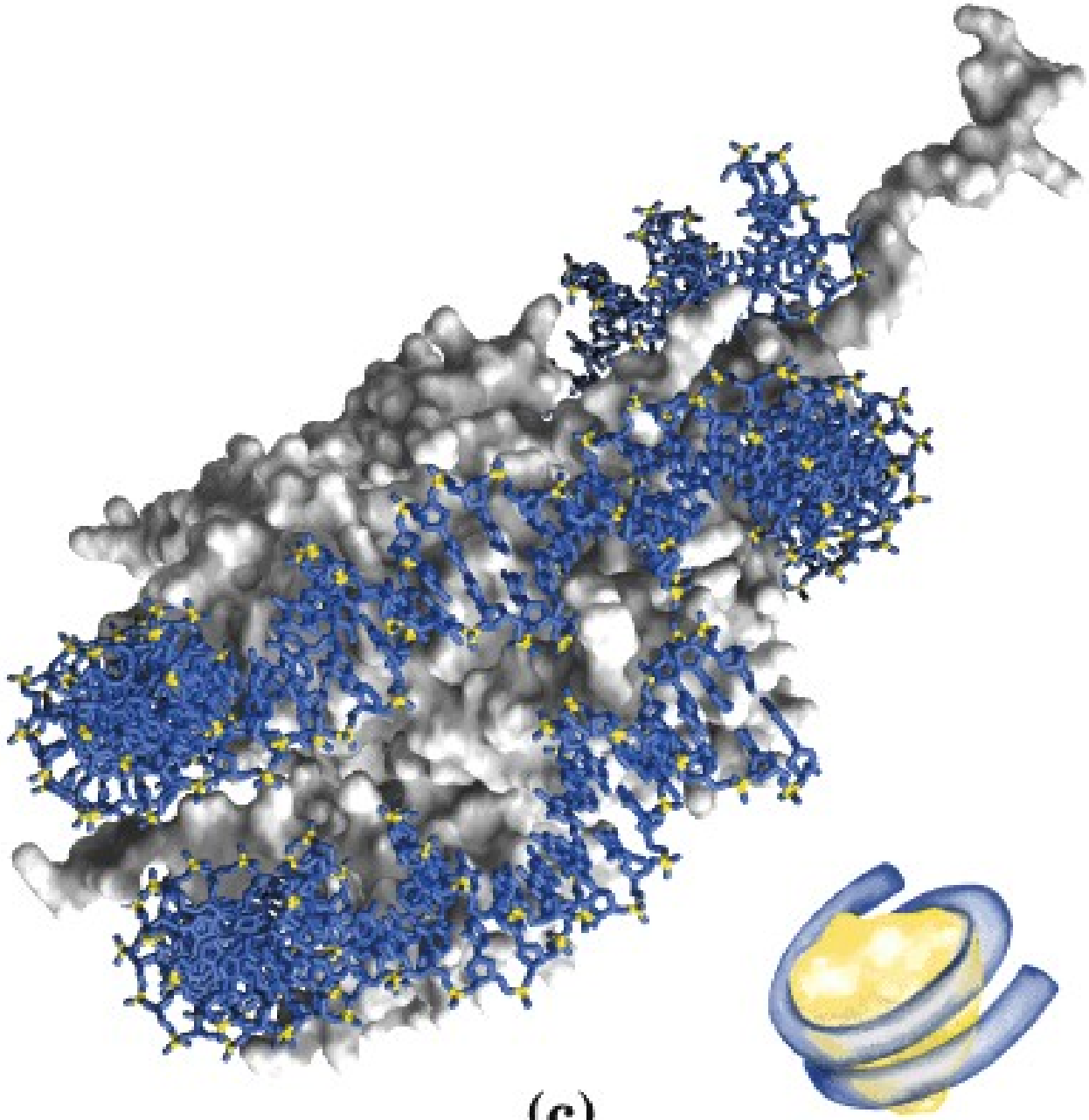




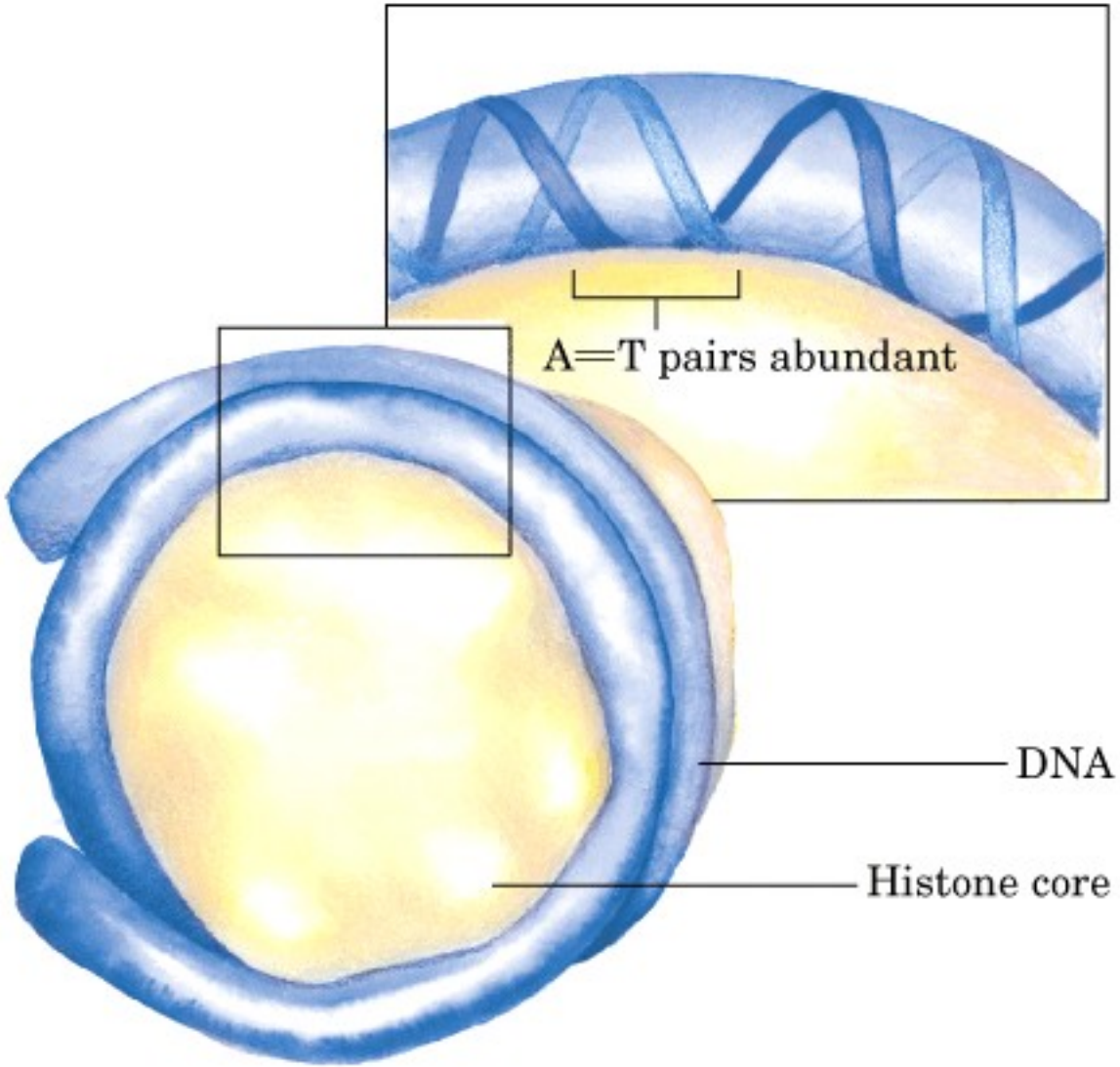


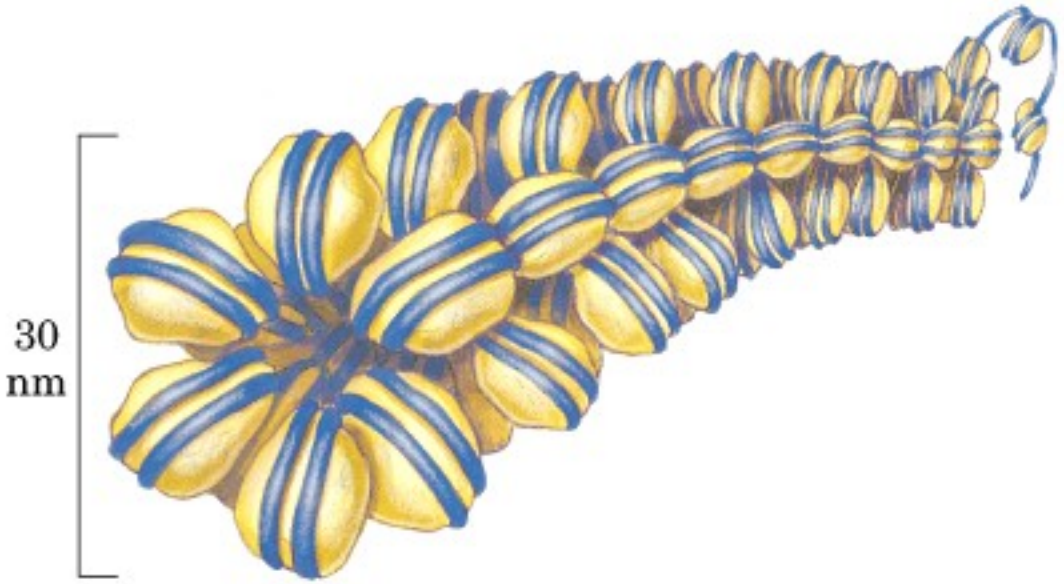
H3 methylation
H3K4,
H3K27
H3K9,
H3K36



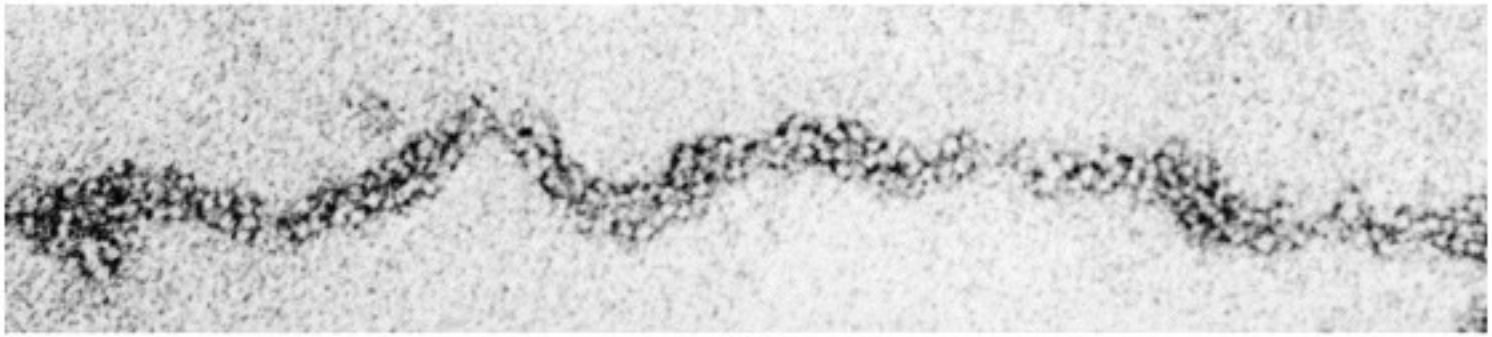


(c)

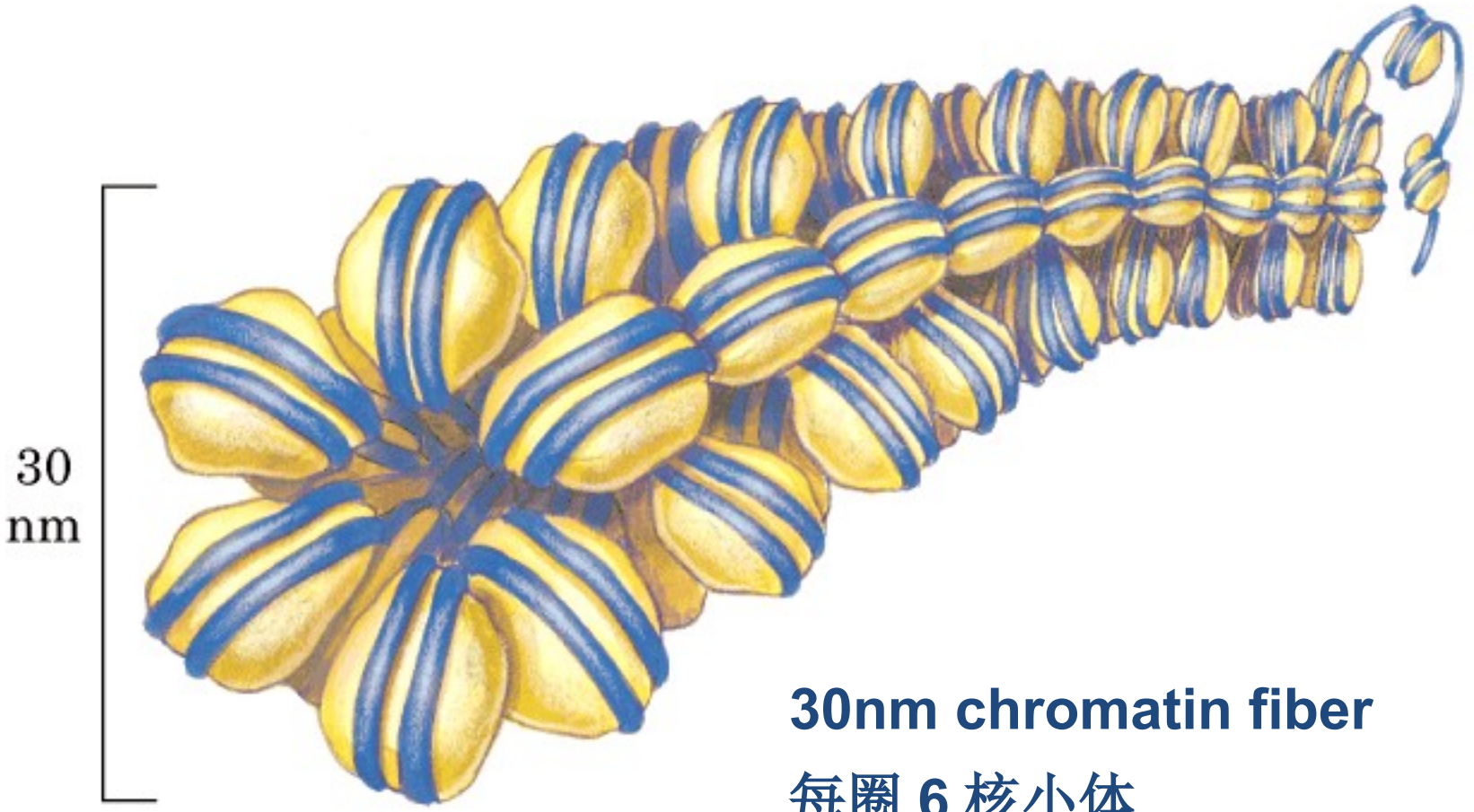

A=T pairs abundant
DNA
Histone core



(a)

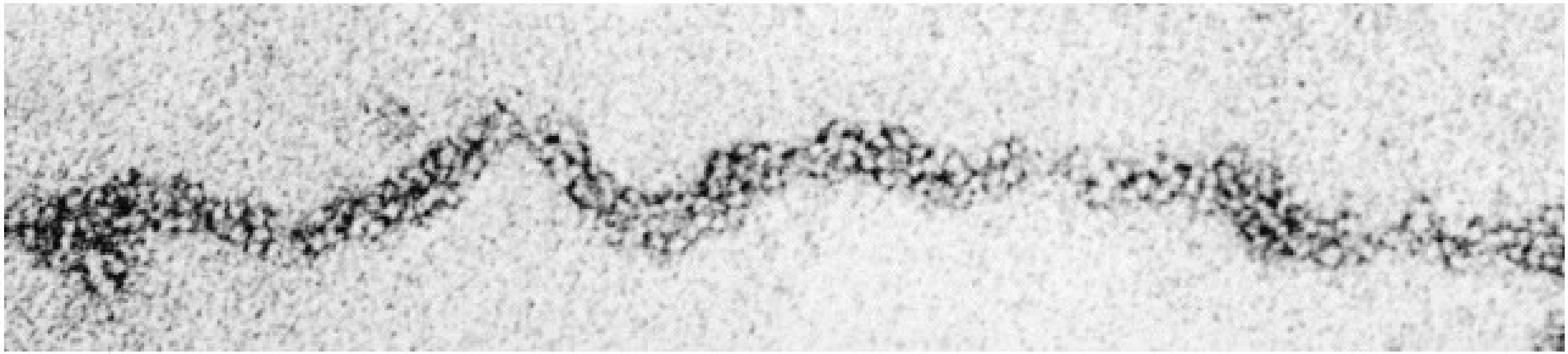


(b)



30nm chromatin fiber

每圈 6 核小体



(b)

Two chromatids
(10 coils each)

One coil (30 rosettes)

One rosette

One loop (-75,000 bp)

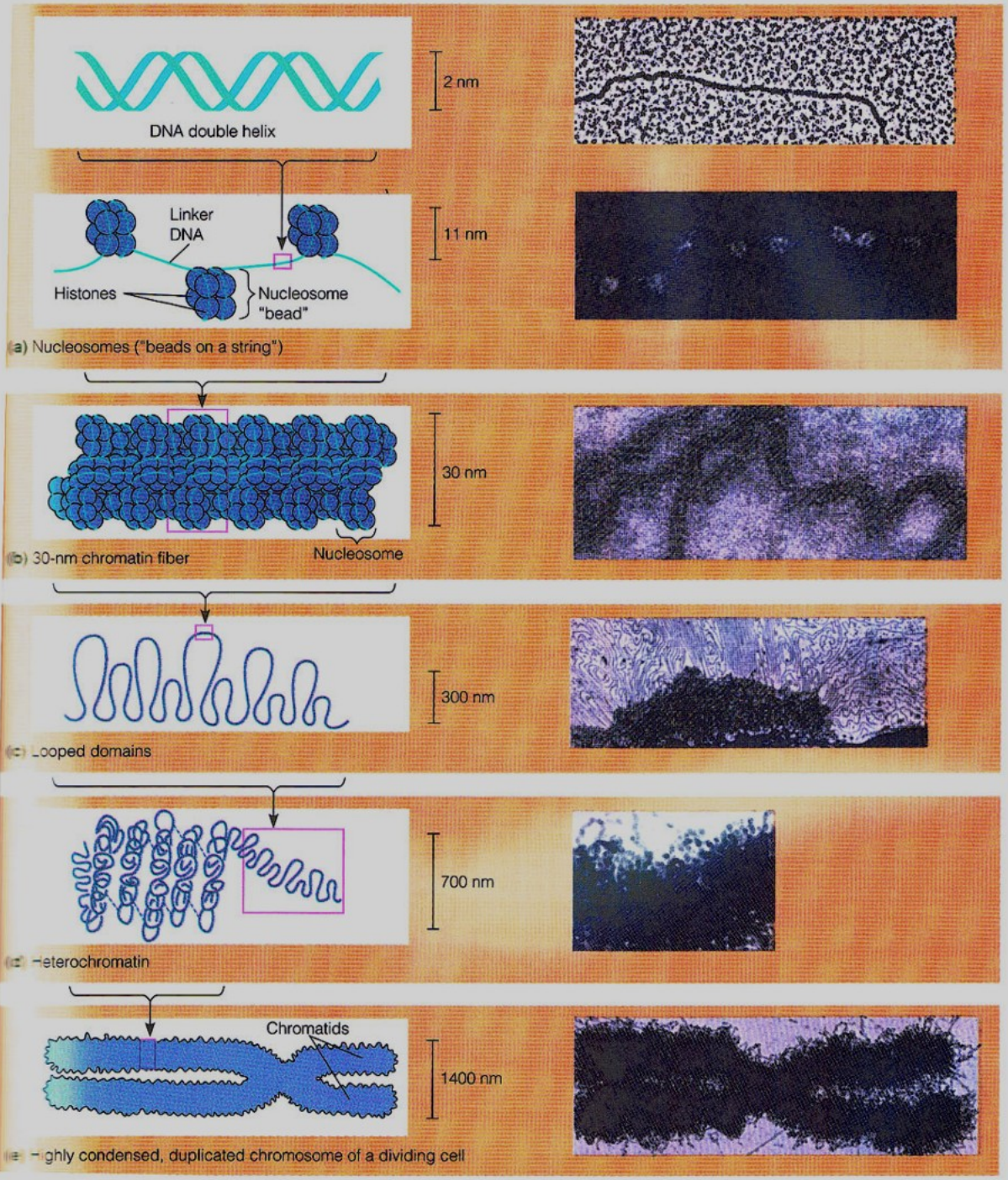
30 nm Fiber

"Beads –on-a-string"
form of chromatin

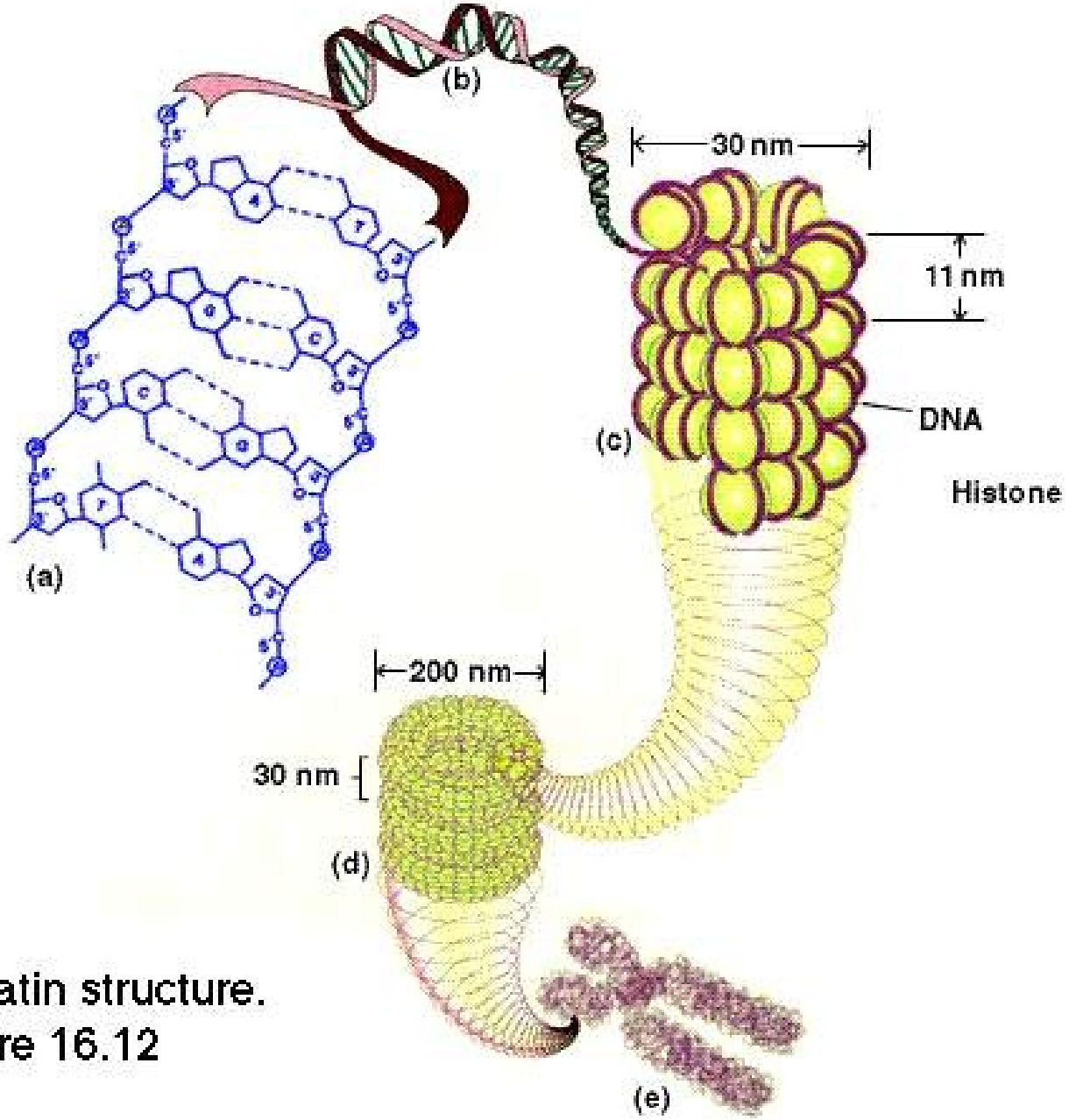
DNA



Nuclear scaffold


DNA double helix
2 nm
Linker DNA
Histones
Nucleosome "bead"
11 nm
(a) Nucleosomes ("beads on a string")
30 nm
(b) 30-nm chromatin fiber
Nucleosome
300 nm
700 nm
Chromatids
1400 nm

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Chromatin structure.
M Figure 16.12

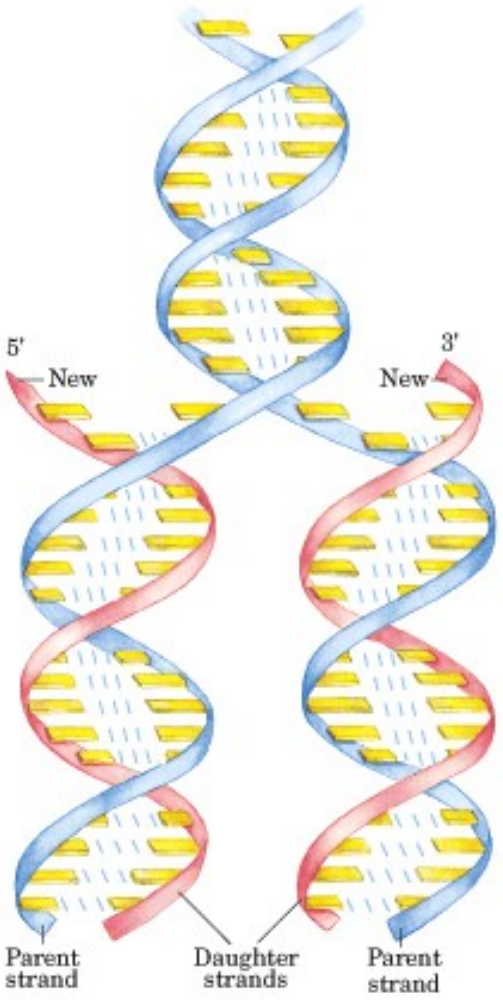
DNA Replication (F3F4)

How does a DNA molecule replicate with high fidelity?



1. The deduced double helix structure of DNA revealed the possible ways for its replication (1953)

The hypothesis of **semiconservative replication** proposed by Watson and Crick in 1953.

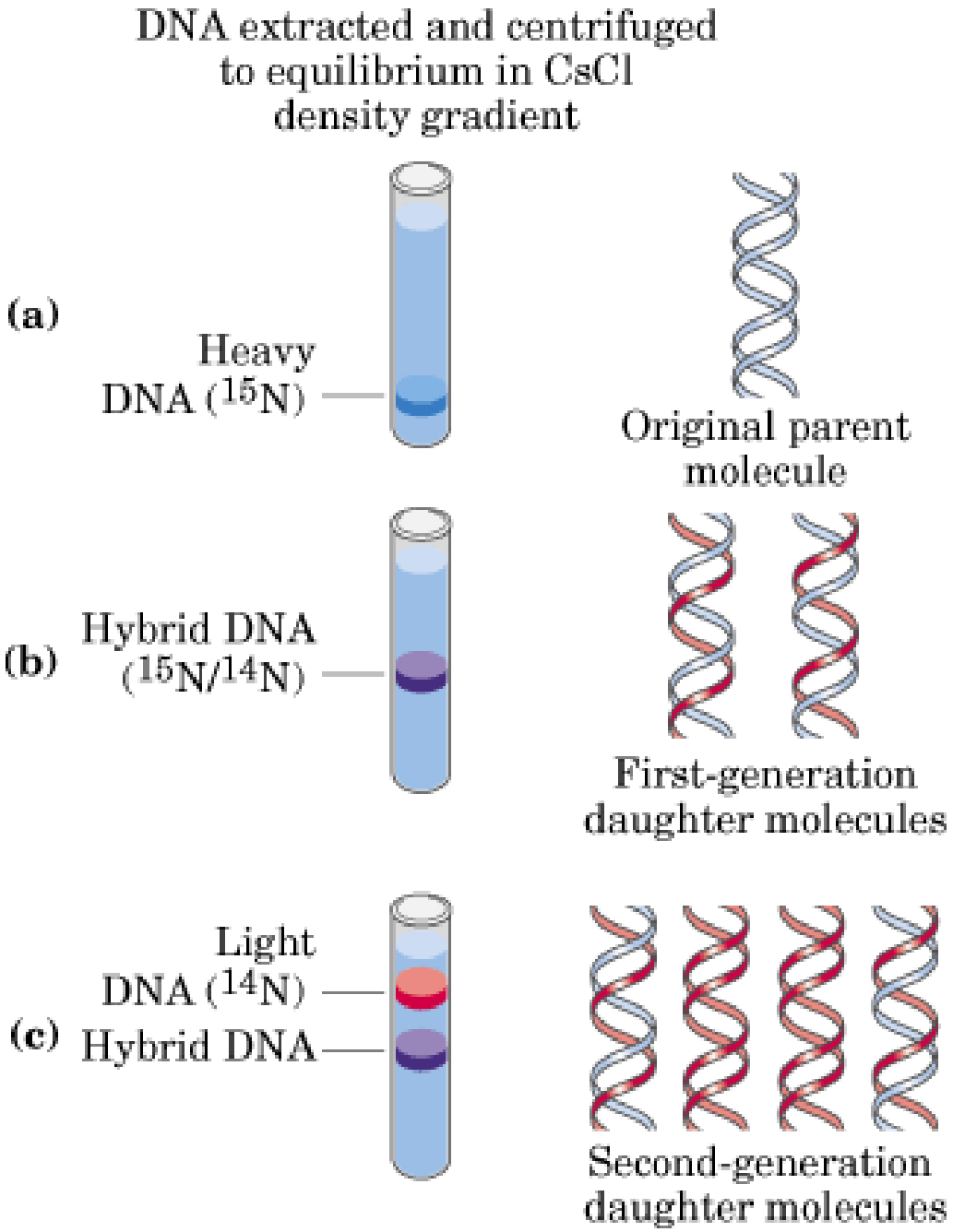


2. DNA replication was proved to be semiconservative by the Meselson-Stahl experiment using *E. coli* cells (1957)

^{15}N (the Heavy isotope) and **^{14}N** (the Light isotope) was used (as NH_4Cl) to label the DNA to distinguish the old and newly synthesized DNA molecules in cells;

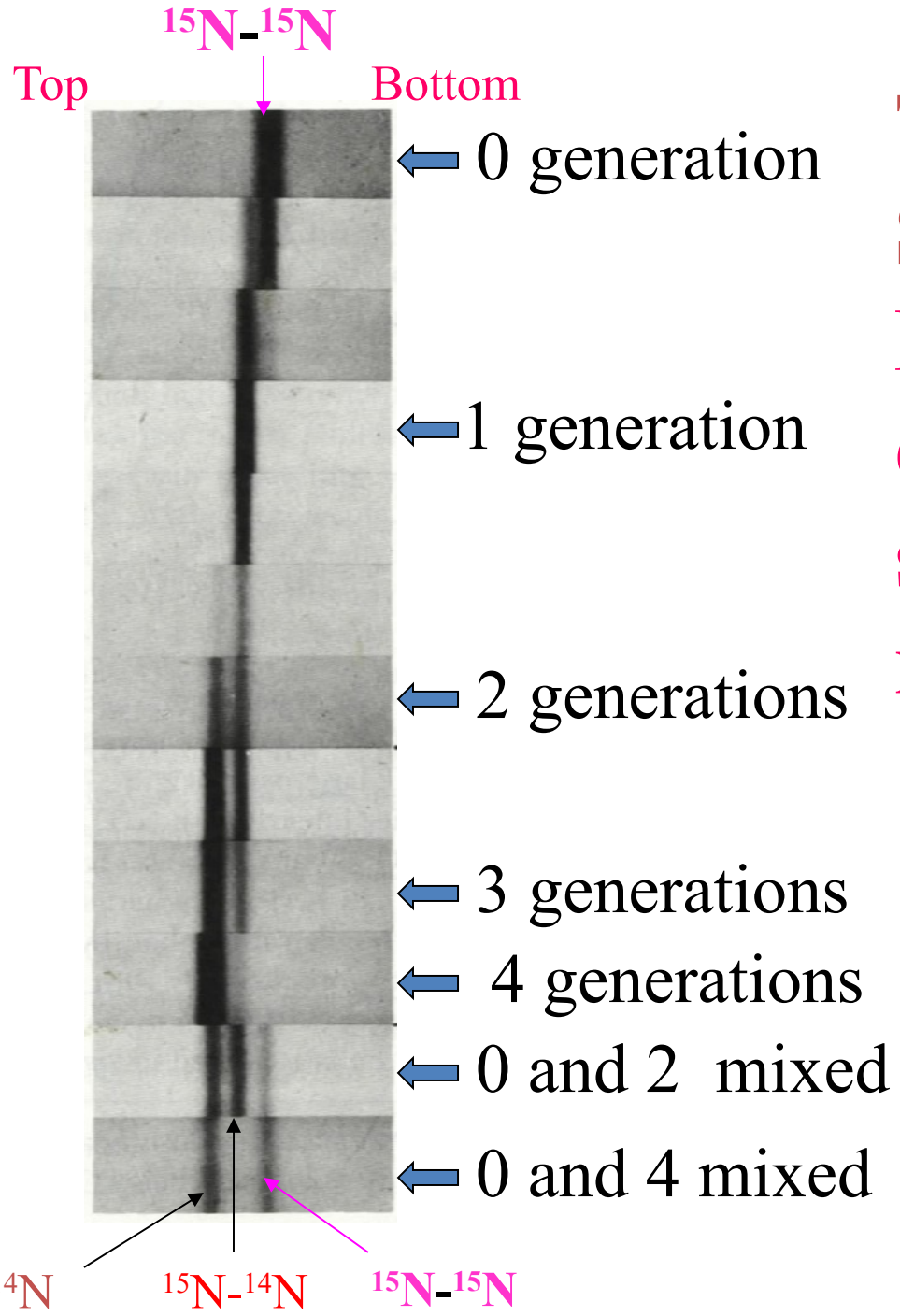
Three types of DNA molecules containing various proportions of **^{15}N** and **^{14}N** (H-H, H-L, L-L) were separated by centrifugation to equilibrium in a cesium chloride (CsCl) density gradient.





Radioisotope labeling and density gradient centrifugation clearly distinguishes replications of semiconservative from conservative.





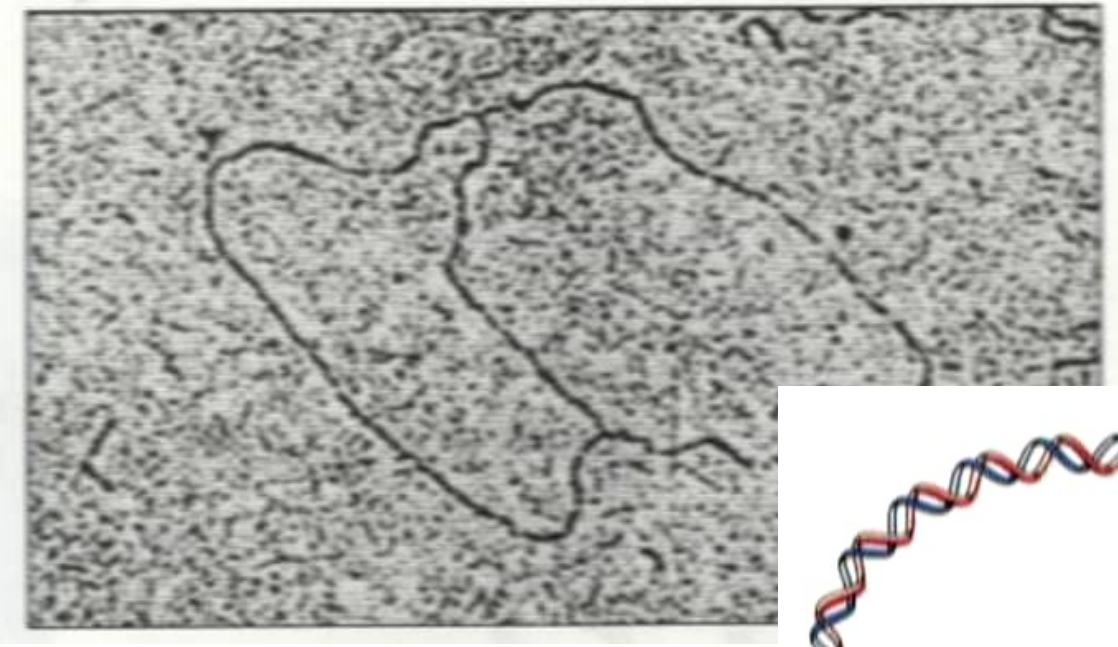
The Meselson-Stahl experiment: DNA molecules duplicate semiconservatively in *E. coli* cells.

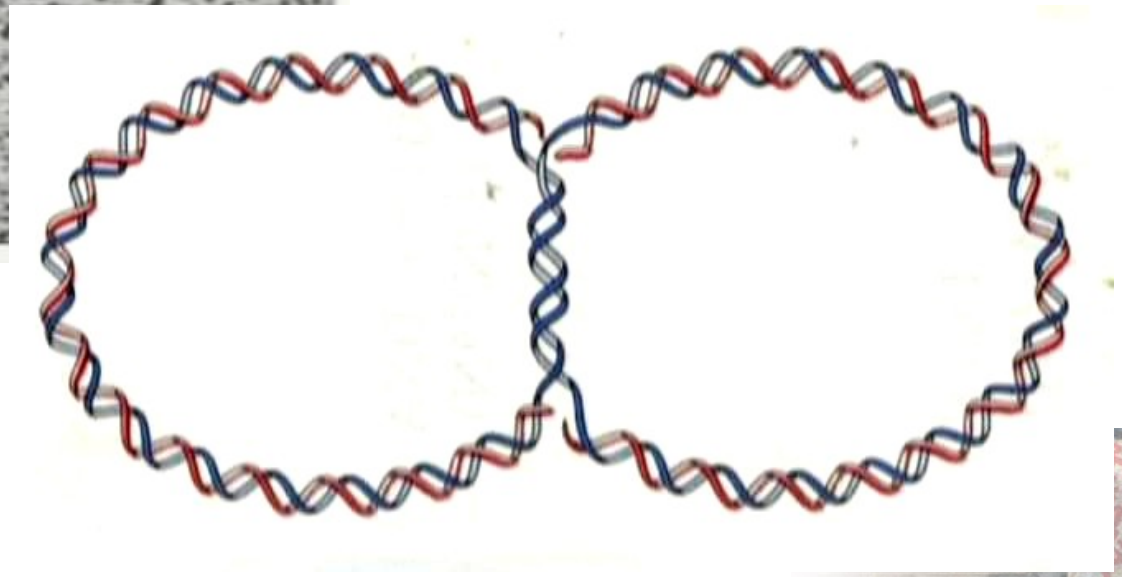
3. A variety of simple questions were asked about DNA replication

Are the two parental strands completely unwound before replication begins?
Does replication begin at random sites or at unique sites?
Does DNA replication proceed in one direction or both directions?
The overall chain growth occurs in 5` → 3`, 3` → 5`, or both directions?
What enzymes take part in DNA synthesis?



4. Autoradiography studies: daughter strands are svnthesized immediately after
ɔarate

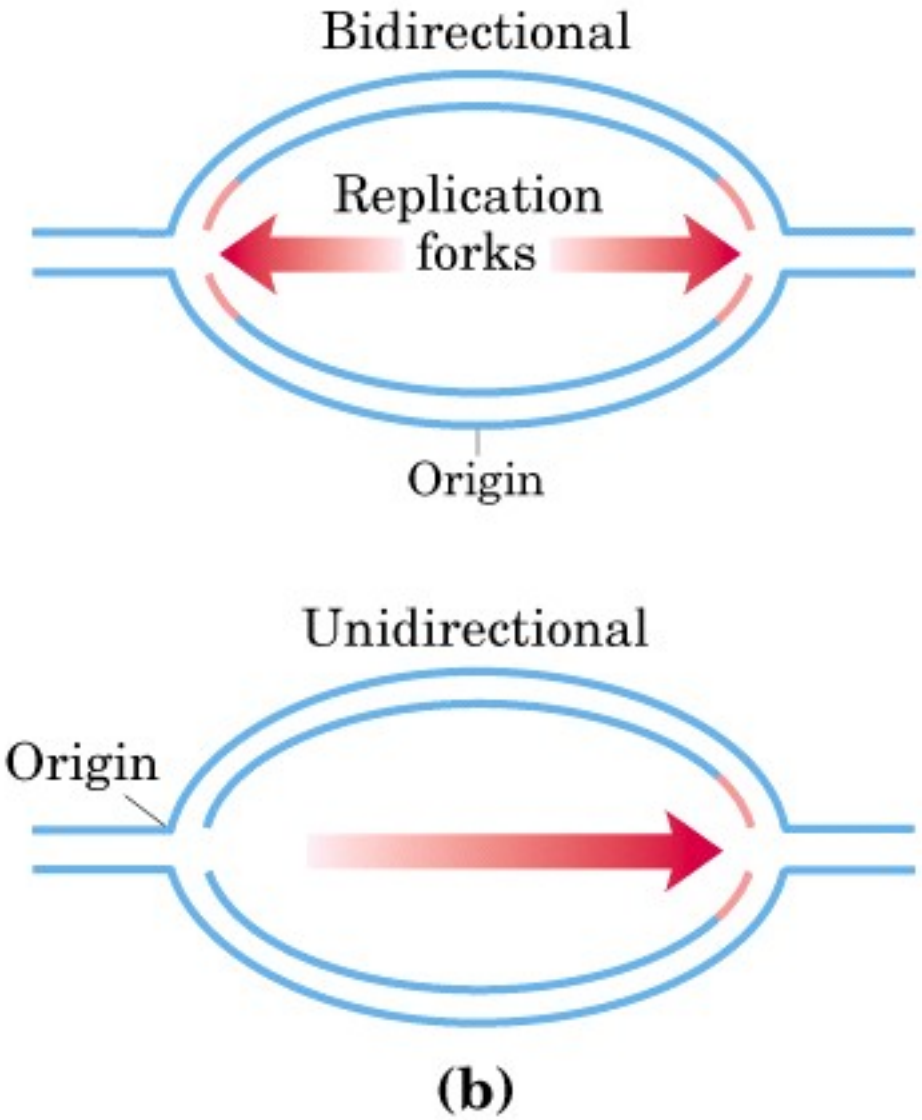




A electron micrograph of the replication intermediate of a plasmid DNA: θ-shaped structures were observed; no single stranded DNA is visible.

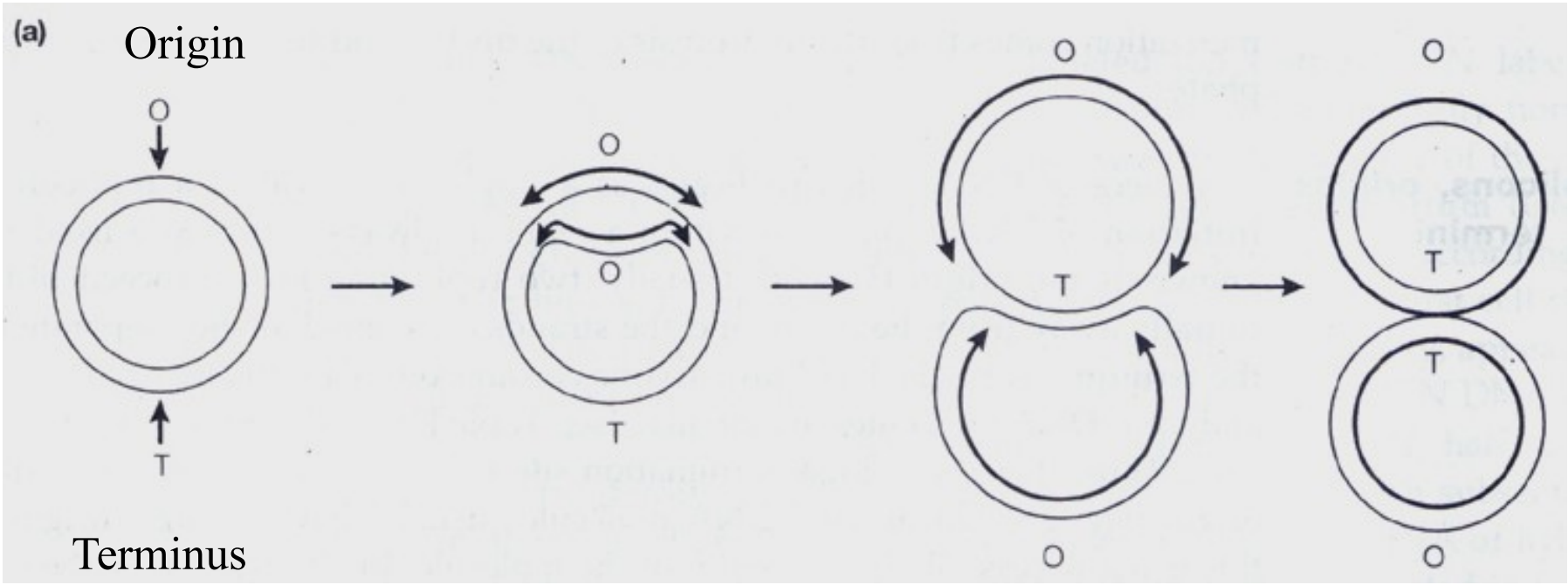
No complete unwinding of the two parental strands occurred before the daughter strands are synthesized




Bidirectional
Replication forks
Origin
Unidirectional
Origin

(b)

Bidirectional replication of a circular bacterial replicon



5. DNA replication was found to begin at specific sites and proceed bidirectionally

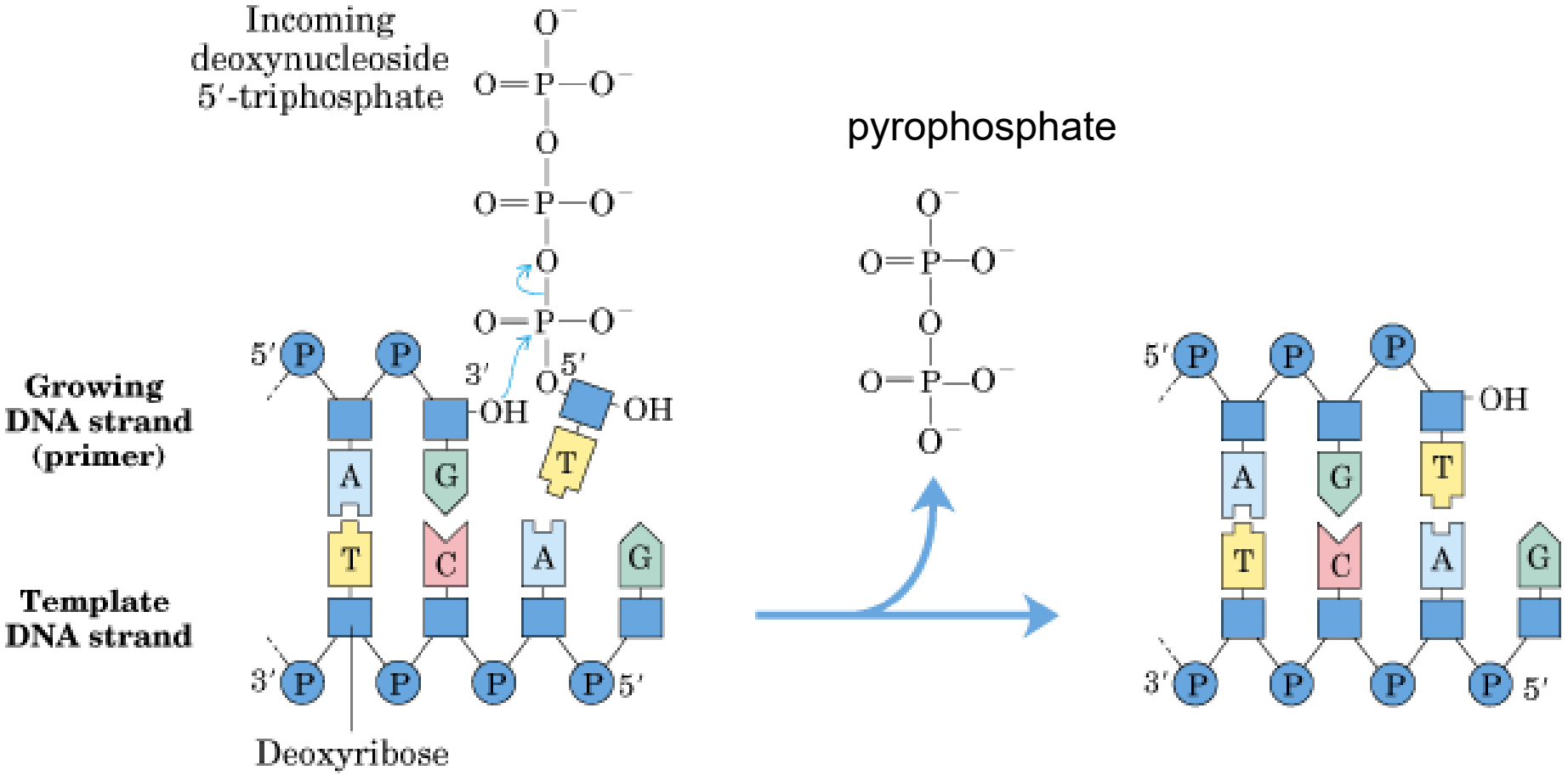
1. Replication always began at a specific internal site (**not from the ends**, **not random**);
2. The replication proceeds **in both directions** (determined by measuring the distance between the replication fork and the ends)



6. The chemistry of DNA polymerization was revealed by *in vitro* studies using a DNA polymerase purified from *E. coli*

DNA polymerase I was found to catalyze DNA polymerization *in vitro* in the presence of a single-stranded DNA template, a preexisting primer with a free 3`-OH group and the dNTPs.




Incoming
deoxynucleoside
5′-triphosphate
pyrophosphate
Growing
DNA strand
(primer)
Template
DNA strand
Deoxyribose
5′
3′
OH
A
G
T
C
P

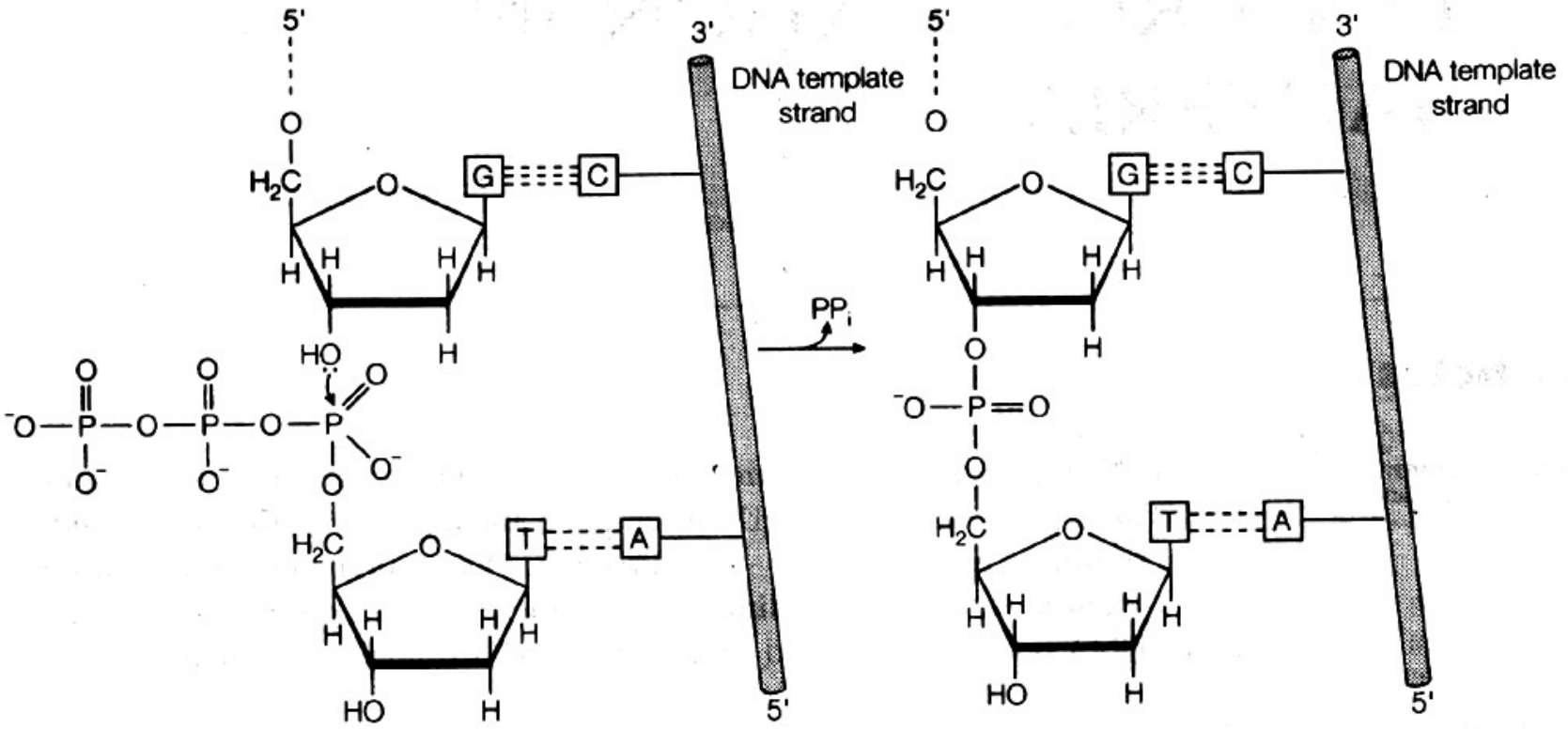


Fig. 1. DNA synthesis. In this schematic diagram, the incoming dTTP hydrogen bonds with the adenine on the template DNA strand and a 3'5' phosphodiester bond is formed, releasing pyrophosphate.

7. DNA polymerase I has 3` → 5`, as well as 5` → 3` exonuclease activities

The 3` → 5` exonuclease activity was also found to be able to remove mismatched base pairs, thus to proofread the newly incorporated nucleotides.

- The 5`→3` exonuclease activity of DNA polymerase I enable it to catalyze the nick translation process: an RNA or DNA strand paired to a DNA template is simultaneously degraded and replaced; an activity used for the removal of RNA primers in DNA replication.



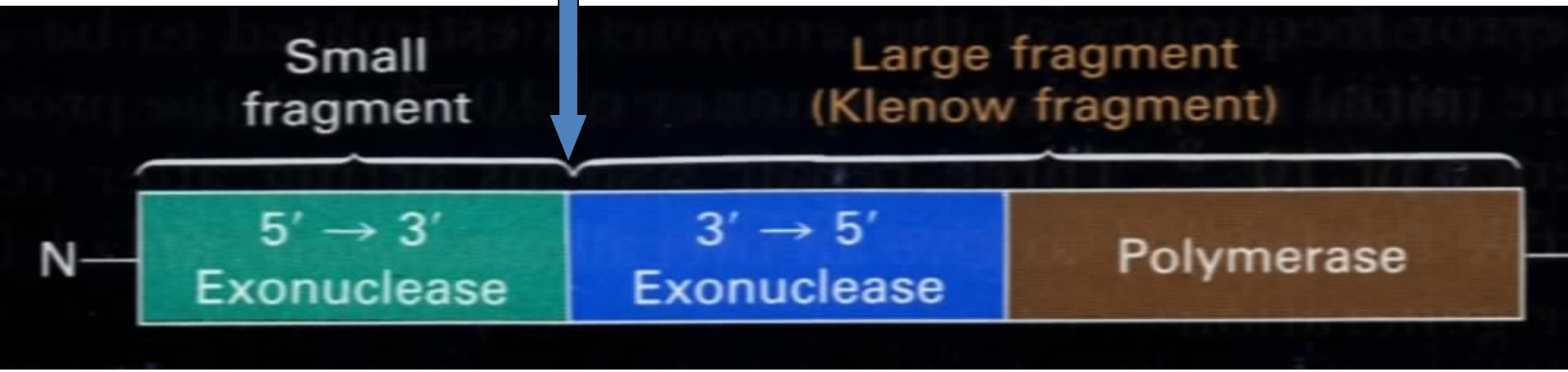
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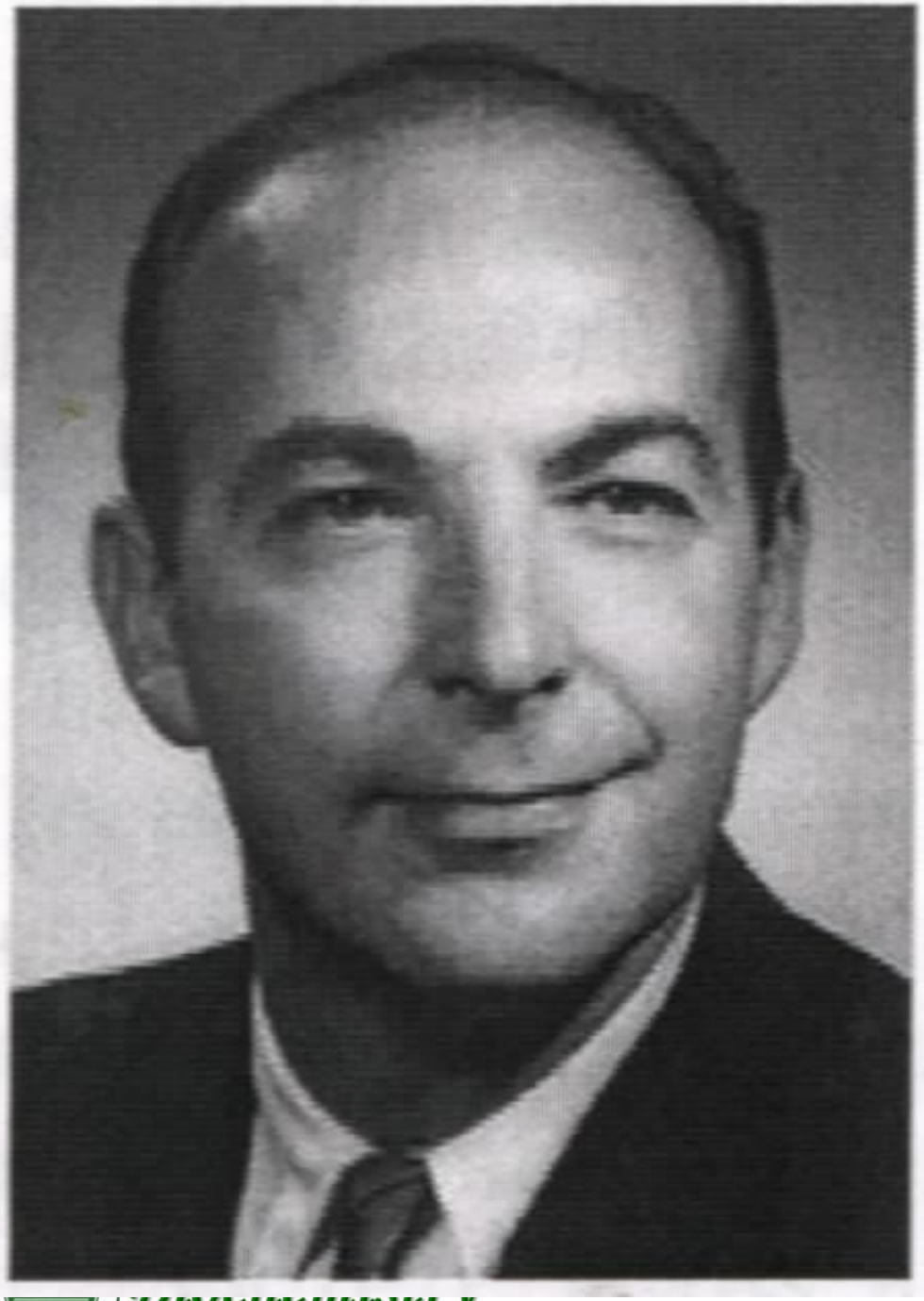
- The 5`→3` exonuclease activity of DNA polymerase I enable it to catalyze the nick translation process: an RNA or DNA strand paired to a DNA template is simultaneously degraded and replaced; an activity used for the removal of RNA primers in DNA replication.



Protease cleavage



DNA polymerase I has three enzymatic activities in a single polypeptide chain, which can be cleaved into two functional parts by mild protease treatment.



Arthur Kornberg won the 1959 Nobel Prize in Medicine for his discovery of the mechanism in the biological synthesis of deoxyribonucleic acid (*before Watson and Crick won theirs*!)



8. DNA polymerase I was found to be not responsible for DNA replication in *E. coli* cells

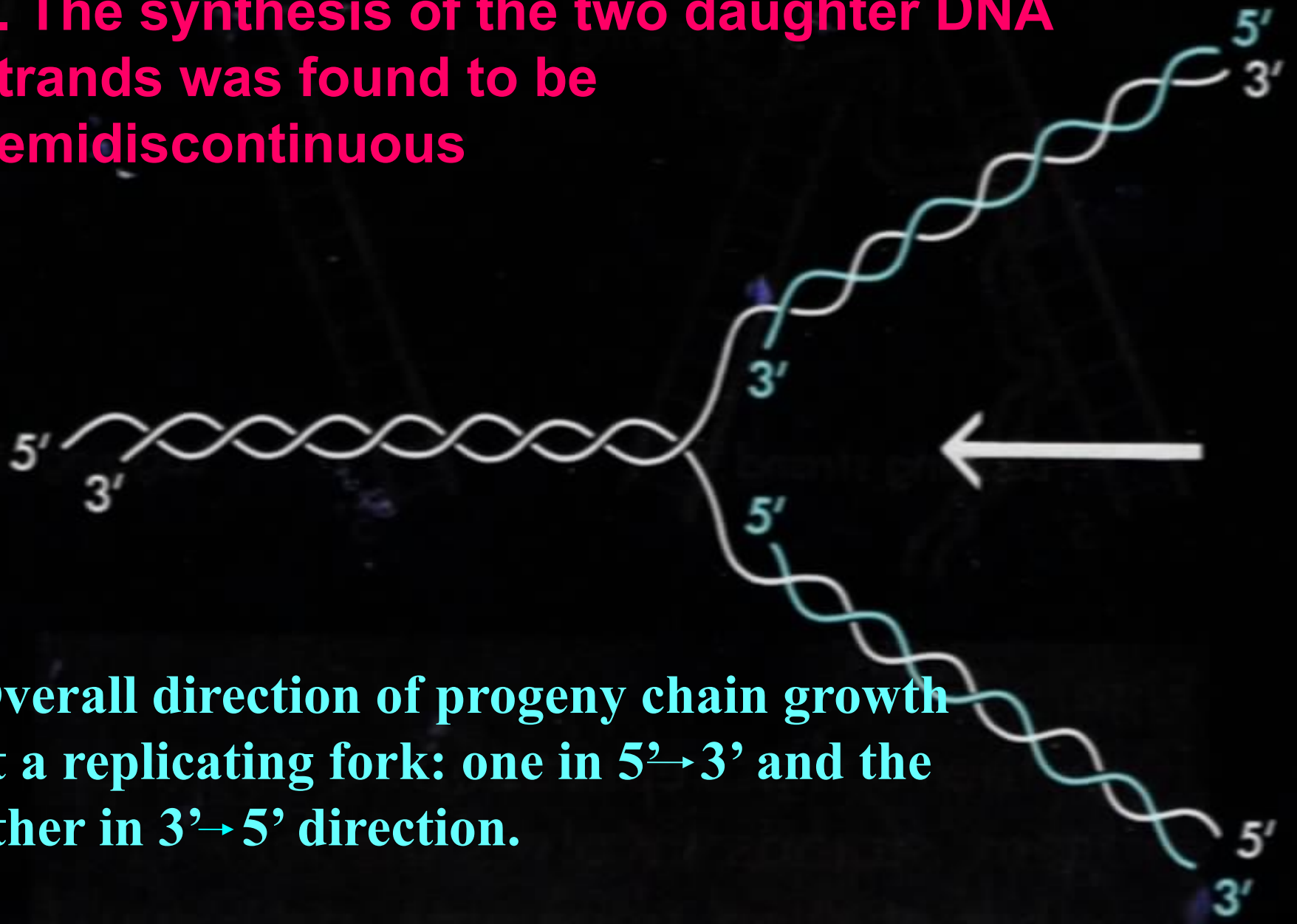
The reaction velocity of this enzyme is too low to account for the observed rates of fork movement.

E.coli cells having a defective DNA polymerase I were found to be still viable, although sensitive to UV light (1969).

Two more DNA polymerases (II and III) were revealed in *E.coli* cells in the early 1970s and two more in 1999.



9. The synthesis of the two daughter DNA strands was found to be semidiscontinuous



Overall direction of progeny chain growth at a replicating fork: one in 5'→3' and the other in 3'→5' direction.

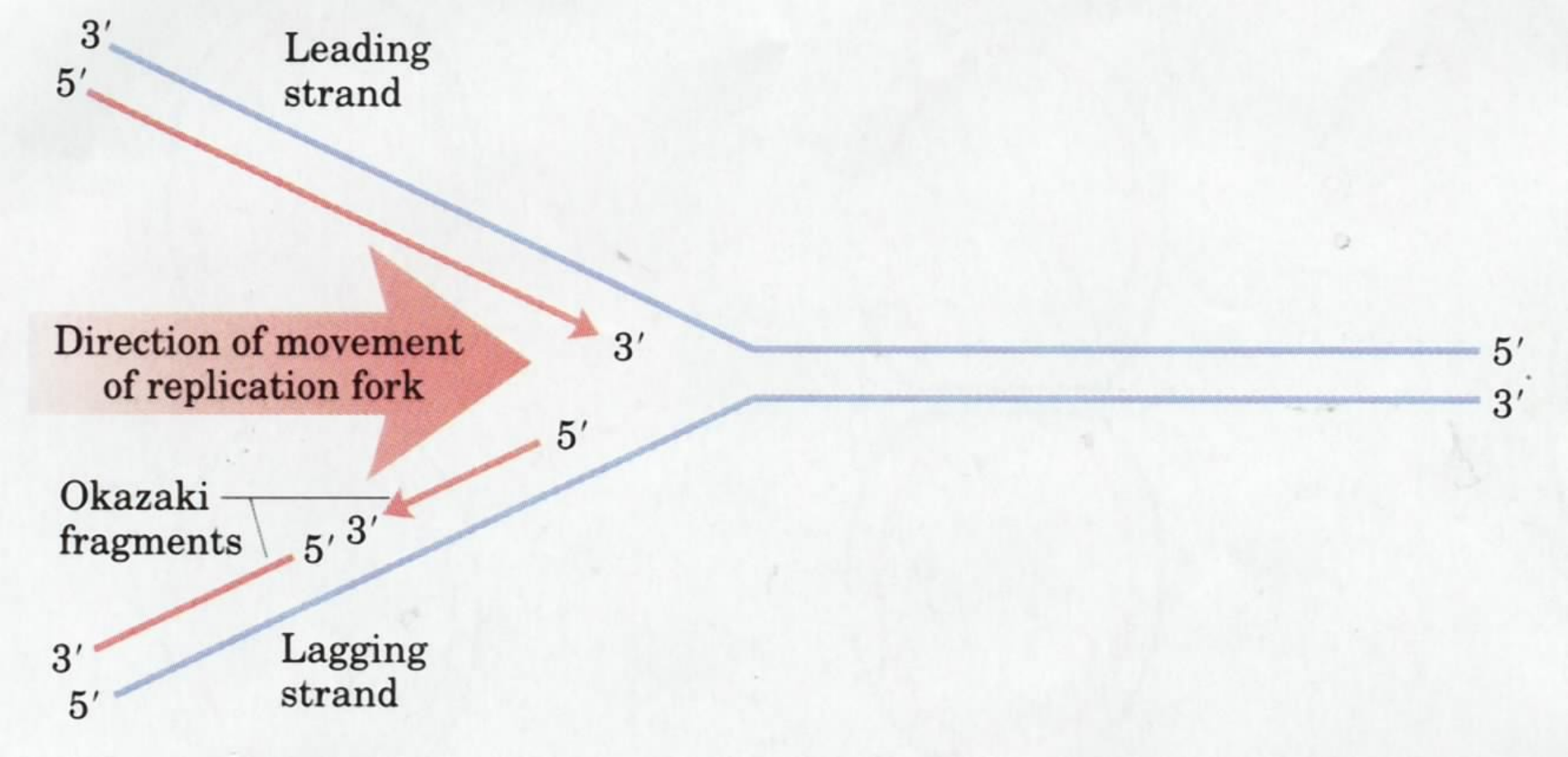
9. The synthesis of the two daughter DNA strands was found to be semidiscontinuous

At a replication fork, the overall elongation direction for one daughter strand is 5` to 3` and 3` to 5` for the other due to the antiparallel features of DNA duplexes. Reiji Okazaki discovered (in the 1960s) that a significant proportion of newly synthesized DNA exists as small fragments!



These so-called Okazaki fragments were found to join together by DNA ligases to form one of the daughter strands;
Thus both daughter strands are synthesized in 5` to 3` direction.
One daughter strand at the replication fork is synthesized continuously and the other discontinuously, called the leading and lagging strands respectively.





Both daughter strands at the replication fork are synthesized in 5’→3’ direction, but one (the leading strand) is synthesized continuously and the other (the lagging strand) discontinuously (synthesized initially as Okazaki fragments).

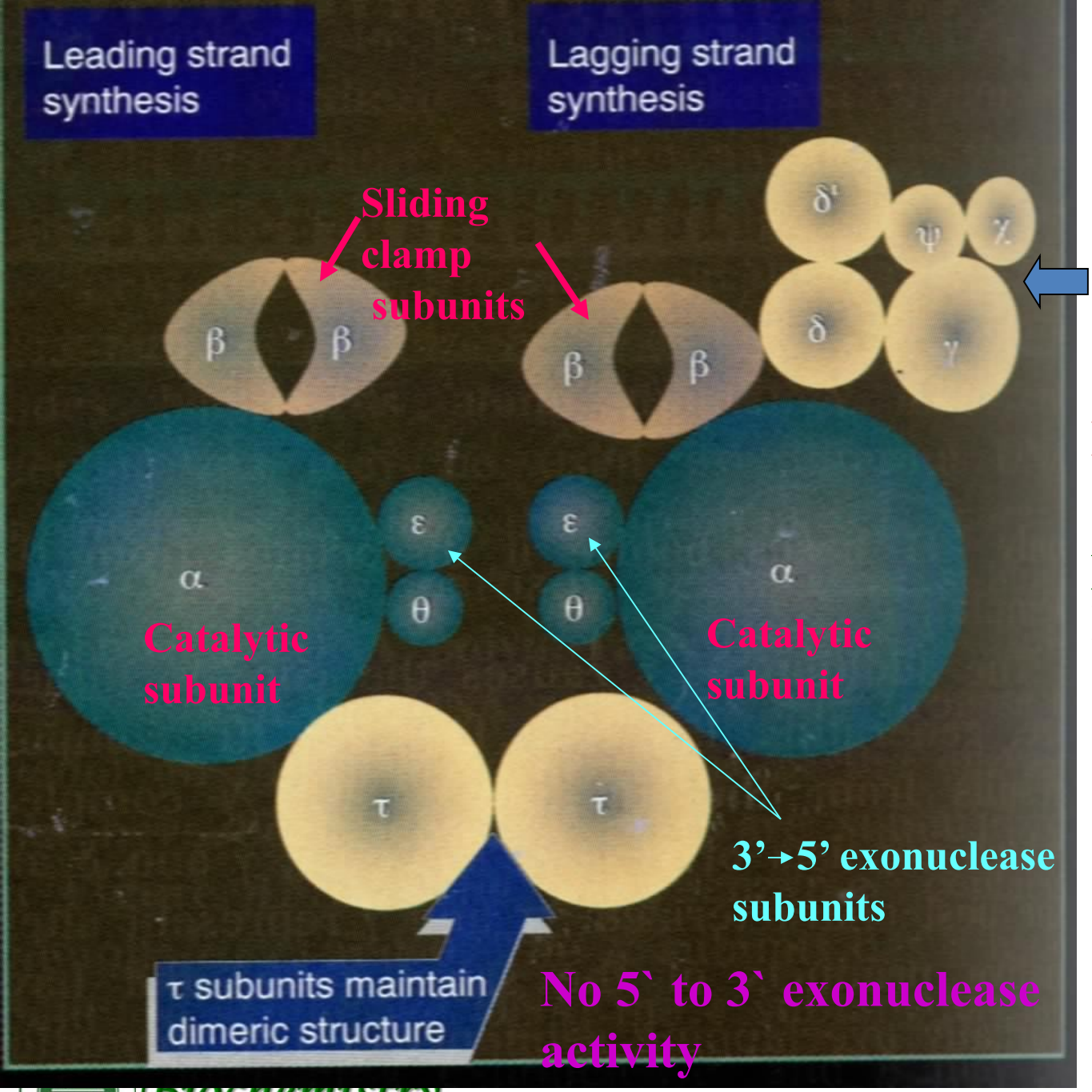
10. DNA polymerase III is responsible for DNA replication in *E.coli* cells

◆ **DNA polymerase III** is the principle replication enzyme in *E.coli*.

TABLE 25–1 Comparison of DNA Polymerases of *E. coli*

	DNA polymerase		
	I	II	III
Structural gene*	*polA*	*polB*	*polC* (*dnaE*)
Subunits (number of different types)	1	7	≥10
M_r	103,000	88,000†	791,500
3′→5′ Exonuclease (proofreading)	Yes	Yes	Yes
5′→3′ Exonuclease	Yes	No	No
Polymerization rate (nucleotides/s)	16–20	40	250–1,000
Processivity (nucleotides added before polymerase dissociates)	3–200	1,500	≥500,000





Proposed architecture of DNA polymerase III holoenzyme: an asymmetric dimer

TABLE 25–2 Subunits of DNA Polymerase III of *E. coli*

Subunit	Number of subunits per holoenzyme	M_r of subunit	Gene	Function of subunit	
α	2	129,900	*polC (dnaE)*	Polymerization activity	Core polymerase
ε	2	27,500	*dnaQ (mutD)*	3′→5′ Proofreading exonuclease	
θ	2	8,600	*holE*	Stabilization of ε subunit	
τ	2	71,100	*dnaX*	Stable template binding; core enzyme dimerization	Clamp-loading (γ) complex that loads β subunits on lagging strand at each Okazaki fragment
γ	1	47,500	*dnaX**	Clamp loader	
δ	1	38,700	*holA*	Clamp opener	
δ'	1	36,900	*holB*	Clamp loader	
χ	1	16,600	*holC*	Interaction with SSB	
ψ	1	15,200	*holD*	Interaction with γ and χ	
β	4	40,600	*dnaN*	DNA clamp required for optimal processivity	

11. DNA polymerase III and many other proteins are part of the replisome for DNA replication in *E.coli* cells

In vitro **studies revealed about 20 proteins are involved in DNA replication in *E.coli* cells.**

Accessory proteins include:

① **the helicase（解螺旋酶） for moving along the DNA and separating the two DNA strands using energy from ATP;**

② **the single stranded DNA-binding (SSB) protein for binding and stabilizing the single stranded DNA generated;**

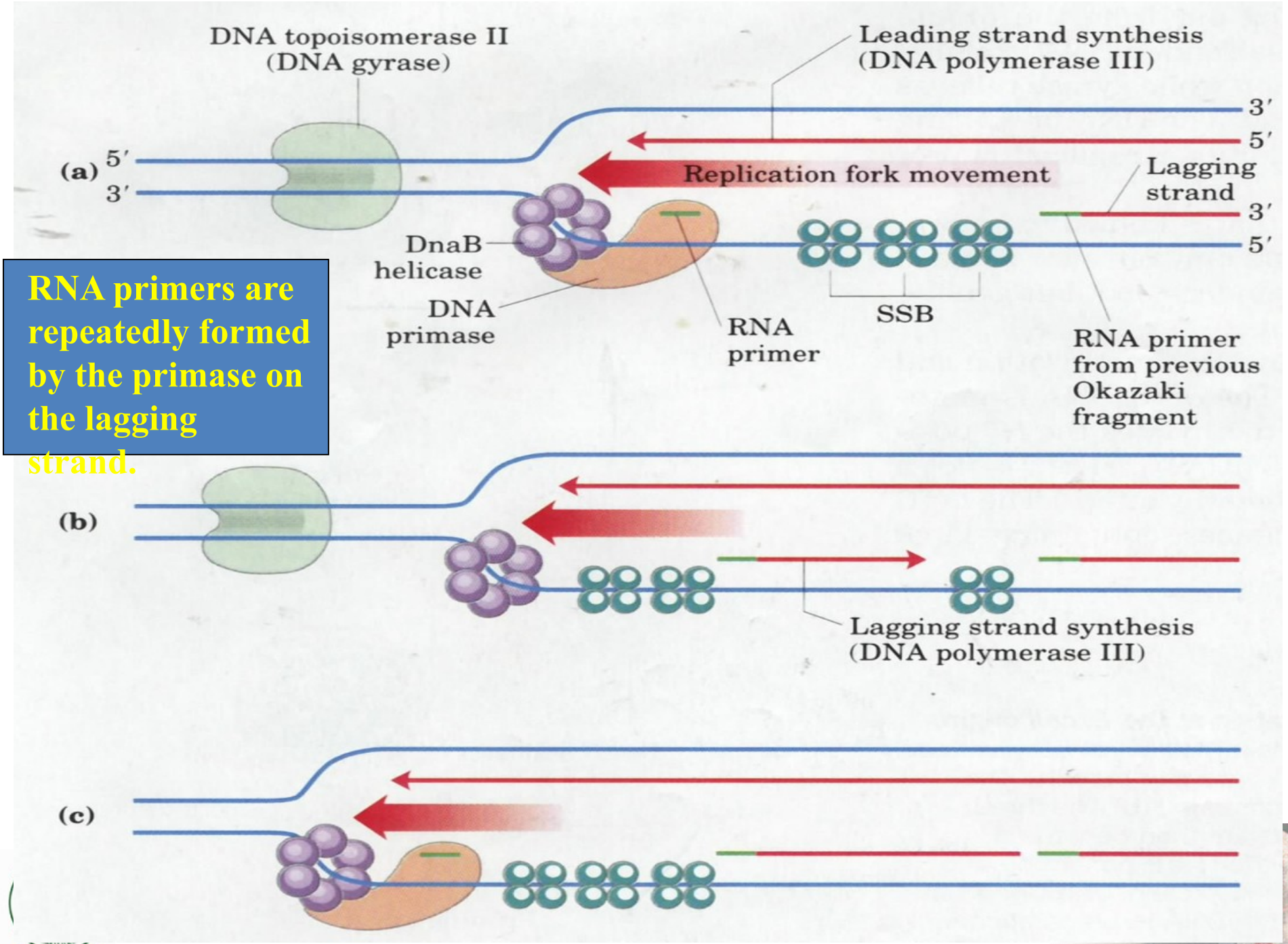


③ the topoisomerases （拓扑异构酶） for relieving topological (torsional) strains in the helical structure (positive supercoils) generated during strand separation;

④ the primase for generating a short RNA primer;

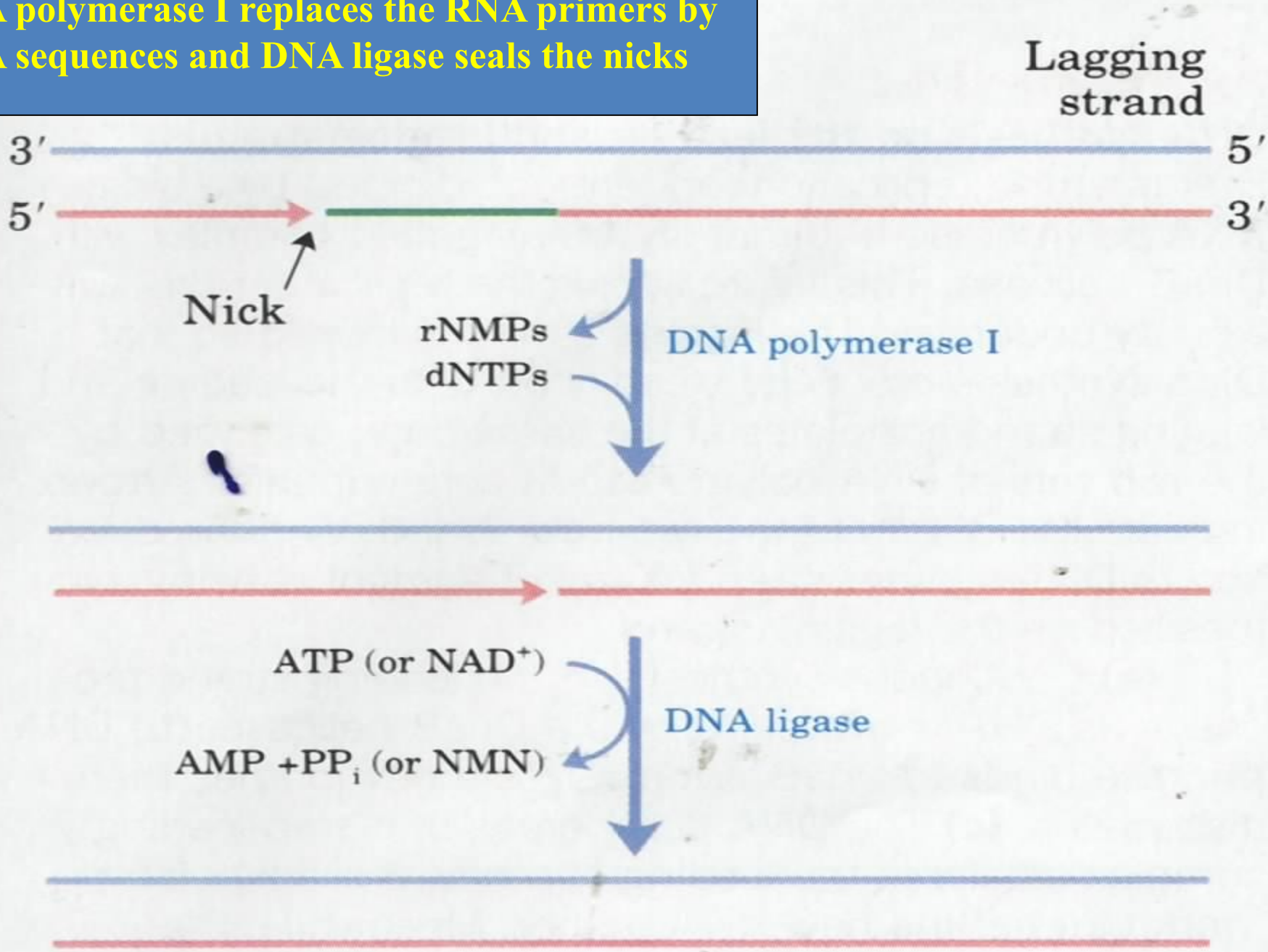
⑤ the DNA polymerase III for polymerizing and proofreading the nucleotides according to the templates;

⑥ the DNA polymerase I for removing the RNA primers and replacing by a DNA

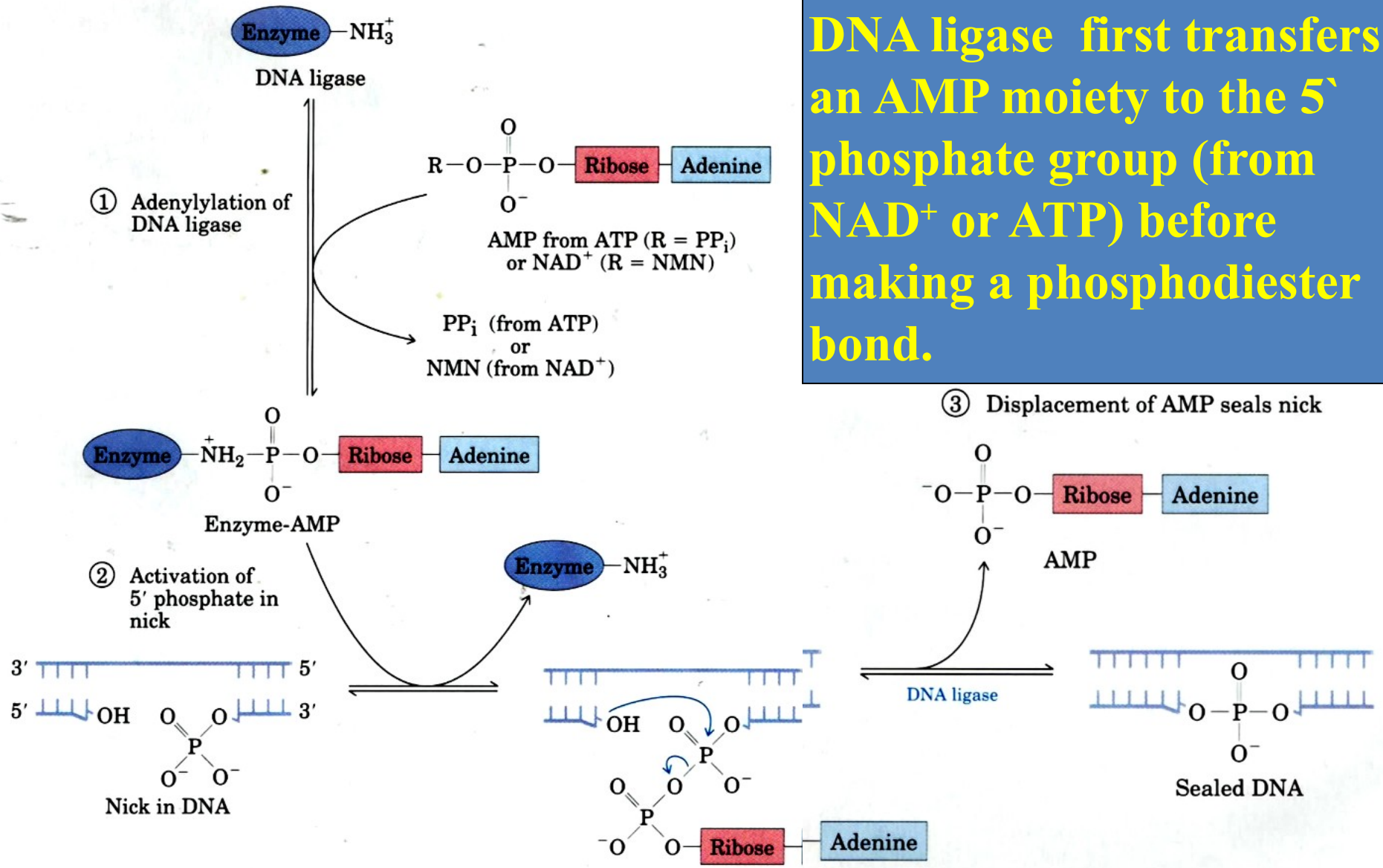


RNA primers are repeatedly formed by the primase on the lagging strand.

DNA polymerase I replaces the RNA primers by DNA sequences and DNA ligase seals the nicks



DNA ligase first transfers an AMP moiety to the 5` phosphate group (from NAD^+ or ATP) before making a phosphodiester bond.



Biochemistry

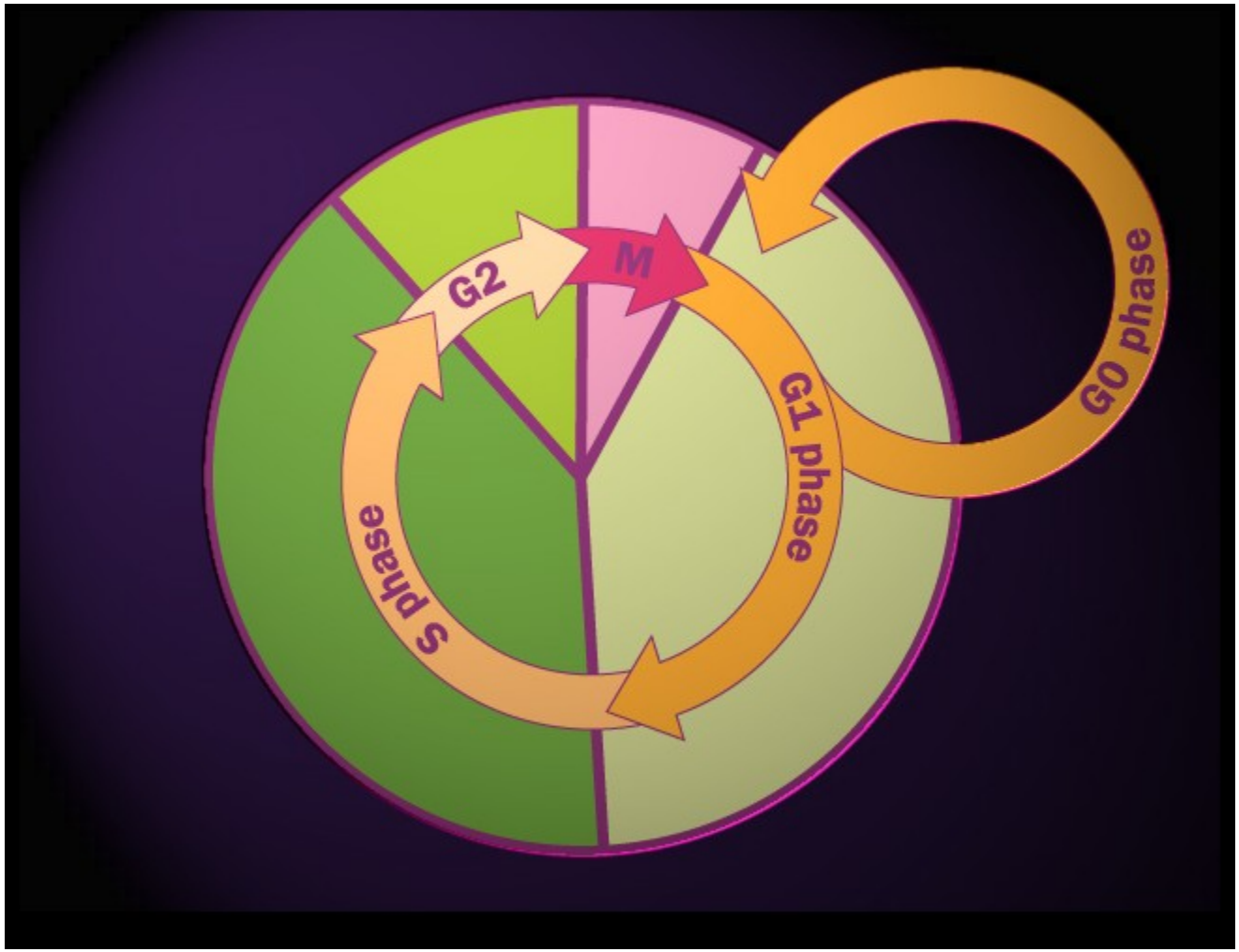
12. DNA replication in eukaryotic cells use essentially the same principles but more complex in the details (F4)

The rate of DNA synthesis in eukaryotic cells is about one tenth of that of the *E.coli* cells.

Replication in the eukaryotic genomes begin at many replication origins.

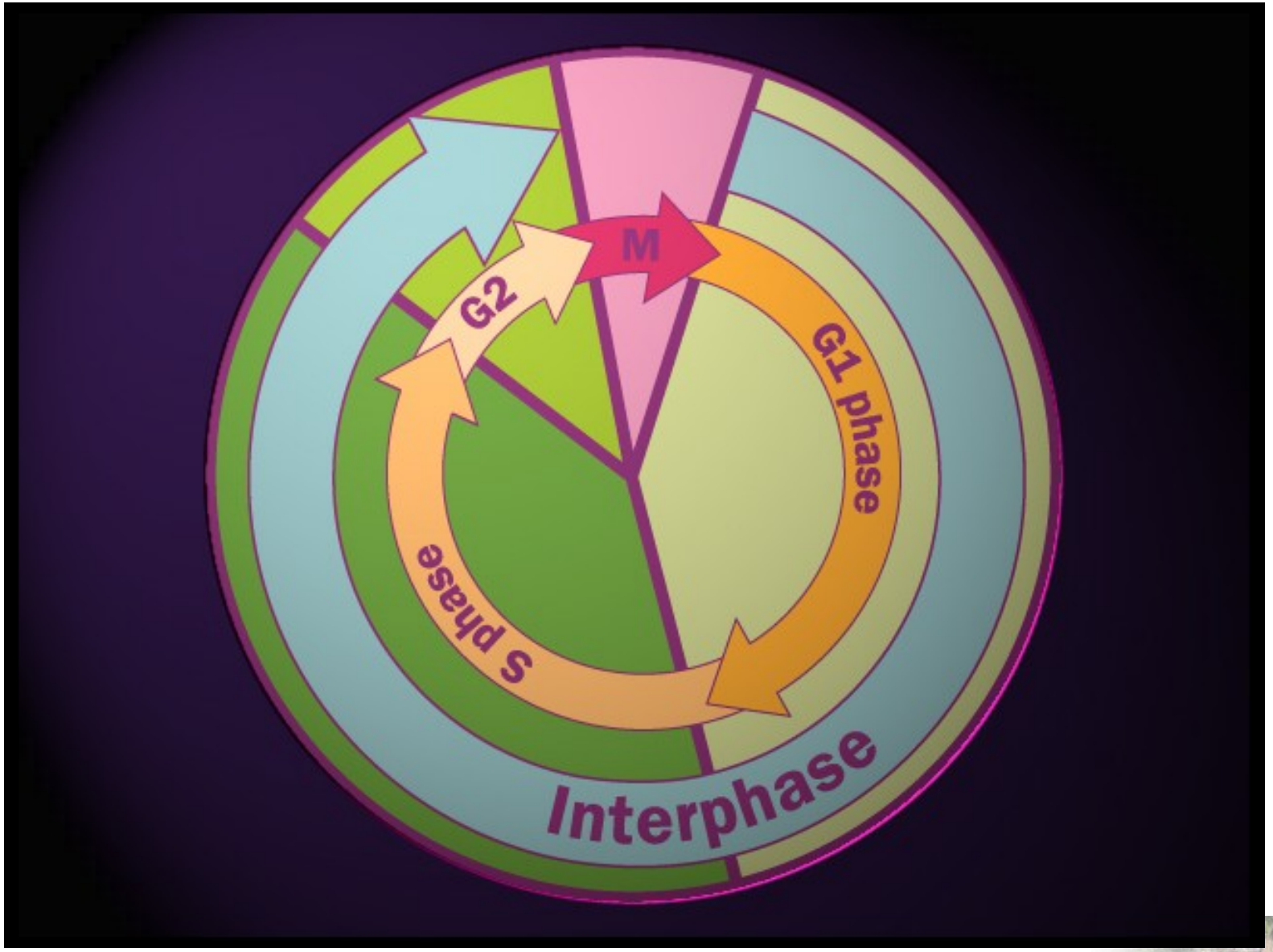


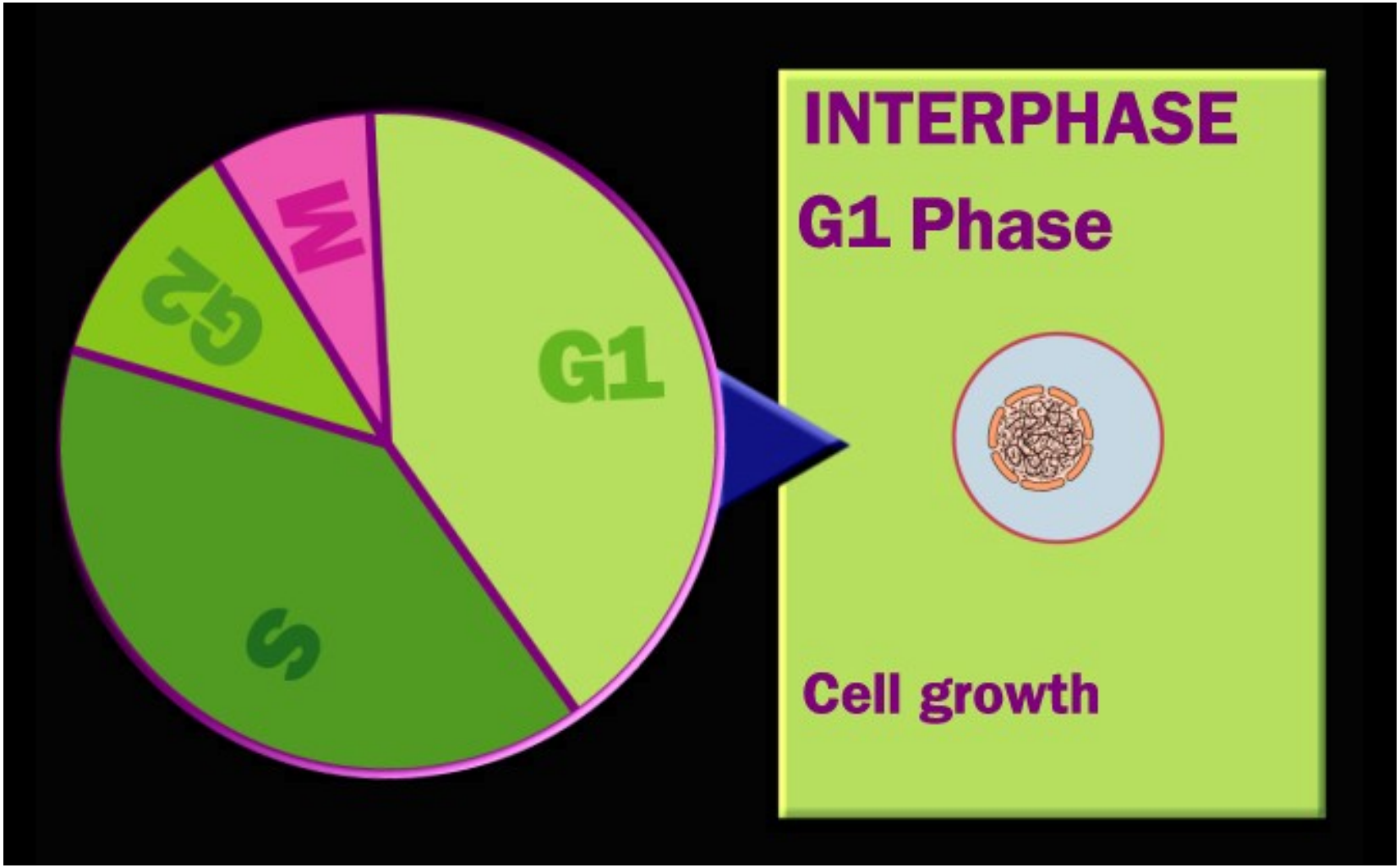
12.1 Cell cycle

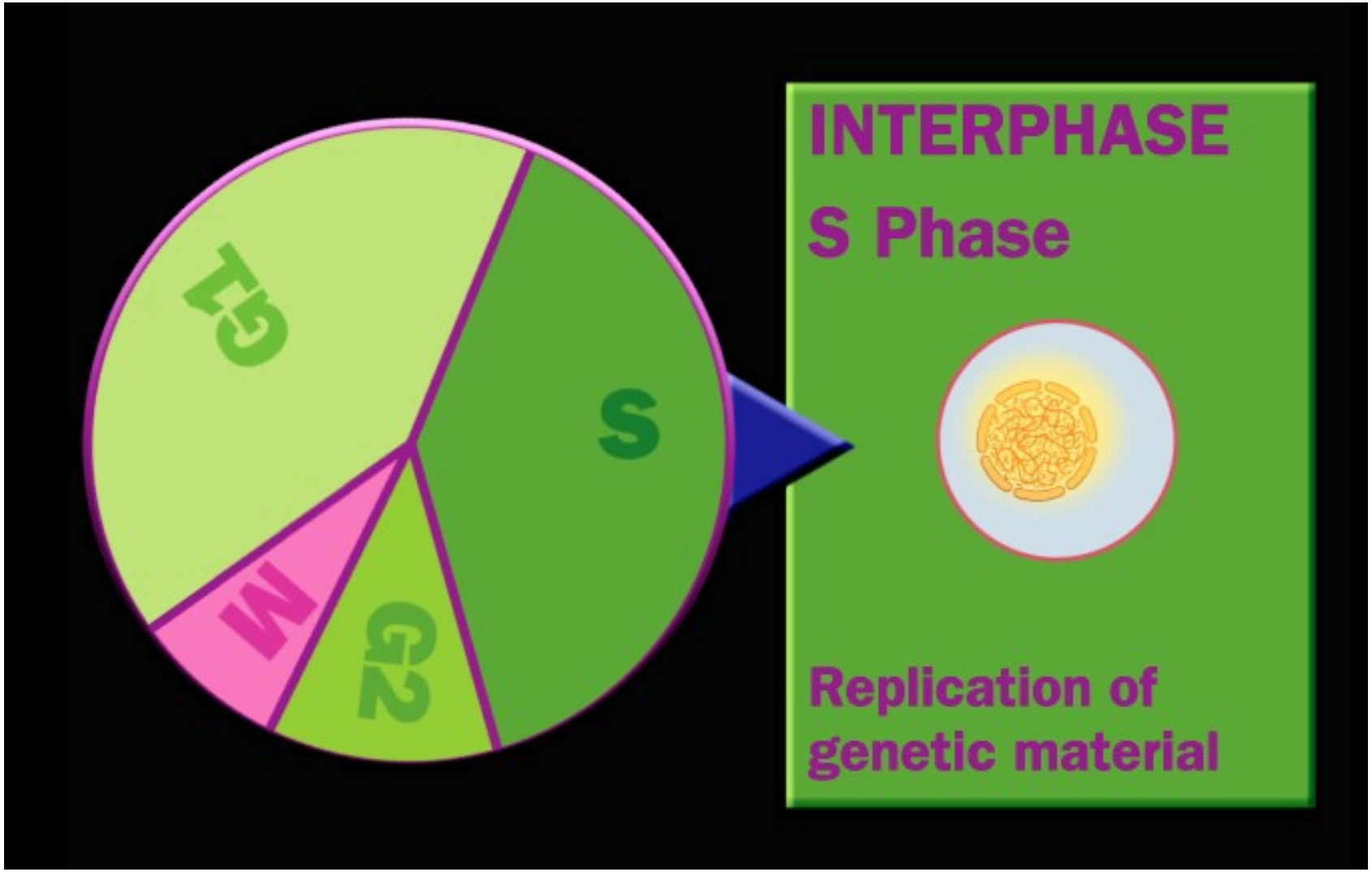


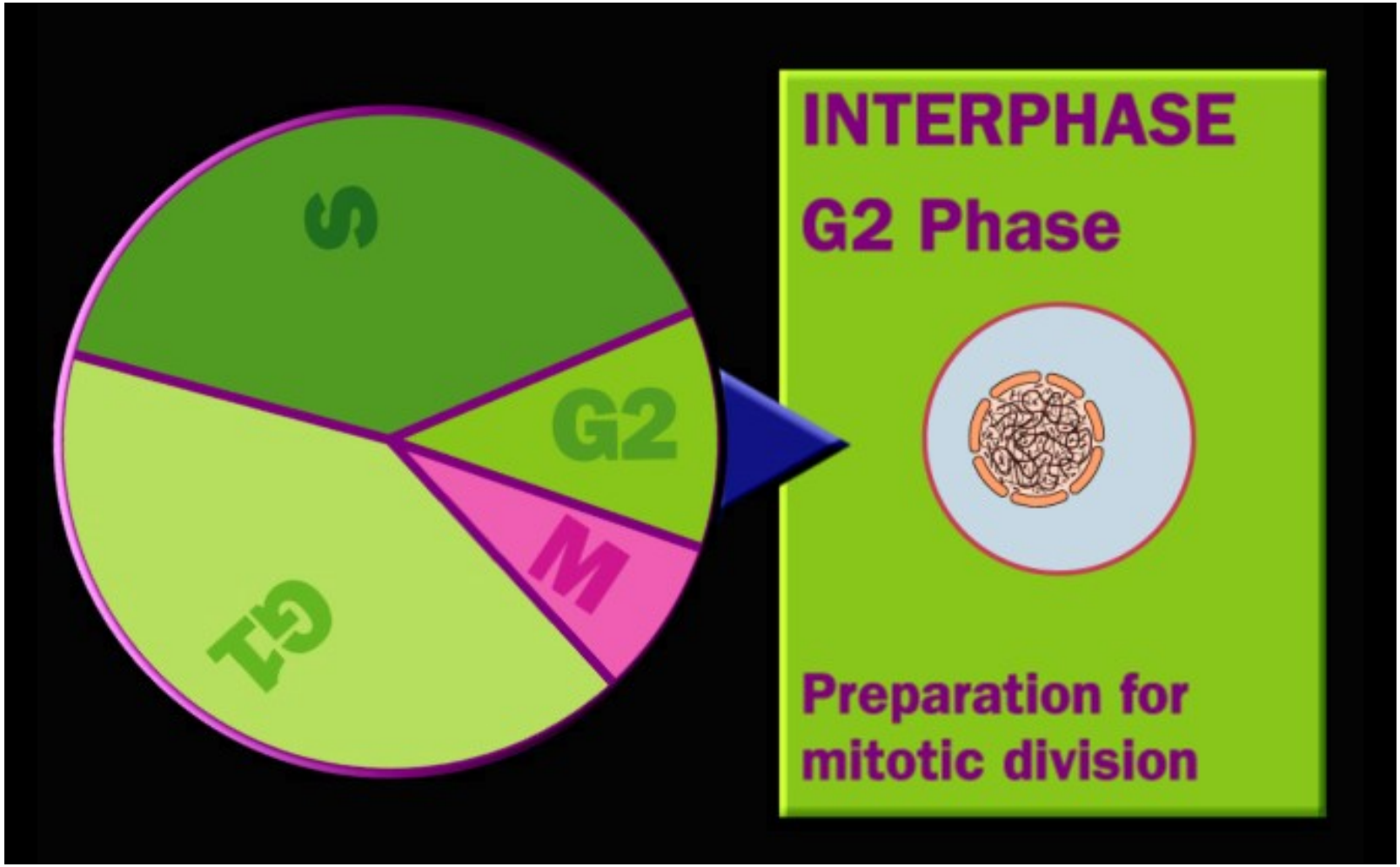
In eukaryotes, the cell cycle consists of G_1, S, G_2 and M phases. Quiescent cells are said to be in G_0 phase.

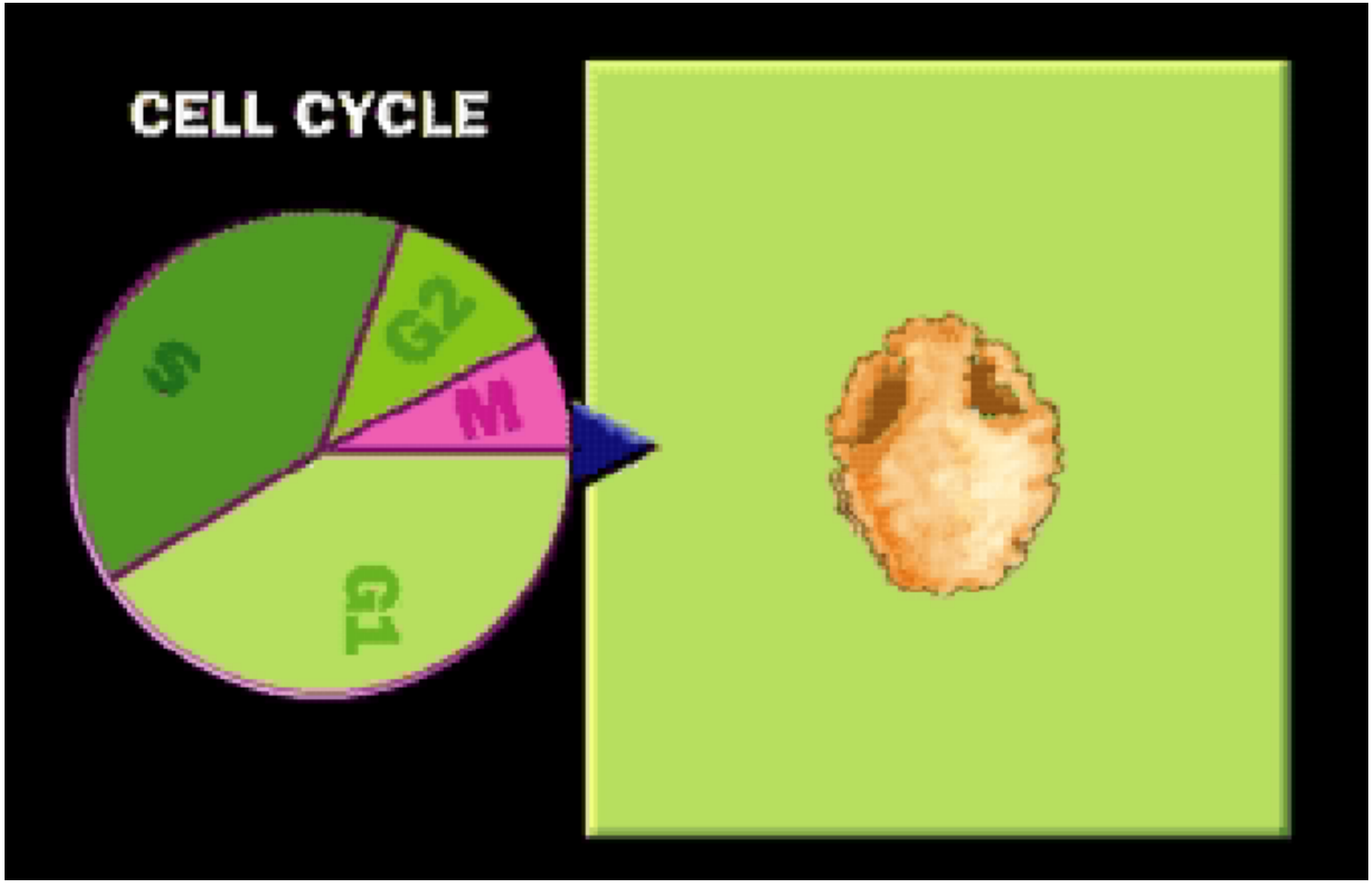
G (gap); S (synthesis); M (mitosis)


M
G2
G1 phase
S phase
Interphase


M
G2
G1
S
INTERPHASE
G1 Phase
Cell growth


G1
S
G2
M
INTERPHASE
S Phase
Replication of genetic material


S
G2
M
G1
INTERPHASE
G2 Phase
Preparation for
mitotic division

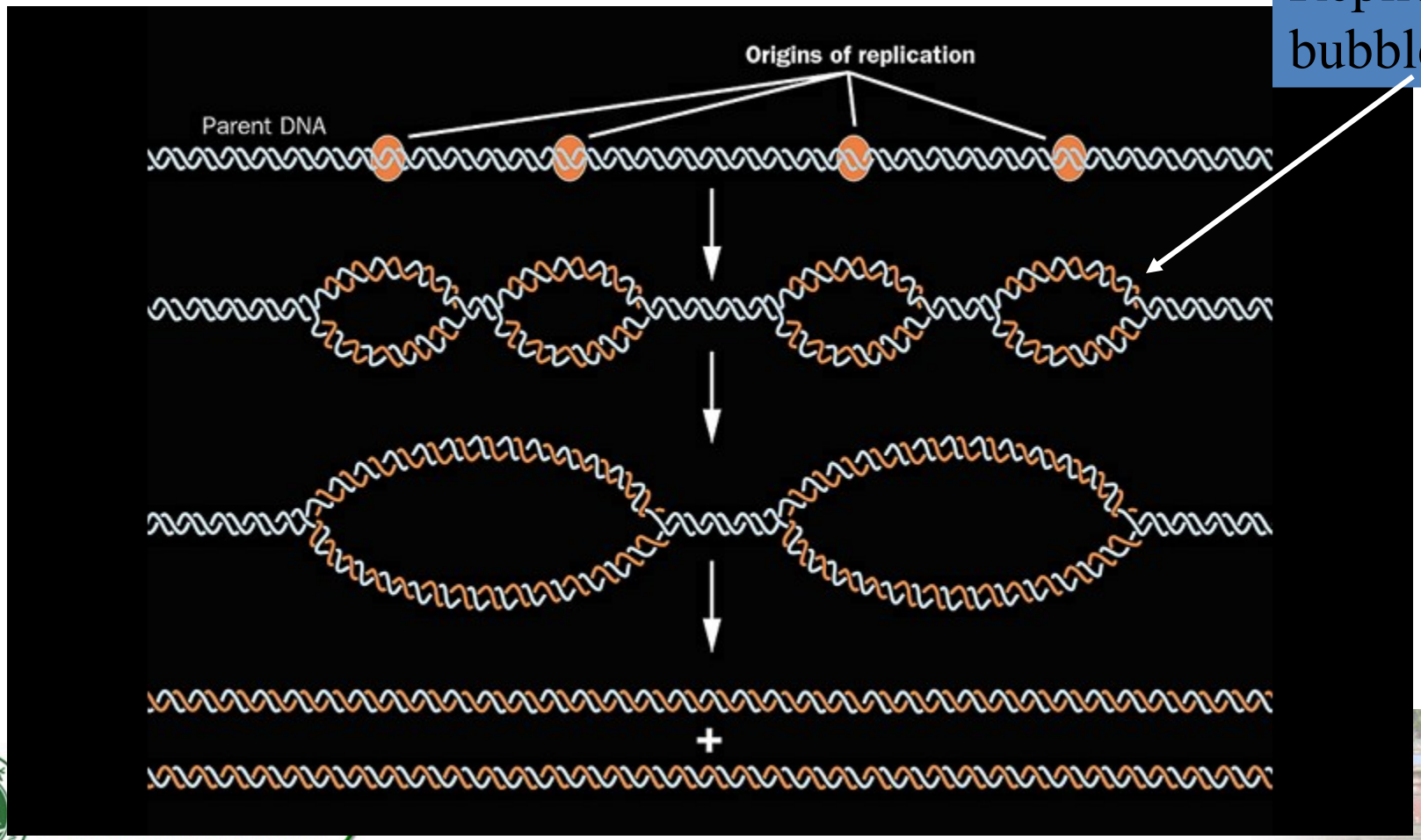


Mitosis and cell division occur in the M phase which lasts for only about 1 h.

12.2 DNA replication occurs only in the S phase.

many chromosomal origins (multiple replicons);

bi-directional; semi-conservative



12.3 Eukaryotic cells contain five different DNA polymerases

DNA polymerases α and δ replicate chromosomal DNA

DNA polymerases β and ε repair DNA

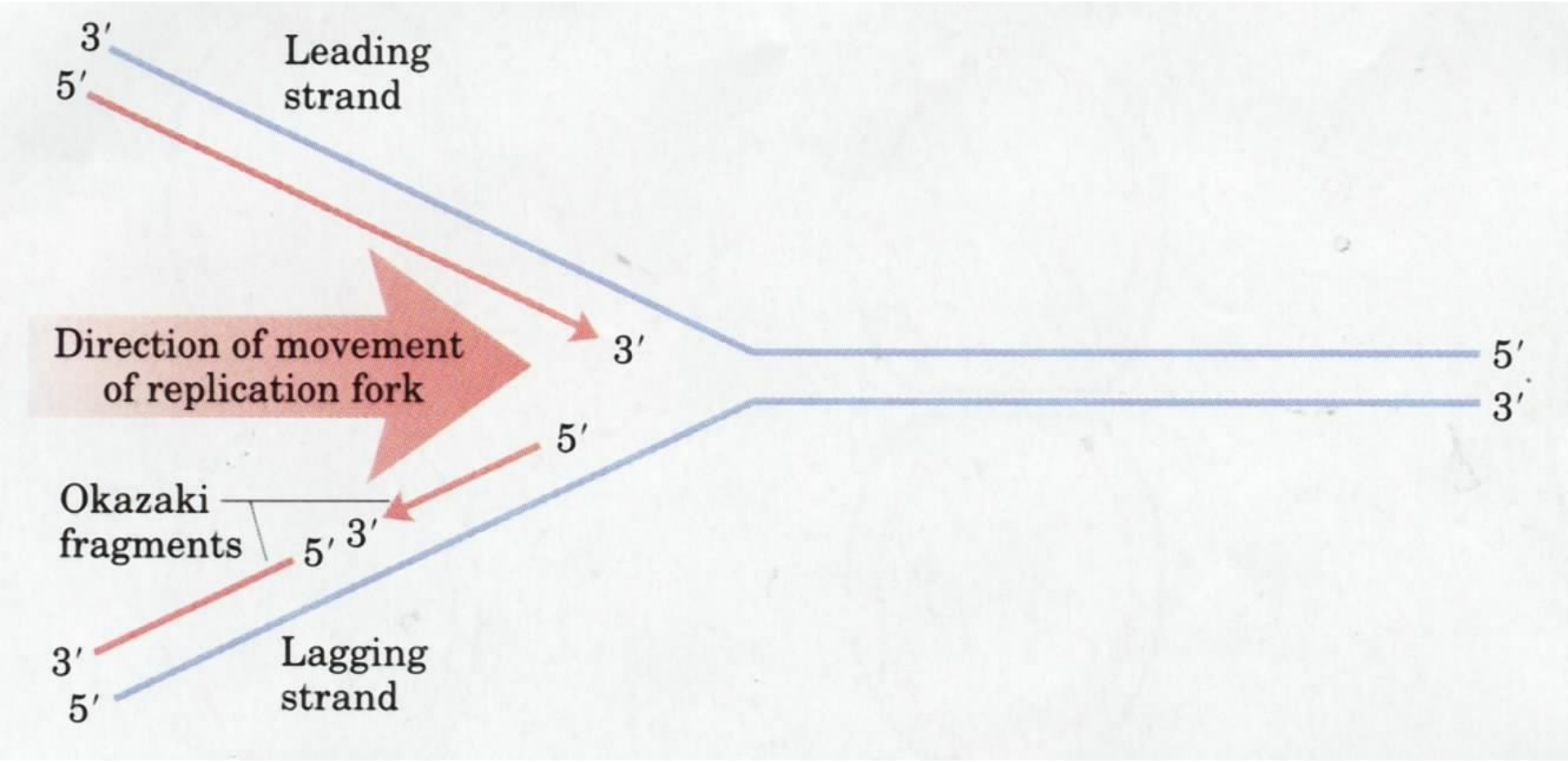
DNA polymerase γ replicates mitochondral DNA.

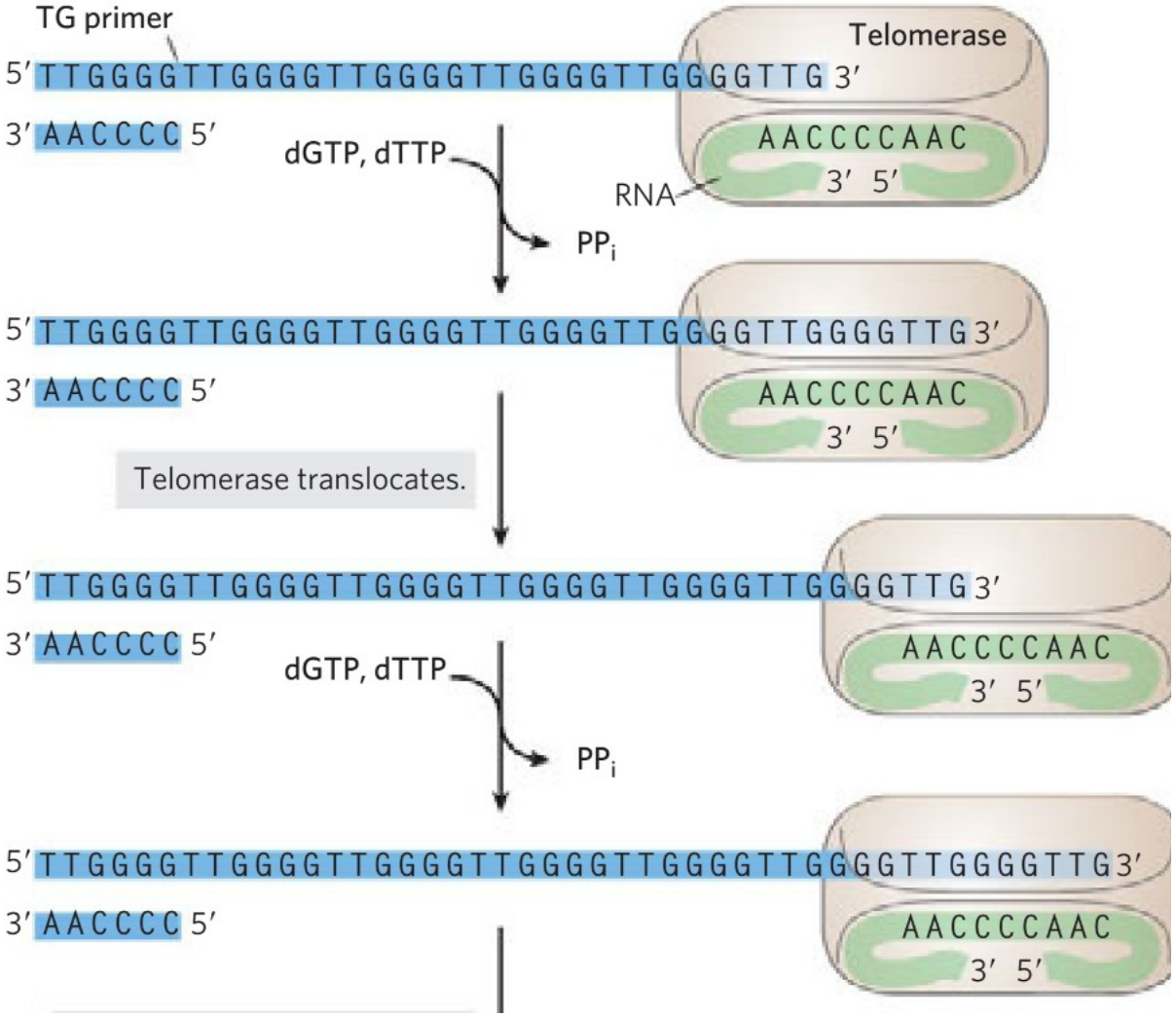


12.4 Telomere replication (端粒复制）

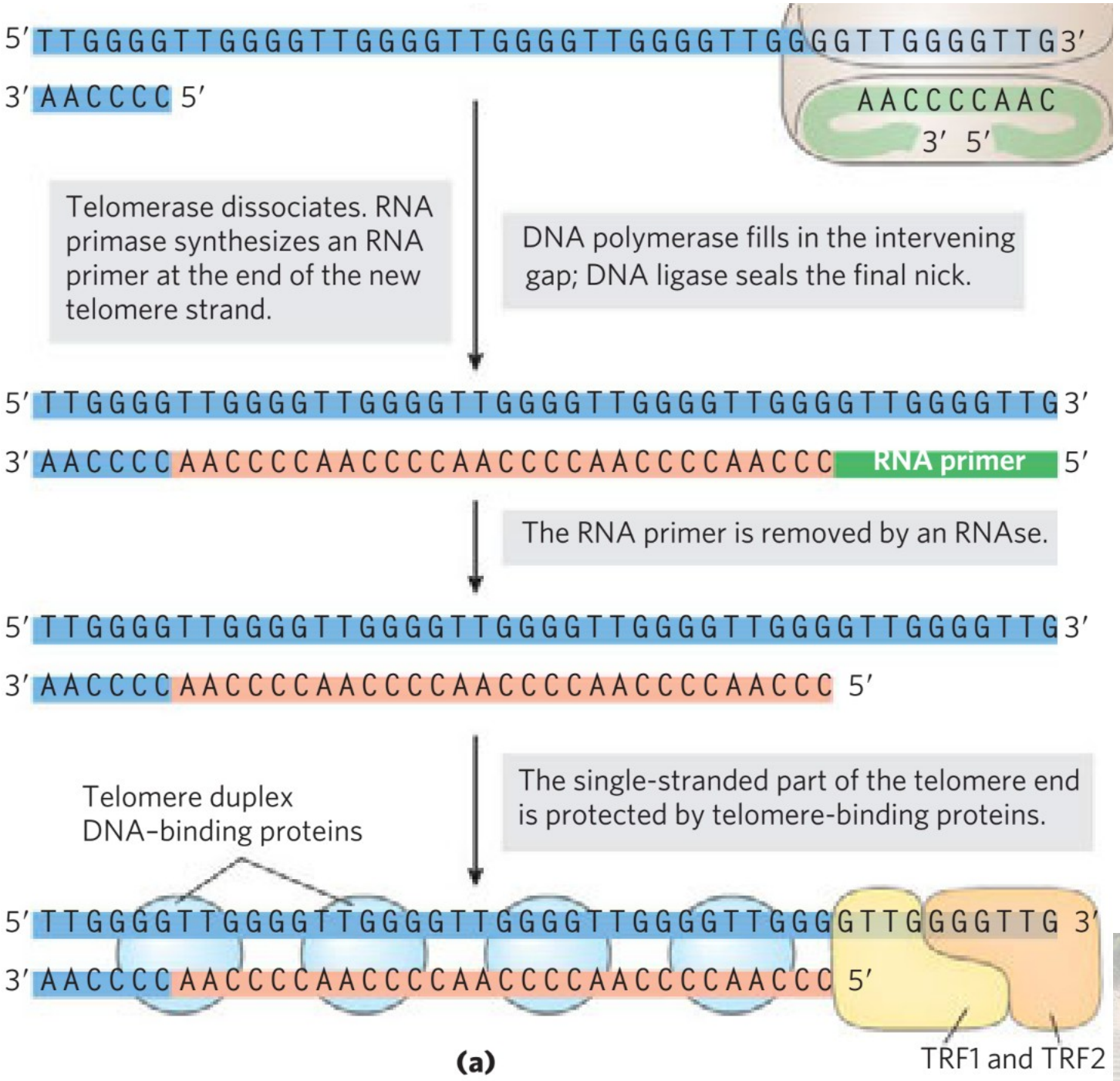
Different with the circular DNA in bacteria, eukaryotic DNA is a linear molecule. According to the mechanism of replication, the 3' end of lagging chain is not replicated. This creates a gap at the end of the chromosome and therefore a shortening of the double-stranded replicated portion. In many organisms the solution is to use an enzyme called telomerase (端粒酶） to replicate the chromosome ends (telomeres).




3′
5′
Leading strand
Direction of movement of replication fork
3′
5′
3′
5′
Okazaki fragments
5′
3′
3′
5′
Lagging strand



Telomerase, a DNA polymerase that contains an integral RNA that acts as its own primer, is used to replicate DNA at the ends of chromosomes (telomeres).


5′ TTGGGGTTGGGGTTGGGGTTGGGGTTGGGGTTGGGGTTGGGGTTG 3′
3′ AACCCC 5′
AACCCCAAC
3′ 5′
Telomerase dissociates. RNA primase synthesizes an RNA primer at the end of the new telomere strand.
DNA polymerase fills in the intervening gap; DNA ligase seals the final nick.
5′ TTGGGGTTGGGGTTGGGGTTGGGGTTGGGGTTGGGGTTGGGGTTG 3′
3′ AACCCCAACCCCAACCCCAACCCCAACCCCAACCCCAACCC RNA primer 5′
The RNA primer is removed by an RNAse.
5′ TTGGGGTTGGGGTTGGGGTTGGGGTTGGGGTTGGGGTTGGGGTTG 3′
3′ AACCCCAACCCCAACCCCAACCCCAACCCCAACCCCAACCC 5′
Telomere duplex DNA–binding proteins
The single-stranded part of the telomere end is protected by telomere-binding proteins.
5′ TTGGGGTTGGGGTTGGGGTTGGGGTTGGGGTTGGGGTTGGGGTTG 3′
3′ AACCCCAACCCCAACCCCAACCCCAACCCCAACCCCAACCC 5′
TRF1 and TRF2

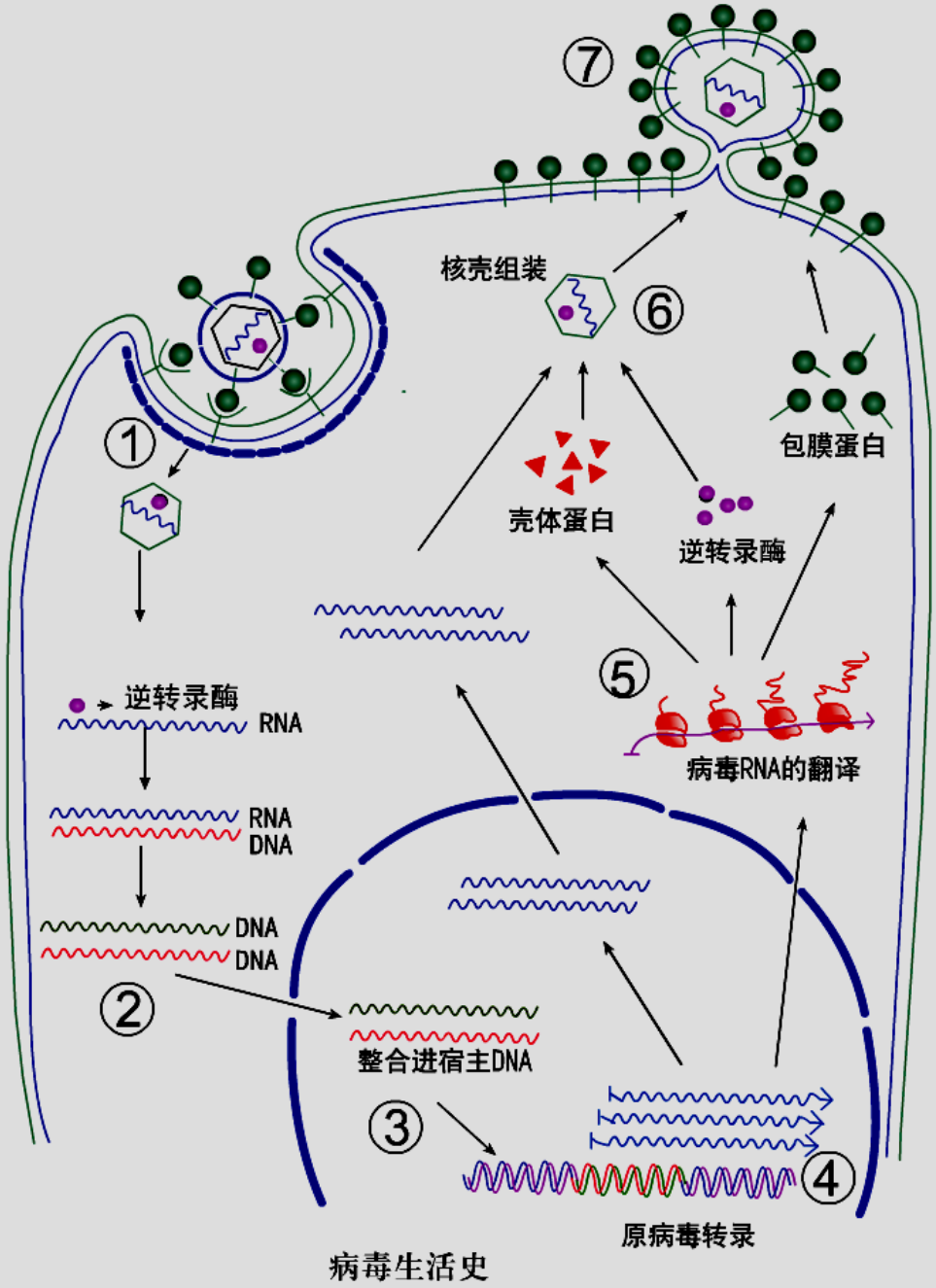
(a)

12.5 Nucleosomes do not dissociate from the DNA during DNA replication.

When a chromosome is replicated, the histones stay in place but somehow must allow the replication machinery to pass through and make new DNA. One suggestion is that the nucleosome histone octamer transiently unfolds into two half-nucleosomes to allow the replication machinery access to the DNA. The new DNA must also be packaged into nucleosomes and so histones are also synthesized during the S phase of the cell cycle.



逆转录病毒


⑦
核壳组装
⑥
包膜蛋白
①
壳体蛋白
逆转录酶
⑤
逆转录酶
RNA
病毒RNA的翻译
RNA
DNA
DNA
DNA
②
整合进宿主DNA
③
④
原病毒转录
病毒生活史

END

Thank you



Date： 2018/10/18

Name： Student number：

Class：

Q: How does a DNA molecule replicate with high fidelity in E. coli ?

A：

